Dermonics

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Here are some high-yield snippets from my soon-to-be released mnemonics-packed dermatology boards review book. Current estimated release date is October 1st, 2024 but definitely no later than December. If this style of learning calls to you then make sure to preorder the book now for half off of the post-release price. Happy studying.

1.1 STRUCTURE AND FUNCTION OF THE SKIN

Structure of the epidermis

- Epidermis is composed of squamous epithelium comprised of keratinocytes connected by Desmosomes, Adherens junctions, Tight junctions and Gap junctions (“**DAT Gap**”)
    - Desmosomes connect keratinocytes by anchoring to **keratin tonofilaments**
        - Mediates **slow but strong** cellular adhesions (“Desmo-slowms”)
        - Consist of **intercellular desmoglein/desmocollin** that binds cytoplasmic **plakoglobin** that binds desmoplakin/plakophilin that binds **keratin**
    - Adherens junctions connect keratinocytes by anchoring to **actin** filaments
        - Mediates **fast but weak** cellular adhesions
        - Consist of intercellular classic cadherins (P-, E-, N-cadherins) that binds cytoplasmic β-catenin/plakoglobin that binds α-catenin that binds actin
            - Notice plako-globin is present in both (“present globally”)
    - Tight junctions are composed of claudins and occludins which form a tight seal
    - Gap junctions allows for intercellular communication and transfer of small molecules between cells; gap junction defects are the MCC of genetic deafness

DAT gap

Desmosome proteins (Cadherin family)

- Desmocollins (associated condition when defected is listed)
    - Desmocollin 1: Subcorneal pustular type of IgA pemphigus
- Desmogleins: **PISS ARV Vulgar ALARM** (“Piss off vulgar alarm”)
    - Desmoglein 1 (PISS): Pemphigus foliaceus, Impetigo (bullous), SSSS, Striate PPK
        - Expressed in all levels of epidermis in skin (superficial > deep)
    - Desmoglein 2 (ARV): Arrhythmogenic Right Ventricular dysplasia
    - Desmoglein 3 (Vulgar): P. Vulgaris (oral PV only Dsg3; mucocutaneus PV is 1 & 3)
        - Expressed in just deep epidermis and all levels of mucosa and fetal skin
    - Desmoglein 4 (ALARM): AR Localized hypotrichosis, AR Monilethrix
        - Expressed in hair follicles

Desmosome proteins (Armadillo family): Armadillo family always **starts** with “plako”

- Plakoglobin (PG): Naxos syndrome (“The movie is not PG if it’s X-rated”)
    - Present in both desmosomes and adherans junctions (“present globally”)
- Plakophillin (PP): ectodermal dysplasia with skin fragility (“your PP is fragile”)

Desmosome proteins (Plakin family): “Plakin family usually **ends** with plakin”

- Desmoplakin 1 (250 kDa) and 2 (210 kDa) (“Plays at his desk in the principle’s CLASSS”)
    - Paraneoplastic pemphigus (antibodies to DSG 1, 3 and entire plakin family)
    - Carvahal syndrome
    - Lethal Acantholytic EB

- Striate PPK (mutation in desmoplakin 1, DSG 1 and keratin 1: “straight like #1”)
- Skin fragility/wooly hair syndrome
- SJS/EM Major (anti-desmoplakin antibodies)

- Remember Plakins and their molecular weight with **PEB DP** mnemonic
  - Periplakin (190 kDa)
  - Envoplakin and Desmoplakin 2 (210 kDa)
  - BPAg1 (230 kDa)
  - Desmoplakin 1 (250 kDa)
  - Plectin (500 kDa)

Adherans junction proteins of importance

- Classic cadherans (E-, P- and N- cadherin) attach to either plakoglobin or β-catenin (armadillo family) which attach to α-catenin which attaches to actin filaments
  - Plakoglobin mutated in Naxos disease
  - β-catenin mutations are seen in pilomatricomas (which are common in Myotonic dystrophy, garners syndrome (APC gene), Turners and Rubinstein-Taybi

Tight junctions have no important disease associations

Gap junction mutations: “**KEV & Deaf Bart** go **HOMe**”

- KEV & Deaf Bart are all **Connexin 26 (GJB2) mutations except** Erythrokeratoderma variabilis which is Connexin 30.3 and 31 (GJB4 and GJB3; “variable/multiple connexins”)
  - KID (Keratitis-ichthyosis-deafness) syndrome
  - Erythrokeratoderma variabilis
  - Vohwinkel syndrome
  - PPK with deafness
  - Bart-Pumphrey syndrome
- HOMe (“say BAC HOM to keep the GJs straight”)
  - Hidrotic ectodermal dysplasia aka Clouston’s: Connexin 30 (GJB6)
  - Oculo-dental-digital dysplasia: Connexin 43 (GJA1)
  - Milroys disease (MC mutation is FLT4): Connexin 47 (GJC2) is 2nd MC

Epidermal layers

Stratum basale

- Keratins 5 and 14 are produced here
- About 10% of cells in the basal layer are stem cells and about 10% are melanocytes
- Expresses ornithine decarboxylase (a/w proliferative states) and catalase
  - Increased by factors such as trauma (Blunt force trauma) and UVB
  - Inhibited by Vitamin D3, Corticosteroids, Retinoids

    “It’s OD (Ornithine Decarboxylase) that the VCR in the basement (basal layer) is not working (inhibited)”

Stratum spinosum: “spines look like a 1 or an l → lame land, lys lip, leg leq, link”

- Keratins 1 and 10 are produced here
- Lamellar granules (Odland bodies) are produced by Golgi bodies in the spinous layer
    - Specialized lysosomes which discharge lipids (most important is **Ceramide**) that extrude their contents to form the intercellular lipids in the stratum corneum
    - Flegel’s disease and Harlequin ichthyosis are due to ↓ lamellar granules
    - X-linked ichthyosis occurs due to absent steroid sulfatase in lamellar granules

Stratum granulosum: “KLIP your basophilic keratohyalin granules”

- Keratin intermediate filaments: binds filaggrin
- Loricrin: Largest component of CCE (mutated in Vohwinkel variant/PSEK; ↓ in psoriasis)
- Involucrin: cross-links with loricrin by transglutaminase (↑ in psoriasis)
- Profillagrin: becomes fillagrin (↑ in lamellar ichthyosis; ↓ in ichthyosis vulgaris and AD)
- Cornified cell envelope (CCE) production primarily takes place in the s. granulosum
- Cross-linking of CCE occurs via Transglutaminase 1 (TG mnemonic is “I’D Peel”)
    - Icthyosis (Lamellar/CIE): Transglutaminase 1 is mutated
    - Dermatitis herpetiformis: Transglutaminase 3 is antigenic
    - Acral peeling skin syndrome: Transglutaminase 5 is mutated

Stratum corneum

- Serves as a water-impermeable mechanical barrier between the epi and environment
- Made of corneocytes which are terminally differentiated keratinocytes that contain a cornified envelope composed of cross-linked proteins (**loricrin**, involucrin, periplakin)
- **Urocanic acid** (“uro-corneum acid”) is a filaggrin biproduct found in the cornified layer
    - **Absorbs UVR** and forms natural moisturizing factor which hydrates s. corneum
- **Major protein component is loricrin**; **major lipid component is ceramide**

Epidermal cells

Keratinocytes and keratin

- Keratinocytes produce keratin filaments which form the cell’s cytoskeletal network
    - Desmin, vimentin, neurofilaments (“neurofinaments”), nuclear lamins and nestin are types of keratin intermediate filaments
- Class 1 keratins: low molecular weight; acidic; K9-28
    - “1st class girls with low body weights and nice ass get higher ratings (9-28)”
- Class 2 keratins: high molecular weight; basic; K1-8
    - “2nd class girls with high body weights are basic and get lower ratings (1-8)”
- There are > 50 epithelial/hair keratins expressed as either type I (acidic), type II (basic), or type I/II coexpressed together as heterodimers (i.e. K5/14) aka **double helix structure** which are bonded together with disulfide bonds utilizing the many cysteine amino acids
- **Basic structure is an α-helical rod consisting of heptad (7) amino acid repeats** (“keratino-sevens”) vs collagen which is a triple helix w/ triple repeat sequence (Gly-X-Y)

    - "Keratino-sevens" also helps you remember that collagen 7 and 17 are produced by keratinocytes, all other collagens (except 8/18) are produced by fibroblasts
- Release and respond to various cytokines most notably **TNF-α** ("keraTiNoFytes")

Keritin filament expression pattern	
Some Girls Can Make Boys Have Nice Puberties	Some Girls Can Make Boys Have Nice Sex
K1: Spinous layer → Ichthyosis Hystrix, Unna thost (non-epidermolytic) PPK, Greithers, EHK (HUGE)	K10: Spinous layer→ Epidermolytic hyperkeratosis (EKH/bullous CIE) and Ichthyosis en confetti
K2: Granular layer → Ichthyosis bullosa of Siemens	K11: Granular layer → Nothing
K3: Cornea → Meesmans corneal dystrophy	K12: Cornea → Meesmans corneal dystrophy
K4: Mucosal epithelium → White sponge nevus	K13: Mucosal epithelium → White sponge nevus
K5: Basal layer → EBS, Dowling-Degos disease	K14: Basal layer → EBS, NFJ, DPR
K6a: Hair, nail, palmoplantar skin → PC type 1	K16: Hair, nails, mouth, palmoplantar skin → PC type 1 and **non-epidermolytic Unna thost PPK**
K6b: Nail, hair follicle, palmoplantar skin → PC type 2	K17: Nails, hair follicles, sebaceous glands → PC type 2
K9: Palmoplantar skin→ Vörner PPK (epidermolytic; "the umlaut in ö looks like keratin clumps lysing")	K19: Stem cells → Nothing

***K5/K14 is the epidermal target proteins for haptens** as this is the heterodimer expressed in basal keratinocytes
*K1/K9 is the keratin heterodimer expressed **exclusively in palmoplantar skin**
*K8/K18 is not in this mnemonic because they are only expressed in epithelial tissues like the liver
***K6/K17 and K7/17 are found in hair/adnexal structures and are upregulated in wound healing**

Melanocytes

- Neural crest-derived melanin-producing dendritic cells found in the stratum basale
    - 1:10 ratio of melanocytes to keratinocytes when viewed in 2-dimensional plane
    - Each melanocyte interfaces with 36 keratinocytes in 3-dimensions
    - Ratio is 1:5 in hair (graying caused by gradual ↓ in # of follicular melanocytes)
- Connected to keratinocytes via **E-cadherin** receptors not desmosomes
- Melanin is produced in melanosomes (lysosome-like organelles) from its precursor tyrosine w/ the **copper-dependent enzyme tyrosinase** ("PHARCK I inhibited tyrosinase")
    - Phenylalanine, Hydroquinone, Azelaic acid, Retinoids, Cysteamine, vitamin C and Kojic acid are competitive inhibitors of tyrosinase
- Melanin absorbs UVR and protects against UV-induced mutations
- Racial variation: melanosomes in darker skinned individuals are larger, more abundant, have more melanin, are more stable and are transferred as individual organelles (vs smaller, less melanin, less stable and transferred in membrane-bound clusters of 3-6 melanosomes in lighter skin individuals); **# of melanocytes is the same between races**
    - "Small white unstable melanosomes must stick together in groups of 3-6"
- Chronic sun exposure causes melanosomes to get larger over time
- Pheomelanin: red color with spherical shape, a microvesicular ("microphesicular") internal structure and increased sulfur ("sulphur") content which is made from cysteine
- Eumelanin: black color w/ oval shape and lamellar structure → You (Eu) are overly lame
- Melanin production is stimulated by melanocyte-stimulating hormone (**product of proopiomelanocortin gene**) on melanocortin 1 receptor (MC1-R) on melanocytes
    - **MC1-R loss of function mutations → ↑ pheomelanin:eumelanin ratio** (phenotype = red hair/fair skin with ↑ risk of melanoma)

        - "MC1-R defect p/w My Color Red"
    - **Promotion of Agouti protein** also causes red phenotype
- Autoimmune destruction of melanocytes: vitiligo
- Defect in **melanocyte migration** (C-Kit): piebaldism ("here kitty makes cat move")
    - C-kit is also mutated in acral/mucosal melanoma and mastocytosis
- Defects in **melanin synthesis** enzymes: oculocutaneous albinism ("white like TP")
    - OCA1A/1B (Tyrosinase), OCA2 (P gene), OCA3 (TYRP), OCA4 (MATP/SLC45A2)
- Defects in **melanosome synthesis** >> transfer: Hermansky-Pudlak syndrome
- Defects in **melanosome trafficking/transfer** → Griscelli and Chediak-Higashi syndrome
    - "Greasy Hash drug traffickers"

Langerhan Cells (LC)
- BM-derived dendritic cell of monocyte-macrophage lineage ($CD34^+$) in s. spinosum
- Recognize and present foreign antigens (haptens) to specific T lymphocytes
    - After APCs like LCs take up antigen, they activate **and drain to nearby LNs**
- Like melanocytes, they are connected to keratinocytes via **E-cadherin** receptors
- During inflammatory states **MMP-9** is secreted which Makes LCs Move out of epidermis
- During non-inflammatory states **BRAK/CXCL14** is expressed homing LCs to the epidermis
    - "BRAK makes Langerhan cell stay BACK in the epidermis"
- Kidney or coffee-bean shaped on H&E (due to folded nucleus)
- Distinct intracytoplasmic organelles (Birbeck granules) w/ tennis racquet-shape on EM
- Positive immunostains: **CD207 (langerin)**, CD1a, S100, CD34, vimentin and actin
- Downregulated by UV, upregulated by TIM (TGF-β, IL-34, MCSF)
    - "TIM lends a hand to Langerhan's cells"
- Langerhan Cell Histiocytosis: Say "Letter Hand Cell" (this terminology may be obsolete)
    - Letter-Siwe: acute disseminated form occurring before 2 years old (yo)
    - Hand-Schuller-Christian: DI, osteolytic bone lesions, exophthalmos; 2-6 yo
    - Congenital self-healing (Hashimoto Pritzker): skin-limited, rapid resolution; birth
    - Eosinophilic granuloma: bone lesions (cranium); 6-12 yo

Merkel Cells
- Ectoderm-derived mechanoreceptor found among basal keratinocytes in areas with high tactile sensitivity (lips, fingers, ORS, oral mucosa) which communicates with neurons
- Markers: "Merkel will Merk a Very Important Person's CNS"
    - MAP-2 (most sensitive and specific neuroendocrine marker)
    - Vasoactive intestinal peptide (VIP)
    - **CK20**, Calcitonin gene-related peptide, Chromogranin A, NSE, Synaptophysin

Basement membrane zone

- Semipermeable barrier b/w epidermis and dermis w/ distinct layers (from sup to deep):
    - Basal keratinocytes (keratin intermediate filaments)
        - K5 (EBS, Dowling-Degos)

        - K14 (EBS, Naegeli-Franceschetti, dermatopathia pigmentosa reticularis)
    - Hemidesmosome-anchoring filament complex: ("Peanut Butter BIL"): **PB** (hemidesmosomal plaque; **connects keratin intermediate filaments to the hemidesmosome**), BI (**lamina lucida;** anchoring filaments), L (**lamina densa**; connects hemidesmosome-anchoring filament complex to anchoring fibrils in lamina densa); because 'L' is below lamina lucida it stains blister floor on SSS
        - Plectin: EBS with muscular dystrophy (only EBS that's AR) and PNP
        - BPAg1: BP, PNP (remember paraneoplastic pemphigus affects all plakins)
        - BPAg2 (Coll 17): "Called 17 times w/ no answer, what a Judgmental BICH"
            - Genetic Disease: Junctional EB (milder phenotype than Herlitz)
            - Autoimmune disease: BP, linear IgA bullous dermatosis/CBDC, Cicatricial pemphigoid, (Carboxy terminus: all others in this mnemonic are N16A terminus of BPAg2), Herpes gestationis
                - N-terminus of BPAg2 is iNtracellular
                - C-terminus of BPAg2 is eCxtracellular
        - Integrin α6β4
            - Genetic disease: junctional EB w/ pyloric atresia has crumpled skinny ears, skinny gastric pylora, ureteral stenosis → renal failure
                - "I is a skinny letter so think skinny stenoses"
            - Autoimmune disease: ocular cicatricial pemphigoid (β4 integrin)
                - "The β looks like glasses for your eyes"
        - Laminin 332 (Laminin 5 or anti-epiligrin): **stains floor on SSS** starting now
            - Genetic disease: Junctional EB (severe version of JEB aka Herlitz)
            - Autoimmune disease: anti-epiligrin cicatricial pemphigoid (a/w malignancy so it is the severe version of cicatricial pemphigoid)
    - Lamina densa: "lamina FEHNsa
        - **Laminin 5** (lamina densa/hemidesmosome-anchoring filament complex)
        - **Type Four collagen**: mutated in Goodpasture and Alport syndrome
        - Entactin/nidogen: connects laminin 1 to collagen 4 (**unique to skin BMZ**)
        - Heparan sulfate proteoglycan (perlecan): gives the BM a negative charge
        - Nidogen: connects laminin 1 to collagen 4 (**unique to skin BMZ**)
    - Sublamina densa: **Type 7 coll** (anchoring fibrils) hook to dermal type 1/3 collagen
        - "Type VII collagen fibers anchor up my BED" (Bullous SLE, EBA, DEB)

Dermis

- Dermal cells: "the Five Massive Giant Mononuclear Dermal cells"
    - Fibroblasts: create extracellular matrix and play a role in wound healing
    - Mast cells: discussed in 1.5
    - Glomus cells: smooth muscle cells which allow for shunting of blood
        - Derived from Sucquet-Hoyer canals on palms and soles ("Suck yer nailes")
    - Mononulcear phagocytes: discussed in 1.5
    - Dermal dendritic cells: highly phagocytic APC in the dermis

Extracellular matrix (ECM)

- Synthesized by dermal fibroblasts and composed of "CLEFT PIG"
    - **Collagens**, Laminins, **Elastins,** Fibulins, Three modifying enzymes
    - Proteoglycans, Integrins, Glycoproteins
- Collagen
    - **Triple helix** configuration composed of 3 amino acids on repeat (**Glycine**-x-y where x is usually proline and y is often hydroxylysine or hydroxyproline)
    - MC component of skin and accounts for 75% of dry weight of the dermis
    - Biosynthesis of collagen
        - Intracellular: lysyl hydroxylase (mutated in kyphoscoliotic EDS and inhibited by minoxidil) and proline hydroxylase catalyze hydroxylation to procollagen; **vitamin C-dependent** process (deficiency → scurvy)
            - Mnemonic "Hy-C"
        - Extracellular: procollagen N-proteinase/ADAMTS2 (mutated in dermatospraxis EDS) cleaves the N-propeptide of procollagen and then lysyl oxidase (mutated in Menkes and occipital horn syndrome; **copper dependent**) catalyzes cross-linking between collagen molecules
    - Collagen (types 1, 2 and 3) is degraded by matrix metalloproteinases (MMPs)
    - Collagen synthesis is ↑ by retinoic acid and doxycycline (inhibits MMPs)
    - Collagen synthesis is ↓ by The Ugly PIIG
        - TNF-α
        - UV irradiation
        - Penicillamine (inhibits lysyl oxidase which crosslinks collagen and elastin)
        - IL-1 (↑ MMP expression)
        - IFN-γ
        - Glucocorticoids
    - Types of collagen: location → disease state
        - Coll 1 (one): #1 most abundant collagen and is found in dermis, scars, tendons, teeth, ligaments and Bone → EDS (arthrochalasic form), OI
        - Coll 2: cartilage ("car-2-lage") → Relapsing polychondritis, MAGIC
        - Coll 3 ("Big Gross Stinky Feet"): Blood vessels (arterial walls), GI tract and Granulation tissue, Scars, Fetal skin → EDS (vascular form)
        - Coll 4: **basement membrane** (of epidermis/BVs) → Alport, Goodpasture
        - Coll 5: skin → EDS (classic form)
        - Coll 6: aorta, placenta → Ullrich Scleroatonic muscular dystrophy
        - Coll 7: anchoring fibrils (attach BM to ECM; BED) → Bullous SLE, EBA, DEB
        - Coll 17 (BPAg2): anchoring filaments of hemidesmosome → JEB, BP, IgA linear, Cicatricial pemphigoid, Herpes gestationis ("Judgmental BICH")
        - Coll 8 (eight) and 18 (eighteen): endothelial basement membrane
    - **All collagens are produced by dermal fibroblasts except coll 7/17 (produced by keratinocytes**; "keritino-sevens") and Coll eight and eighteen (endothelial cells)
- Elastin

    - Elastic fibers are highly cross-linked proteins which provide resilience from stretching and account for 4% of dry weight of the dermis
    - 90% elastin (core) and 10% fibrillin (surrounds elastin)
    - Elastin contains **desmosine** and **isodesmosine** which crosslink with fibrillin via lysyl oxidase (same copper dependent enzyme which crosslinks collagen)
        - Enzymes that require copper ("never TALC to cops")
            - Tyrosinase (mutated in OCA type 1)
            - ATP7α (Menkes and XLR cutis laxa aka occipital horn syndrome)
            - Lysyl oxidase (LMNOP): Lysyl oxidase is ↓ by Lathyrism from sweet Pea ingestion and Penicillamine; mutated in Menkes/Occip horn)
            - Cystathione β-synthase (homocystinuria)
    - Elaunin fibers run horizontal in reticular dermis and oxytalan fibers run vertical
        - "Stand up straight (vertical), high (high in dermis) and tall with oxytalan"
    - UV radiation damages elastic fibers (dermal elastosis hallmark of photodamage)
- Microfibrils
    - Connects perlecan in the BMZ to horizontal elaunin fibers in the reticular dermis
    - Consists mainly of fibrillins ("Do not be a fibber and tell on me to MS. C")
        - Fibrillin 1: linear Morphea (autoAbs), Marfans, Stiff skin syndrome
        - Fibrillin 2: mutated in Congenital Contractural arachnodactyly
    - Also consist of fibulins: fibulin 4 and 5: mutated in AR cutis laxa
    - Loss of function of the microfibril-associated glycoprotein (MFAP) Latent-TGF-β binding protein 4 (LTBP4) also causes AR cutis laxa and activating mutations in TGF-β receptors 1 and 2 cause Loeys-Dietz syndrome
- Extrafibrillar matrix
    - Amorphous material that the dermal fibril network and cells are embedded into
    - Consists of proteoglycans/GAGs, glycoproteins, hyaluronic acid and water
    - Glycosaminoglycans (GAGs) are negatively charged sugars that bind water
        - GAGs bind proteins via serine hydroxyl groups forming proteoglycans
        - The most ubiquitous protein-free GAG (non-sulfated) is hyaluronic acid
        - Chondroitin/dermatan/keratan/heparan sulfate are protein-bound GAGs
    - Glycoproteins: fibronectin, vitronectin, matrilins, tenascins and ECM proteins
        - Tenascin X deficiency: classic EDS and hypermobile EDS
        - ECM-1: lipoid proteinosis (gene mutation), LS&A (auto-antibody)
- Integrins
    - Mediates attachment to either ECM or to other cells
    - "JIL" has an integrin mutation
        - JEB with pyloric atresia: α6β4 integrin (MMP has autoAbs to β4 integrin)
        - Juvenile hyaline fibromatosis: CMG-2 gene (encodes an integrin receptor)
        - Infantile systemic hyalinosis: CMG-2 gene ("Can't Move Good")
        - Leukocyte adhesion deficiency: β2 integrin
        - Lupus erythematosus: integrin α_M

Cutaneous vasculature

- **VEGF** is the primary mediator of vasculogenesis
- Lymphatic markers include VEGFR-3, Prox1, D2-40 (Podoplanin), LYVE-1
    - "Vegan Proxies are Down 2 Live in lymph"
- Anatomy
    - Veins, arteries and arterioles are composed of smooth muscle and elastic tissue surrounding the basement membrane (BM) and endothelial cells
        - The smooth muscle layer is just much thicker in arteries
    - Venules and capillaries are made of endothelial cells surrounded by **pericytes**
        - "The venue has a cap of 100 persons"
    - Lymphatics: endothelial cells and BM w/o pericytes or smooth muscle

Cutaneous neurology

- Sensory nerves are divided into free nerve endings and corpuscular nerve endings
    - Free nerve endings include small unmyelinated C-type fibers (dull pain) and larger myelinated Aδ fibers (sharp pain and temperature)
        - "The δ sign is a sharp symbol"
    - Specialized nerve receptors (corpuscular nerve endings; be able to recognize!)
        - Krause end bulbs: found on genitalia, perianal region, and vermillion lips
            - "Krazy Krause ends on erotic areas"
        - Meissner corpuscle: **on dermal papillae of digits** for pressure/light touch
            - "Well-Mannered Meissner lightly touches fingers"
        - Pacinian corpuscle: onion shaped mechanoreceptor of palmoplantar skin, nipples and genital region; suited for vibration located in deep dermis/fat
            - "Women always pack their vibrators deep in their suitcases"
        - Ruffini corpuscle: on fingernails in deep fat; suited for sustained pressure
            - Ruff sex = fingernails deep in your back with a sustained pressure
        - Merkel nerve endings: found on fingertips, lips, and external genitalia in the basal epidermis (superficial); suited for pressure

Adnexal structures

- Eccrine glands
    - Secretory gland of merocrine secretion (exocytosis) w/ thermoregulatory effects
    - They produce a clear, odorless substance consisting mostly of **sodium** and water
    - Eccrine Everywhere Except External auditory canal and Euphoric areas (lips, glans penis, clitoris and labia minora)
    - Most numerous on palms and soles "sweaty palms"
    - No relationship to pilosebaceous follicle
    - Innervated by sympathetic fibers that have Ach (not NE) as the neurotransmitter
        - Neurotoxins (botox) blocks Ach release and treats hyperhidrosis
    - Delivers hypotonic sweat to skin but the faster you sweat the more isotonic it is
        - "Your skin gets salty when you exercise intensely"
    - ECCrine Gland Stains: EMA, CEA, Cytokeratin, GCDFP-15 (apocrine>eccrine), S100
- Apocrine gland (ABCDE)

- Secretory gland of apocrine/**decapitation (part of cell) secretion** located on…
  - **Anogenital area, Axilla**
  - Breast (areola/nipple), **Belly button**
  - Canal of external ear, Dividing line of lip, Eyelid margin
    - C,D and E are all the facial orifices except nose
- **Connect to infundibulum of hair follicles** except 3 named ectopic apocrine glands which empty directly onto surface (mammary glands of breast, Moll's gland of eyelid margin and ceruminous glands of external auditory canal)
- Innervated by β-adrenergic receptors stimulated by circulating catecholamines
- Secretory products include cholesterol, cholesterol esthers and fatty acids
- Bromhidrosis: smelly sweat from apocrine secretions mixing w/ surface bacteria
- Chromhidrosis: pigmented sweat due to its rich lipofuscin content
- Stains for CEA, keratins (same stains as eccrine glands)

- Sebaceous glands
  - Secretory gland of holocrine (whole cell lyses off) secretion located mainly on the scalp, face and chest ("seborrheic areas") but NOT on palms and soles
  - Associated with terminal hair follicles except the ectopic sebaceous glands
    - Ectopic seb-eye-ceous glands all have the "eye" sounds
      - Zeis glands on eyelid margin next to Molls glands ("Z-eye-s")
      - Meibomian glands on eyelid tarsal plate ("M-eye-bomian")
        - More deep (more letters) in eyelid than Zeis and Molls
      - Fordyce spots on vermillion lip/penis ("Ford-eye-ce")
      - Montgomery tubercles on areola/nipples ("Montgom-eye-ry")
      - Tysons glands on labia minora/prepuce ("T-eye-sons")
        - "Mike Tyson beats up the genitals"
  - Composition of secretions is "Three Wacky Sebaceous Contents"
    - **Triglycerides** (#1 component; makes up > 50% of sebaceous secretion)
      - Of importance, in acne vulgaris, resident microorganisms like P. acnes secrete lipases which split triglycerides into free fatty acids which alter the pattern of keratinization in the infundibulum (comedones) and which attracts neutrophils (inflammatory acne); pubertal androgens stimulate sebaceous glands which produce more free fatty acids from triglycerides in the presence of P. acnes
    - **Wax esters** (2^{nd} MC), **Squalene** (3^{rd} MC), Cholesterol/Cholesterol esters

Hair

- Infundibulum: from epidermis to opening of sebaceous gland duct (**apocrine gland connects to the follicle above the sebaceous gland in the infundibulum**)
  - No IRS but contains an **ORS with normal epidermal keratinization** (has **keratohyalin** granules (KHG)/granular layer)
- Isthmus: from sebaceous gland duct to arrector pili insertion (bulge)
  - No IRS but contains an **ORS with trichilemmal keratinization** (no granular layer)
    - "Pilar cysts are also called trichilemmal cysts and isthmus-catagen cysts"

- **Bulge**: part of ORS at level of arrector pili (**major site of hair follicle stem cells**)
- Lower hair follicle: from arrector pili insertion to hair bulb
  - **Contains an IRS** and a **non-keratinized ORS**
    - IRS is not present in telogen hairs ("The IRS won't rest until you pay tax")
    - IRS only present in lower hair follicle ("The IRS stoops so low")
    - IRS has **trichohyalin** granules found in Huxley's layer ("The IRS is tricky")
  - Critical line of Auber is the widest area where most mitotic activity occurs
    - "Auber is pronounced like Ober which sounds like Obese (widest part)"
  - Hair bulb is the lowest part of follicle which contains melanocytes for hair color
    - 1 melanocyte: 5 keratinocytes (melanocytes only produce pigment in anagen phase because melanocytes in matrix apoptose in catagen phase)
- Hair anatomy in cross section
  - Layers (inner to outer): hair shaft (medulla, cortex and cuticle), IRS (cuticle, Huxley's layer and Henle's, layer), ORS, glassy membrane
    - Just remember that the Huxley layer hugs the cuticle, and from there it's easy to remember that the Henley's layer is next followed by the ORS
    - Cuticle from IRS and hair shaft fuse → split ends (trichoptilosis)
- Hair growth (ACT)
  - Anagen: **hair grows 0.4mm/day or 1cm/month**; lasts **3 years** (89% of scalp hair)
    - Hair (1cm/month), toenails (1mm/month), fingernails (2-3mm/month)
    - **Hairs look like a hockey stick** (anagen effluvium hairs will be surrounded by a clear sheath whereas loose anagen hair syndrome hairs lack this)
  - Catagen: regression phase where IRS of lower hair follicle is lost and melanocytes in matrix apoptose; lasts **3 weeks** (1% of scalp hair at any given time)
    - Can be identified by a very **thick vitreous sheath** (just outside the ORS)
  - **T**elegen: Rest (**T**ired phase); **Club-shaped**; high in dermis; lasts **3 months** (10%)
    - "Night Clubs give you a high but then get you tired and need a rest"
- ShApe of follicle (oval → curly; round → straight) and Disulfide Bonding via Cysteine residues determines Curliness of hair (these bonds brake when hair is straightened)

Nails

- Nail anatomy
  - Proximal nail fold is contiguous with/in proximity to both skin and the matrix
  - Eponychium aka cuticle ("Every pony is cute") which is b/w the plate and matrix
  - Nail matrix: proximal matrix forms Dorsal nail plate; Distal matrix forms ventral nail plate ("just remember that the Ds are opposite")
    - Nail Matrix has Melanocytes and **lacks a granular layer**
      - Proximal matrix: forms 80% of nail Plate (biopsy → nail dystrophy)
      - Distal matrix (lanula is visible distal portion): 90% of melanocytes
    - Nail thickness is determined by the length of the nail matrix
  - Lunula (distal nail matrix) → nail bed → nail plate
    - Nail bed extends from the lunula to the onychodermal band (reddish pink transverse band) and has minimal contribution to nail plate synthesis

        - Nail bed **lacks a granular layer** (like isthmus of hair follicle)
- Nail growth
    - Fingernails: 2-3 mm/month (0.1 mm daily; 6 months to grow out)
        - Beau's lines 1/3rd up the fingernail plate → insult occurred 2 months ago
    - Toenails: 1 mm/month (12-18 months to grow out)

2.5 ONCOLOGIC AGENTS IN DERMATOLOGY

Melanoma

Dacarbazine

- Cell cycle non-specific chemotherapeutic alkylating agent that halts DNA replication
- Use: metastatic melanoma but fallen out of favor w/ more tolerable and effective meds
- SE: GI (vomiting), pancytopenia, allergic reactions (anaphylaxis) liver injury, neuropathy

Interferon alpha

- Works as a cytokine that signals and modulates immune responses
- Uses: FDA approved for melanoma, hairy cell leukemia and RCC
- SE: flu-like symptoms, injection site reactions, N/V/D, pancytopenia, liver injury, depression and rare but fatal **rhabdomyolysis** w/ multiple organ failure

Taliomogene laherparepvec (T-VEC)

- Genetically engineered **herpes virus type 1** that selectively infects and replicates in tumor cells producing granulocyte-macrophage colony-stimulating (GM-CSF) which promotes anti-tumor immunity capable of destroying tumor at distant sites
- NCCN: intralesional T-VEC is used as a 2nd line agent in combo w/ an immune checkpoint inhibitor or targeted therapy in those that have progressed on initial 1st line therapy

Interleukin-2 (IL-2)

- MOA: activates immune response to kill tumor cells
- Initially used systemically in 1998, but now NCCN recommends intralesional IL-2 (due to severe toxicity of IV) and only as a 2nd line therapy for those who have progressed on targeted or immune checkpoint therapy and only when T-VEC is unavailable
- SEs (IV): **capillary leak syndrome** (oliguria, generalized edema, tachyarrhythmias, hypotension), fever, N/V/D, catheter-related sepsis and death
- SEs (IL): pyrexia, flu-like symptoms, GI symptoms, injection site pain/redness/swelling

BRAF inhibitors (VEmurafenib, dabrafenib, encorafenib)

- MOA: serine/threonine signal transduction kinase that inhibits BRAF (targets the V600E mutation of BRAF) in the MAPK pathway; given PO
    - Substitution of valine (V) for glutamic acid (E) at amino acid codon 600
- **MAPK pathway: BRAF is a phosphorylating serine/threonine kinase** that with the GTPase activity of NRAS, leads to the phosphorylation of MEK and ERK → phosphorylated ERK activates **nuclear cyclin D1** ("nucl-ERK cyclin D1") which complexes w/ CDK4 and phosphorylates Rb → phosphorylated Rb releases the transcription factor E2F which leads to progression through the S phase of the cell-cycle
- **NCCN guidelines**: use in pts w/ stage 3 melanoma or higher and a known BRAF mutation

- **Quicker onset** of action vs immunotherapy (use BRAF/MEK combo over immunotherapy if pt has organ failure/symptoms); benefits are not as long lasting as w/ immunotherapy
    - "Use a BRAF to give Rapid relief"
- SEs: **Verrucous keratoses**/KP (MC skin lesions), **Eruptive KAs/SCCs/AKs** (blocking BRAF leads to new SCCs by paradoxical activation of RAF), Elevated temperature/pyrexia (MC SE), **Erythema nodosa (neutrophilic)**, palmoplantar Erythrodysesthesia, new BRAF wild-type melanomas/nevi (blocking BRAF pathway causes new melanomas by shunting to the MEK pathway), photosensitivity, dysgeusia, muscle aches, alopecia, **prolonged QT**, seb-derm-like rash, acneiform eruption, **KP-like rash,** photosensitivity, N/V/D, arthralgia
- **Resistance** develops after 6 months of therapy unless combined w/ MEK inhibitors
    - There are currently 3 combinations: "Biden was Tested for COV-19"
        - Binimetinib/ encorafenib
        - **Trametinib/dabrafenib**
        - Cobimetinib/Vemurafenib
- Sorafenib also targets the BRAF V600E mutation (in addition to PDGFR, c-KIT, VEGFR)
    - SEs: **hand-foot-skin reaction**, alopecia, rash/desquamation, facial erythema, nail changes, pruritis, mucositis, eruptive benign nevi, **KAs/SCCs**

MEK inhibitors ("MECK": trametinib, cobimetinib, binimetinib, selumetinib)

- Serine/threonine/tyrosine kinase that inhibits MEK1/2 of the MAPK pathway; given PO
- Uses: **neurofibromatosis (selumetinib)** and late-stage melanoma (commonly Combined w/ BRAF inhibitors which is more effective and ↓ risk of BRAF-induced KAs/SCCs)
    - 2-year survival goes from 30% w/ BRAF monotherapy to over 50% w/ combo
- SEs: **MC SE is acneiform reactions** (Rx w/ doxycycline), photosensitivity, pruritis, alopecia, peripheral edema; Cardiomyopathy and retinal vein occlusion were reported

Ipilimumab and tremelimumab

- **Blocks CTLA4** (cytotoxic T-lymphocyte associated antigen 4) on cytotoxic T-cells; given IV
- Use: metastatic melanoma (commonly used in combination w/ PD-1 blockers)
    - 2-year survival w/ ipilimumab is 30% and if combined w/ PD-1 inhibitors is 65%
- Unlike BRAF inhibitors, responses take long (3 months) to develop, but are more durable
- SEs: immune-related SEs → **Diarrhea, Dermatitis (MC)**, and **Deadly colitis (most severe)**
    - Treat grade 3 or 4 colitis toxicity w/ high dose steroids (infliximab given if no improvement after 3 days) and pt should permanently discontinue ipilimumab
    - Other SEs remind me of DRESS: rash, facial swelling, hypothyroidism, hepatitis
        - Treat these immune-related side-effects w/ systemic steroids
    - **SE of vitiligo-like leukoderma is correlated w/ improved clinical response**
- SE onset: rash/pruritis (4 weeks), diarrhea/colitis/liver toxicity/hypophysitis (6 weeks)

PD-1 (on T-cells) inhibitors (Cemiplimab, Nivolumab, Pembrolizumab, Pidilizumab; "Cem Nice Pembroke Pines PD"): PD-1L (on tumor cells) inhibitors (atezolizumab, durvalumab, avelumab)

- Checkpoint inhibitors that target the Programmed cell death-1 (PD-1) receptor; given IV
    - PD-1 inhibit T-cells from attacking tumor cells (these meds stop this inhibition)

- Uses: late-stage melanoma, Merkel cell carcinoma and NMSC
  - Avelumab/pembrolizumab are FDA-approved for locally advanced Merkle
  - **Cemiplimab** is FDA-approved for unresectable NMSC (BCCs and SCCs)
- **NCCN guidelines**: preferred adjuvant treatment for stage 3 melanoma after primary treatment and preferred monotherapy in stage 4 (unresectable/metastatic) disease
  - Pembrolizumab is approved adjunctly for stage 2B/2C melanoma after excision
- SEs: common SEs include fatigue, diarrhea, nausea, **pruritis**, **rash**, hypothyroidism
  - Pruritis Rx ("PD"): Potent topical steroids (grade 1), Pregabalin/gabapentin (grade 2), Dupilumab/omalizumab (grade 3)
  - Rash ("PD-1L"): the two MC reaction patterns are **Lichenoid (MC; 50%)** and maculo-Papular (spongiotic; 40%) but others include **bullous Pemphigoid**, Photo-distributed vitiligo, Psoriasiform and vacuolar interface Dermatitis
    - SJS/TEN, sarcoid, AI-CTD like eosinophilic fasciitis are much less common
- Efficacy improved when combined w/ ipilimumab (pitfall is ↑ immune-related SEs)
- The combination of **Relatlimab (LAG-3 blocker)** and Nivolumab (PD-1 blocker) was FDA approved in 2022 for unresectable and metastatic melanoma
  - SE: 5% get hepatitis and 5% get adrenal insufficiency
  - "The RN Lagged 3 steps behind the PD (Program Director)"
- Management latter: if they fail a PD-1 add a CTLA-4, if they fail that try a PD-1 + LAG3
- Development of **vitiligo-like leukoderma** is correlated w/ **improved clinical response**
  - And spongiotic (eczematous/psoriasiform), lichenoid and BP to a lesser degree
- Relative contraindication: **solid-organ transplant patients**

Immunotherapies for the treatment of skin cancer

- Class of treatments that use persons own immune system to help kill cancer cells
- "Immunotherapies make you itch" (PICA is itch in Spanish")
  - Pembrolizumab (PD-1 inhibitor): used in advanced Merkel, melanoma and NMSC
  - Poly ICLC (toll-like receptor 3 agonist): used in advanced SCC (not FDA approved)
  - Ipilimumab (CTLA-4 inhibitor): used in advanced melanoma
  - Imiquimod (toll-like receptor 7/8 agonist): used topically in subsets of BCC/SCCs
  - Cemiplimab (PD-1 inhibitor): used in unresectable NMSC
  - Avelumab (PD-1 inhibitor): used in metastatic Merkel cell carcinoma

Imatinib mesylate (Gleevac®)

- Inhibits multiple tyrosine kinases (Bcr-Abl, c-Kit, platelet-derived growth factor receptor)
- Derm uses: CML (Bcr-Abl), DFSP (COL1A-PDGFR), Hypereosinophilic syndrome (FIP1L1-PDGFR), acral/mucosal Melanoma (C-Kit), Mastocytosis (C-Kit and FIP1L1-PDGFR)
  - "CBDC's are a Highway to Financial Prison and will Make the Masses Controlled"
- SE (GLEEvac): **Gingival/tooth/nail hyperpigmentation**, **Loss of pigment** (via inhibition of c-Kit), Lichenoid eruption (usually oral), **Eye edema (periorbital)**

BCC

Hedgehog pathway (Hp) inhibitors (vismodegib, sonidegib, taladegib, saridegib, patidegib)

- Pathway: in the normally functioning Hp, patched (PTCH) normally binds and inhibits smoothened (SMO); when this pathway is activated, Hp proteins bind to the transmembrane PTCH receptor which ↓ the inhibition of PTCH on SMO, allowing SMO to activate a downstream cascade (i.e. GLI) that results in cell proliferation
    - Even without smoothened, GLI is normally inhibited in the cytoplasm by SUFU
- MOA: acts as a cyclopamine-**competitive antagonist of smoothened (G-protein coupled receptor)** in the sonic hedgehog pathway → ↓ gene transcription of various cancers
    - "The 3 vegetarian PA's (Erivedge, PATCH, Pategedib gel, cyclopamine) always flirt w/ the smooth doctor and inhibit him (SMO) from doing his work in cancer"
    - Smoothened is a G-protein coupled receptor ("the smooth doctor is such a G")
- Uses: metastatic (vismodegib only), locally advanced BCCs and pts w/ Gorlin's syndrome
    - Vismodegib has a 30% response rate in pts w/ metastatic BCC and 45% in those w/ locally advanced inoperable BCC; medium duration of treatment is 10 months
    - Cemiplimab (PD1) is used if BCC got worse on Hp inhibitors or if can't tolerate SE
- SEs ("MAD side effects"): **Muscle spasms (68%) > Alopecia (63%), > Dysgeusia (51%)**
    - Sonidegib has less muscle spasms (taking w/ L-carnitine can lessen this SE)
    - Other SEs: teratogenicity (**cyclopia**), fatigue, **SCC**, amenorrhea/azoospermia, DVT
- Females and males need contraception x 24 months and 3 months off drug, respectively for Vismodegib and for 20 months (females) and 8 months (males) for sonidegib
- Females and males must avoid donating blood for **24 months** off drug (20 for sonidegib)

CTCL

Brentuximab vedotin ("Brent is 30 years old MALE Nurse Practitioner")

- Antibody-drug conjugate of monoclonal **anti-CD30** conjugated to an antimicrotubule
- Uses ("30-year-old MALE"): MF (large cell change), Anaplastic large cell lymphoma, LyP
    - Use is restricted to multifocal or refractory cases
- SE: **peripheral sensory Neuropathy (Numbness)**, N/V, Pneumonia, Pyrexia, Neutropenia
- Black box: JC virus infection resulting in Progressive multifocal leukoencephalopathy

Alemtuzumab

- Humanized monoclonal antibody **against CD52** surface Ag on T- and B-cells
    - "Its's not **all** (alemtuzumab), its actually just slightly more than half (52%)"
- Use: CTCL w/ peripheral blood disease (i.e. Sézary)
- Black box warning for cytopenias, infusion reactions and infections

Romidepsin, Remetinostat and Vorinostat

- Histone deacetylase inhibitors ("he was treated for CTCL in his RV")
- Use: FDA-approved for CTCL pts w/ recurrent disease refractory to 2 systemic therapies
    - Vorinostat also shows promise for patients w/ advanced head and neck SCC
    - Topical remetinostat may be effective in treating early localized cutaneous SCC
- SEs: GI SEs, fatigue, cytopenias, infections and T-wave flattening

Denileukin diftitox

- Binds **IL-2α** or **CD25** and introduces diphtheria toxin into cells → cell death
    - "Di means 2 which helps me remember it inhibits CD25 component of IL-2"
- Use: CTCL whose malignant cells express the **CD25 component of the IL-2 receptor**
- SEs: **capillary leak syndrome** (oliguria, generalized edema, arrhythmias, hypotension)
- Black box: fatal infusion rxns, capillary leak syndrome, loss of visual acuity/color vision

Lenalidomide

- **Thalidomide analog:** ↑ Th1-cytokines and enhances T and NK cell-mediated killing
- FDA approved for myelodysplastic syndrome, MM, mantle cell lymphoma, CTCL, Sézary
- SEs: cytopenias and fatigue/malaise

Mechlorethamine hydrochloride and Carmustine

- Nitrogen Mustard alkylating agents used for patch/plaque stage MF
- SEs (mechlorethamine): contact dermatitis (MC), anaphylaxis and SCCs
- SEs (carmustine): local reactions (MC) and myelosuppression

Topical oncologic agents used in dermatology

- 5-fluorouracil ("PTSD"): **Pyrimidine analog which binds to Thymidylate Synthetase** (thus **Deoxyurine cannot convert to thymidine**) and results in ↓ DNA synthesis
    - Uses: AKs and superficial BCCs (SCCIS is not mentioned in label)
    - SEs: erythema, erosions and blistering at application site
    - **Severe seborrheic dermatitis which p/w burning (simulates 5-FU reaction) can be seen weeks after 5-FU treatment (pearl: pretreat w/ topical ketoconazole)**
    - Do not use in patients w/ **dihydropyrimidine dehydrogenase** (DPD) deficiency → even topical application can cause severe toxicity → pancytopenia → death
    - Pregnancy category X
- Imiquimod: **activates TLR7** > TLR8 → activates **NFkB** transcription factor → ↑ cytokines (**TNF-α, IFN-γ**, IFN-α, IL-12) which trigger an antiviral and antitumoral T-cell mediated adaptive immune response (can only use in immunocompetent adults)
    - Uses: AKs on face/scalp (2x per week for 16 weeks), superficial BCC on trunk/extremities/neck (5x per week x 6 weeks), SCCIS/lentigo maligna (off label)
    - SEs: local reactions like 5-fluorouracil, flu-like symptoms, eruptive milia/cysts
    - **Topical steroids make imiquimod less effective but not 5-fluorouracil**
    - 10% of people have ↓ toll-like receptors in skin and don't respond to imiquimod
    - Pregnancy category C
- Diclofenac: ↓ cyclooxygenase enzymes → ↑ apoptosis
    - Uses: AKs (apply BID x 2-3 months)
    - SEs: more mild local reactions, photosensitivity, headache, ↑ LFTs
    - Pregnancy category B
- Tirbanibulin: inhibits microtubules (↓ tubulin polymerization) and **inhibits Src kinase**

    - Uses: AKs
    - SEs: local side effects
- Pategedib: inhibiting smoothened receptor in the sonic hedgehog pathway
    - Uses: BCCs
- Remetinostat: histone deacetylase inhibitor
    - Uses: superficial BCCs

Radiation therapy (RT)

- Ionizing electromagnetic radiation in the form of high energy photons produces **double-stranded breaks in DNA** when absorbed by tissues (esp in rapidly dividing cells)
- The MC form of cell death induced by RT is **mitotic catastrophe** (mitotic cell death)
- Main dermatologic indications ("MAD DUCKS"): Merkel cell, Angiosarcoma, Desmoid tumor, DFSP, Desmoplastic melanoma, UPS, CTCL, Keloids, Kaposi sarcoma, SCC/BCCs
    - Alternative treatment for skin cancers in nonsurgical candidates
    - Adjuvant treatment for aggressive tumors to decrease risk of recurrence
        - BCC/SCC: PNI of nerves ≥ 1mm, multiple nerves or cranial extension
    - Palliative treatment to provide symptomatic relief for incurable disease
- The unit of radiation dose is Gray (Gy): absorbed energy (J) per unit mass of tissue (kg)
- Malignant tumors: **50 Grey** ("50 shades of Grey") in 20 daily fractions is typical
- Success rate of keloids w/ surgical excision followed by RT is 75% (12 Gy in 3 fractions)
    - RT should be initiated within 24 hours post-op to induce apoptosis of fibroblasts
- CI: **genetic syndromes** predisposing patients to skin cancers (i.e. XP/Gorlin's syndrome)
    - Relative CI: **AI-CTD**, poorly vascularized tissue (**lower leg**, dorsum of hand), edematous tissue, trauma, thermal burn, ulcer, prior RT to site
        - "Don't hang out with Christian Grey if you are already swollen, traumatized, ulcerated or if you have genetic syndromes or CTD"
- SEs
    - Early: erythema, epilation, desquamation or ulceration
    - Late: telangiectasias, fibrosis, ulceration, necrosis or secondary malignancies
- Forms done by dermatologists in the outpatient clinic for thinner tumors
    - Brachytherapy: radiation source is applied on the skin surface
    - Superficial x-ray therapy (SXRT): source applied at a distance and aimed at site
- Forms done by radiation oncologists for thicker tumors
    - Electron beam RT: source is an electron delivered through a linear accelerator

3.5 CONNECTIVE TISSUE AND SCLEROSING DISEASES

Antibodies in autoimmune connective tissue diseases

Antibodies in Lupus

- Antinuclear antibodies (ANAs): most sensitive test for SLE (considered positive if > 1:40)
    - ANAs also (+) in "ANA loves SPIRM ("I imagine Anasthesia Steele in 50 Shades of Grey") → SLE (99%), SCLE (70%), SSc (95%), Sjögren's, PNGD > IGDA (50%), Polymyositis/DM (40%), Idiopathic juvenile arthritis, RA, Morphea, MCTD
    - ANAs may be detected via IIF utilizing HEp-2 tumor cells as substrate or via **ELISA**
        - SLE: rim pattern (IIF) or anti-dsDNA (ELISA/multiplex immunoassays)
        - SLE and drug-induced SLE: homogenous/diffuse pattern (IIF) or anti-dsDNA and anti-histone respectively (ELISA/multiplex immunoassays)
            - "SLE can be diffuse or rim but DI-SLE is always Diffuse pattern"
        - Non-specific but a/w Sjögren's, SCLE, NLE or MCTD: speckled pattern (IIF) or anti-Ro/SS-A, La/SS-B, anti-U1RNP, anti-Smith, and anti-Scl-70 (ELISA)
        - Systemic sclerosis and polymyositis (PM): nucleolar pattern (IIF) or RNA processing molecule fibrillarin aka U3RNP (ELISA/multiplex immunoassay)
            - "You 3 (U3) fibbers lied and said you were going to systemically sclerose Panama with a nuclear bomb"
        - CREST syndrome: Centromere pattern (IIF) or anti-Centromere (ELISA)
- Other antibodies (+) in lupus (prevalence in parentheses; "ABCD KRUSH")
    - ANA: discussed above
    - β2 glycoprotein (25%) and Cardiolipin (50%): antiphospholipid syndrome in SLE
    - C1q (60%): a/w renal dz in SLE and "hypoqomplementemic urtiqarial vasqulitis"
    - dsDNA (60%): very specific for lupus; monitors Disease Activity in **lupus Nephritis**
        - Crithidia IIF is used as a confirmatory test after positive anti-dsDNA ELISA
    - Ku (10%): overlap syndrome w/ SLE, DM/PM and systemic sclerosis
        - "When I Kiss u, our lips overlap"
    - rRNP (10%; 40% in Asians): specific for SLE; a/w neuropsychiatric LE
    - U1RNP (50%): overlapping features with other AI-CTDs (100% in MCTD)
    - SS-A/Ro (50%): MC in neonatal lupus (99%), SCLE (85%) and Sjögren's (70%)
    - Smith (20%): highly specific for lupus
        - Previous 4 Abs targeted ribonucleoproteins: "ribonucleoproteins R'USS"
    - ssDNA (70%): risk for developing SLE in DLE patients; also seen in linear morphea
    - Histone (40%): drug-induced SLE

Antibodies in DM/PM ("MAPS")

- Mi-2 (helicase): a/w **classic** DM (myositis w/ classic skin findings); good prognosis
- MDA5 (CADM-140): a/w rapidly progressive ILD (adults and kids); amyopathic w/ unique skin findings (skin/oral ulcers, painful palmar papules, mechanics hands and panniculitis)
- Aminoacyl tRNA synthetase (Anti-**Jo**, -PL7, -PL12, -EJ/OJ): anti-synthetase syndrome (**rapidly progressive ILD**, **myositis**, mechanics hands, Raynaud's, arthritis)

    - "I imagine 2 guys named CAD ("MD") and Jo running a race so they are both SOB (ILD) but whereas CAD falls down early in the race onto his hands (skin ulcers, painful palmar papules), Jo finishes the race making his muscles very weak"
- ANA: MC IIF pattern is nucleolar; **most sensitive antibody in DM**
- **p155 (anti-TIF1γ): amyopathic DM a/w cancer** and psoriasiform lesions, **palmar hyperkeratosis** and verrucous papules in adults; MC antibody in juvenile DM (35%)
- p140 (NXP-2): a/w cancer (adults) and calcinosis cutis (kids)
- SRP: poor prognosis from severe cardiac involvement ("Severely Rapid Pulse")
- SAE: severe skin disease, cancer and myositis

Antibodies in systemic sclerosis ("Hard fibrotic CARS")
- Fibrillarin (U3RNP): a/w limited SSc which presents at a younger age
- Centromere: a/w limited SSc/**CREST** (a/w pulmonary HTN and digital ulcers) > SSc
- ANA: MC IIF pattern is **nucleolar**
- RNA polymerase (I and III): a/w diffuse SSc (**Renal** and severe skin disease) > CREST
- Scl-70 (DNA topoisomerase I): a/w diffuse SSc (**pulmonary fibrosis**) > CREST

Antibodies in morphea ("more hard HATS")
- Histones: linear and generalized morphea
- ANA
- Topoisomerase IIα
- ssDNA: linear morphea ("single stranded aka linear"); also ↑ risk of SLE in DLE

Antibodies in rheumatoid arthritis
- RF (Fc portion of IgG): high levels c/w severe erosive RA and mixed cryoglobulinemia
- Cyclic citrullinated proteins: a/w severe RA

Sjögren's syndrome ("ASS": in order of decreasing sensitivity and specificity)
- **Anti-α-fodrin** (Actin-binding protein): most sensitive and specific
- ANA: speckled pattern on IIF
- SS-A/Ro (hyRNP): also important in neonatal lupus and SCLE
- SS-B/La (hyRNP): least sensitive and specific

Lupus erythematosus

Chronic cutaneous lupus (CCLE)
- MCC is discoid lupus (MC in African Americans)
    - Most pts w/ SLE will get discoid lupus; only 10% w/ discoid lupus will get SLE
- Subtypes
    - Discoid lupus erythematosus (DLE)
        - p/w atrophy, scarring (cicatricial alopecia), **central hypopigmentation w/ hyperpigmented periphery** typically on the scalp, face and **conchal bowls**
        - Labs: ANA positive in 10% of cases

            - Widespread (above and below neck) and childhood DLE regularly has SLE
            - Path: vacuolar interface, epidermal atrophy, BMZ thickening, follicular plugging, superficial and deep PV/PA lymph infiltrate w/ ↑ dermal mucin
            - DIF: LBT+ in 75% of DLE cases: higher sensitivity in **old lesions ("disc-old")**
    - Chilblains lupus erythematosus
        - p/w Circumscribed, dusky-red papules on acral sites precipitated by Cold
        - May be sporadic (adults; a/w anorexia, lymphoma, pregnancy) or familial (child; AD mutation in TREX1 gene which encodes a DNA exonuclease)
        - Chronic relapsing course and persists year-round (vs non-lupus chilblains)
        - Ddx: pernio (chilblains) is less chronic (appears 12-24 hours after cold exposure and is **a/w AI-CTD like SLE,** drugs, CML; Rx: CCBs.), covid-toes (pseudo-chilblains) occurs as a late feature of covid, and ADHD stimulant toes (resolves after discontinuation of amphetamines)
        - Path: features of both chilblains (pap dermal edema + lymphs) and DLE
            - DIF: LBT (+)
        - Rx: topical Corticosteroids, CCBs, Cigarette Cessation, rewarming
    - Tumid lupus erythematosus (TLE)
        - p/w edematous, indurated, erythematous, annular plaques w/ central clearing on the trunk > face; lesions heal in weeks, but relapses can occur
        - Jessner's, TLE and reticular erythematous mucinosis are on a spectrum
            - Rx: treat all 3 with antimalarials
        - Path: similar to DLE but w/ massive dermal mucin and no epidermal changes (lacks follicular plugging, vacuolar interface, BMZ thickening)
    - Lupus profundus (panniculitis)
        - p/w indurated, non-tender SQ nodules/plaques that heal with atrophy
        - Located on the **upper arms (MC)**, face, upper trunk, breasts, butt, thighs
        - May have overlying discoid changes (scaling, follicular plugging, scarring)
        - Overlying skin may be "tethered" down creating surface depressions
        - **Path: lymphocytic lobular panniculitis with hyalinized fat necrosis**
    - **Hypertrophic (verrucous) LE**
        - p/w thick, hyperkeratotic **verrucous** scaling plaques on sun-exposed skin (upper half of body vs lower half in hypertrophic LP) **w/ features of DLE**
        - ↑ risk of SCC (like hypertrophic LP)
        - Path: orthohyperkeratosis w/ endophytic buds of hyperplastic follicular epithelium; pseudoepitheliomatous hyperplasia (often mistaken for SCC)
- Labs: ANA and other serologic abnormalities more common w/ widespread CCLE lesions
- Rx (all types of CCLE except chilblains which was discussed above): "SHaQ does Meth"
    - Sun-protection, Steroids (systemic steroids if lesions are widespread), **HCQ**
        - If no response add Quinacrine, if still no response add Methotrexate
        - **HCQ takes 3 months to work so wait 4 months until adding quinacrine**
        - Future directions: anifrolumab (inhibits IFN-1 receptor)

Tumid lupus related diseases

- Jessner's lymphocytic infiltrate

- Photosensitive eruption on a spectrum w/ tumid lupus and REM in adult females
  - p/w **annular red plaques** (may see central clearing) on the face, neck, upper back
  - Path: similar to tumid lupus and REM (superficial and deep PV/PA lymphocytic infiltrate w/ mucin; absent vacuolar interface) but w/ **CD8⁺ T-cells** and less mucin
    - "The word Jessner's has 8 letters in it"
  - Rx: resolution in 1-2 months with sun protection and antimalarials
- Reticular erythematous mucinosis
  - Photosensitive eruption on a spectrum w/ tumid lupus and REM in adult females
  - p/w red papules and plaques in a reticular configuration on the upper back/chest
  - Path: same as tumid lupus but DIF is negative
  - Rx: resolution in 1-2 months with sun protection and antimalarials

Subacute cutaneous lupus erythematosus (SCLE)

- More common in females (4:1) and more common in Caucasians (DLE is MC in AAs)
- Nearly half of patients will eventually meet criteria for SLE but will have mild disease
- Pathogenesis: **UVB** ("say luBus") > UVA-induced apoptosis → release of nuclear antigens (Ro/La) plus reduced Clearance of these apoptotic cells (due to Complement deficiencies) → loss of immune tolerance and production of proinflammatory Cytokines
- Genetic associations: HLA-B8-DR3; hereditary Compliment deficiencies
  - Whereas Bullous lupus and EBA have DR2 (B is 2nd letter of alphabet), Childhood DM and SCLE have HLA-B8-DR3 (C is 3rd letter of alphabet)
- Complement: SCLE is a/w Complement deficiencies in early intrinsic pathway (C1/C2/C4)
- p/w **Chronic, relapsing** course of psoriasiform or scaly annular (Central Clearing) plaques on lateral face/Chest (sun-exposed skin) → heals with hypopigmentation but no scarring
  - "Looks like a scaly chronic version of Jessners lymphocytic infiltrate/tumid lupus"
  - Ddx of annular scaly lesions: annular LP (purple plaques w/ white Wickman striae scale), EAC (trailing scale), tinea (peripheral rim of scale), PLC (lesions appear in crops and have central scale), guttate psoriasis (hundreds of small drop-like lesions w/ thicker scale), secondary syphilis (copper-colored lesions involving palms/soles), PR (herald patch, collarette of scale w/ Christmas tree orientation)
- Half of patients have severe **photosensitivity** (a/w the presence of anti-Ro/La Abs)
- MC systemic finding (70%) is **arthritis/arthralgias** ("joints Crack")
- Labs: **anti-Ro** (75-90%), anti-La (30-40%), ANA (70%; usually a speckled IIF pattern)
- Path: hyperkeratosis, epidermal atrophy, vacuolar interface, BMZ thickening, PV/PA lymphoid aggregates (superficial +/- deep dermis) w/ plasma cells and ↑ dermal mucin
- DIF: LBT (+) in 75%
- Rx ("SHaq does meth"): Sun protection, Steroids and (HCQ/chloroquine)
  - Counsel to avoid smoking w/ HCQ as this makes it less effective
- Before treating, consider drug-induced causes as 25% of SCLE cases are drug-induced (anti-Ro/SS-A+); unlike DI-SLE, there is no specific antibody that indicates DI-SCLE
  - 'Greasy PATCH' (as the mnemonic progresses, the drugs become more common)
    - Griseofulvin
    - Piroxicam, **PPIs**

        - ACE inhibitors, Anti-histamines, Anti-convulsants, **Aldactone**
        - Terbinafine (**2nd MC**), TNF-α inh. (etanercept is MC), Taxanes (docetaxel)
        - CCBs (Diltiazem is Delayed → has the longest latency until rash onset)
        - **Hydrochlorothiazide** (MCC): careful because the MCC of drug-induced SLE (Hydralazine and procainamide) and DM (Hydroxyurea) all start w/ "Hydr" but HCTZ and SCLE both have 4 letters

SCLE-like syndromes

- Neonatal lupus erythematosus (NLE)
    - Pathogenesis: transplacental passage of maternal anti-Ro (99%) autoantibodies
    - In a healthy Ro (+) mom, baby has 1-2% chance of NLE; in Ro (+) mom w/ CTD, baby has 15% of NLE; mom w/ prior NLE child, **next baby has 25% of NLE**
        - In all cases the MC outcome is healthy baby, **but still order serial fetal US**
    - 50% of mothers w/ NLE baby are asymptomatic at time of baby's birth but half of these mothers later develop SCLE, SLE or Sjögren's
    - Like SCLE, it is MC in Caucasians and p/w a photosensitive papulosquamous annular rash with prominent facial involvement which heals with Color Changes but no scarring (SCLE affects lateral face but NLE affects the central face around the eyes → "**raccoon eyes**"; racoon eyes in neuroblastoma are more bruise-like)
    - Systemic findings
        - Cardiac: MC systemic finding; 70% overall; 33% with Congenital 3rd degree (Complete) heart block which p/w irreversible bradycardia
            - "Ro makes the heart go slow"
        - Hepatobiliary disease (50%): elevated LFTs and bilirubin on CMP
        - Hematologic abnormalities: Cytopenias (esp **thrombocytopenia**) on CBC
    - 1st test: ECG (electrocardiogram)
    - Path: same as SCLE but DIF only shows (+) LBT in 50% (vs 75% in SCLE)
    - Labs: anti-Ro/SS-A, ↑ LFTs, Cytopenias (↓ platelets is MC)
    - Rx ("SHaq does meth")
        - Skin: Sun protection + topical Steroids
        - Cardiac disease ppx (mom): HCQ + systemic Steroids in pregnancy
            - Consider in Ro (+) moms w/ CTD or if prior baby had NLE
        - Cardiac disease treatment (baby): pacemaker for complete heart blocks
    - Prognosis
        - Skin: lesions resolve by 6 months with atrophy, dyspigmentation and telangiectasias which persists for years (but no scarring)
        - Cardiac: complete heart block is irreversible and has a 33% mortality
    - Stats to know
        - If mom had prior NLE baby, next baby has a 1 of 4 chance of getting NLE
        - 1 of 3 NLE babies will get 3rd degree heart block (1 of 3 of these will die)
        - 1 of 2 NLE babies will get hepatobiliary disease
        - 1 of 2 moms with a NLE baby will have no CTD symptoms at time of birth
- Complement deficiencies

- Pathogenesis (like SCLE): UV-damaged apoptotic keratinocytes express autoantigens (esp. Ro) which cannot be cleared due to complement deficiencies
- p/w ↑ risk of SLE and ↑ risk of encapsulated bacterial infections (recurrent PNA)
- **C2 def (MC but least severe)** > C4 > C1s > C1r > **C1q** (least common; most severe)
  - C2 deficiency-associated lupus: p/w **adult-onset** lupus w/ prominent photosensitivity, SCLE-lesions, ↑ bacterial infections but **no renal disease**
  - C1q/r/s and C4 deficiency-associated lupus: p/w **childhood onset** lupus w/ photosensitivity, SCLE/CCLE-lesions, ↑ bacterial infections w/ **severe, recalcitrant renal disease** +/- palmoplantar keratosis (C4 deficiency only; "2 palms + 2 soles are keratotic = 4")
- Anti-C1q autoantibodies (acquired): arises in nearly ½ of SLE patients (a/w renal disease) and nearly all of "hypoqomplementemic urtiqarial vasqultitis" patients
- Labs: low or (-) ANA, **(+) anti-Ro** in most, **↓ complement levels** (screen w/ CH50)

Compliment deficiencies: "Compliment Jessica ALBA"
↓C1q: Acquired Angioedema and severe SLE (C1q auto-Ab in SLE and 'hypoqomp urtiq vasq')
↓C2: MC compliment deficiency in Lupus
↓C3: Barraquer Simons syndrome (acquired partial lipodystrophy) w/ glomerulonephritis
↓C4: seen in all forms of acquired or hereditary Angioedema and in severe SLE w/ PPK

Acute cutaneous lupus erythematosus (ACLE)

- Of the 3 major lupus subtypes (DLE, SCLE, ACLE), ACLE is most strongly a/w SLE
- p/w **transient erythema on the nasal bridge and B/L malar eminence ("butterfly rash")** after sun exposure; **spares NL folds (vs DM)** as sun cannot get into those deep dark folds
  - Variable morphologies: scaling, papules, poikiloderma, atrophy, dyspigmentation
  - Rash may involve other sun-exposed sites such as neck, upper back, arms, legs and dorsal hands classically **sparring the knuckles (DM favors knuckles)**
- Path: vacuolar interface, sparse PV lymph infiltrate limited to upper dermis
- DIF: LBT (+)
- Labs: same as in SLE (ANA, dsDNA, CBC, CMP etc.)
- Rx: ACLE skin lesions correlate w/ systemic disease activity; treat systemic symptoms

Cutaneous lupus variants

- Bullous SLE
  - MC in females and African Americans (Black people)
  - Pathogenesis: Abs vs NC1 and NC2 domains of type VII collagen (same as EBA)
    - "VII" symbol looks like a BED frame: Bullous lupus, EBA, Dystrophic EB
  - HLA-DR2 (Bullous lupus and EBA have a B which is the 2nd letter of the alphabet)
  - p/w widespread, symmetric eruption of tense, subepidermal Bullae on an urticarial Base +/- mucosal lesions in patients with SLE
  - Path: subepidermal bullae with neutrophils at DEJ and in dermal papillae
  - DIF: continuous deposition of IgM/G/A and C3 along BMZ; IgG stains floor on SSS
  - Labs: same as in SLE (ANA, dsDNA, CBC, CMP etc.)

- Rx: dapsone (differentiates from EBA: "Extremely Bad Antidotes")

- Rowell's syndrome
  - Anti-Ro (+)
  - EM-like lesions arising in the setting of ACLE, SCLE or DLE
    - ROWELLL: Really Odd Wacky EM-Like Lesions in Lupus
- TEN-like lupus erythematosus
  - TEN triggered by excessive UV exposure in pts w/ preexisting ACLE or SCLE

Systemic lupus erythematosus (SLE)

- MC in females and African Americans
- Pathogenesis: complex but simply put it is a defect of phagocytes to clear apoptotic cells
  - ↑ plasmacytoid dendritic cells (**CD123+**) → ↑ IFN
- Genetics (SLE): multiple genes encoding STAT4, LYN and Early compliment components
- Triggers (SLE): Sunlight (UVB; "say LuBus"), Smoking, Low vitamin D levels, Estrogen
- High yield cutaneous manifestations in SLE include:
  - Sneddon syndrome: APLS + livedo reticularis + ischemic strokes
  - Loops of wandering and dilated glomeruloid-like vessels around the nails
    - vs symmetric dilation and vessel dropout seen in DM and SSc
  - LCV (and other forms of vasculitis like HUV or PNGD) → ulcers, purpura, Dego's
  - **Eruptive dermatofibromas**; malar or palmar Erythema; Erosions in mouth
- Fetal complications (SLE): Slow heart (congenital heart block), Loss of fetus, Early birth
  - Rx: treat mother w/ systemic steroids and HCQ
- Labs: ↑ ESR/CRP, cytopenias, UA, ↓ compliment levels, antibodies to "ABCD KRUSH"
- Path: from ACLE to SCLE to DLE, the vacuolar interface becomes more obvious, lymphoid infiltrate becomes deeper, and epidermis becomes thinner w/ more follicular plugging
- Rx ("SHaQ CRAB"): Sun protection/topical/IL/systemic Steroids, HCQ +/- Quinacrine
  - Recalcitrant disease: Cyclophosphamide, Rituximab, Anifrolumab, Belimumab
- Prognosis: younger onset has worse prognosis; 90% 10-year survival (MCC of death):
  - In First Five years of disease: MCC of death is Inflammatory lesions or Infection
  - Beyond five years: MCC of death is **thromboses** (DVT/PE/MI)
- Workup for SLE: ANA (dsDNA, Sm) UA, CBC, BMP, ESR/CRP, C3/C4, antiphospholipid Abs
- SOAP BRAIN MD criteria (4 of 11): Serositis, Oral ulcers, Arthritis, Photosensitive rash, Blood/Renal disease, ANA, Immunologic disease, Neurologic disease, Malar/Discoid rash

Things that erupt: "When coronavirus erupts, Gates HASSS D VAX"
Gryzbowski (eruptive keratoacanthomas)
Generalized eruptive Histiocytosis
Eruptive Aquired nevi (EBA and LS&A)
Eruptive Syringomas (Asians, Downs); Sign of leser trelat (SKs); eruptive SCCs/KAs (BRAF inh)
Eruptive Dermatofibromas (SLE)
Eruptive Vellous hair cysts
Eruptive Acne (conglobata)
Eruptive Xanthomas

Drug-induced SLE (DI-SLE)

- Begins > 1 year after drug initiation and resolves within 1 month of discontinuation
- Usually p/w no skin lesions and only mild lupus symptoms: arthritis/arthralgias, myalgias, fevers and serositis (no severe internal involvement like renal or CNS disease)
  - vs DI-SCLE which only has cutaneous findings and no systemic findings
- Most important implicated drugs: "Dads Tiny Mini Pirate SHIP"
  - D-penicillamine, **TNF-α inh** (Dads Tiny): looks like true SLE (↑ skin involvement)
  - Minocycline, Methimazole, PTU (Mini Pirate): a/w vasculitis and internal disease
  - Sulfonamides, **Hydralazine**, **INH**, **Procainamide** (SHIP): MCC of DI-SLE
- Labs: (+) anti-histone antibodies
  - DI-SLE from D-penicillamine or TNF-α inh: **(+) ds-DNA** and (-) anti-histone
  - DI-SLE from PTU, methimazole, minocycline: **(+) p-ANCA** and (-) anti-histone
  - DI-SLE from sulfonamides, hydralazine, INH, procainamide: **(+) anti-histone**
    - PTU and hydralazine can cause both (+) histone and (+) p-ANCA

Other autoimmune connective tissue diseases and sclerosing dermopathies

Dermatomyositis (DM; Polymyositis)

- F > M; bimodal Peaks (5-14 yo and 45-64 yo)
- Pathogenesis: immune-mediated process triggered by outside factors (malignancy)
  - HLA alleles: DR3, -B8 (juvenile DM), DR52 (anti-Jo Abs), -DR7, -DRw53 (anti-Mi-2)
  - **TNF-α** 308A polymorphism (a/w juvenile DM) → ↑ thrombospondin-1 → clots
  - **Type-1 interferons** are uniquely elevated in skin lesions of dermatomyositis
- p/w symmetric Proximal muscle weakness (lacks muscle pain) mainly affecting the shoulders and the hip girdles (difficulty brushing hair/teeth or walking upstairs)
- Skin findings (compared to lupus, lesions are more **Purpuric, Pruritic and Psoriasiform**)
  - Gottron's sign: macular erythema over joints such as elbows, fingers and knees
  - Gottron's papules: less common but pathognomonic; Purple Paps on knuckles
  - Symmetric confluent macular violaceous erythema (CMVE) of face involving NL folds (vs lupus), eyelids (heliotrope sign), lateral thigh (Holster sign), joints (Gottron's sign; erythema of lupus spares joints) and extensor tendons of hands
  - Photodistributed CMVE ("V-sign"/Shawl sign) and Photodistributed Poikiloderma
  - Mechanics hands: rough hyperkeratosis/fissuring of the lateral and Palmar digits
    - The hyperkeratosis is more prominent on ulnar side of thumb and radial side of index finger (where a mechanic would turn to tighten on knob)
    - Similar lateral hyperkeratosis is seen on the feet ("hikers feet")
    - a/w anti-synthetase syndrome (anti-Jo-1): "Jo is a mechanic"
  - Nail changes: ragged (hypertrophic) cuticles (Samitz sign), Periungual erythema, dilated capillary loops alternating w/ areas of Proximal nailfold vessel dropout
  - Scalp: Psoriasiform dermatitis and severe Pruritis (vs psoriasis/lupus with no itch)
  - Calcinosis cutis: seen in juvenile DM; favors elbows, knees and buttocks

        - a/w anti-p140 (NXP-2) autoantibodies
    - Painful Palmar Papules (inverse Gottron's sign): p/w Painful red Palmar Papules
        - a/w anti-MDA5 (CADM-140)
    - Vasculitis: a/w malignancy (in adults) or severe course in kids (Bankers variant)
    - Flagellate erythema (BDSM): Bleomycin, Docetaxel, Dermatomyositis, Diode laser, Stills disease, Mushrooms (shitake), Man-o-war/jellyfish
    - Wong-type DM has overlap features of DM and Pityriasis Rubra Pilaris
        - "Nothing Wong with sliding in her DMs but People Recognize Players"
- Non-muscular systemic findings: diffuse ILD (a/w anti-Jo and anti-CADM-140), heart disease (a/w anti-SRP), arthralgias and non-erosive arthritis
- Subtypes based on autoantibody
    - Anti-CADM-140 (MDA5): vasculopathic DM skin lesions (painful palmar papules, skin ulcers) w/ rapidly progressive ILD but no muscle disease (can affect kids too)
    - Anti-synthetase (anti-Jo): mechanics hands, severe ILD, arthritis and myositis
        - "I imagine an MD named CAD and mechanic named Jo (mechanic hands) in a race making them both SOB (ILD), but whereas CAD is an unathletic MD doctor and falls down early onto his hands (painful palmar papules and skin ulcers) and doesn't finish the race (so his muscles are not weak), Jo the athletic mechanic finishes the entire race (weak muscles)"
    - Anti-p155 (TIF-1): MC Ab in juvenile DM (a/w Gottrons papules, photosensitivity, linear extensor erythema) and a/w amyopathic and **cancer-associated DM** w/ severe skin disease (psoriasiform w/ **palmar hyperkeratotic papules**) in adults
        - MC cancers (PLONC): Pancreatic, Prostate, Lymphoma, **Lung**, **Ovarian**, Nasopharyngeal (MC in Asia), Colorectal; also gastric, breast, melanoma
            - **Ovarian (#1) and Lung (#2) are the two MC cancers a/w DM**
                - "The closer to the center of PLONC the more common"
        - **RFs a/w cancer (PLONC): males (Penis), LCV, Old age, skin Necrosis and evidence of Capillary damage on muscle biopsy**
        - **ILD and Raynaud's indicate less than average risk of malignancy**
        - 10-25% of adults w/ DM are diagnosed w/ cancer within 3 years
        - Risk for cancers returns to normal 5 years after diagnosis except the two at the ends of PLONC (Pancreatic and Colorectal stays ↑ beyond 5 years)
    - Anti-p140 (anti-NXP2): juvenile DM with calcinosis (peds DM is never a/w cancer)
    - Anti-Mi2 (helicase): classic DM skin findings with no visceral disease or cancer
    - Anti-SRP: fulminant myopathic DM w/ severe cardiac involvement
        - "Symmetric Rapid Proximal muscle weakness and Severely Rapid Pulse"
    - Anti-SAE: classic skin findings, dysphasia and HCQ-associated skin eruption
    - Anti-U1-RNP: DM overlaps with MCTD
    - Anti-Ku: DM overlaps with SLE, Sjögren's or scleroderma
    - ANA: although not specific, ANA is still the most sensitive antibody in DM
- Juvenile DM: average age is 7 yo; never a/w cancer
    - Brunsting variant: gradual onset of classic skin and muscle disease w/ calcinosis cutis which favors sites of trauma (fingers, elbows, knees); steroid responsive

        - Banker variant: rapid onset of vasculitis (ulcers, livedo reticularis, GI perforation) and severe muscle involvement; poor response to steroids/poor prognosis
- Path: subtle vacuolar interface, ↑BMZ, sparse PV/PA lymph infiltrate, ↑ dermal mucin
    - Looks just like lupus on path but epidermis tends to be more atrophied in DM
- Dx: after confirmatory biopsy evaluate for systemic disease
    - Evaluate for myositis: ↑ muscle enzymes (CK, **aldolase**), EMG, MRI or US of proximal muscles, muscle biopsy (gold standard; triceps muscle is highest yield)
    - Evaluate for cancer (PLONC): CA19-9 (pancreatic), PSA (prostate), CT (lung), **CA125/transvaginal US** (Ovary), FOBT/colonoscopy (colorectal)
    - Evaluate for ILD: **PFTs w/ CO diffusion** (if abnormal → high-resolution chest CT)
    - Evaluate for cardiac (EKG) and esophageal (barium swallow, manometry) disease
- Rx (similar to SLE, we start w/ sun protection and topicals then systemics and rituximab)
    - Non-vasculopathic skin disease: sun protection + topical CS/CNIs
        - IVIG or HCQ (morbilliform drug eruption in 31% of DM pts) if recalcitrant
            - IVIG has great efficacy for esophageal complications (dysphagia)
    - Skin and muscle disease: systemic CS (Prednisone) +/- steroid sparing agents
        - IVIG if patient failed therapy w/ steroid and DMARD
    - Vasculopathic skin disease (painful palmar papules; anti-CADM): rituximab + ILK
    - Future directions: JAK3 inhibitor (tofacitinib) is demonstrating great efficacy
- Prognosis: 5-year survival is 63% (MCC of death: cancer, Pulmonary/heart disease, PNA)

Periungual telangiectasias: "These conditions have ↑ ODSS of having periungual vessels"
Osler-Weber-Rendu: ectasia of half of capillary loop
DM: **symmetric dilation of vessels w/ vessel dropout** (↓ capillary density)
Systemic sclerosis: same as DM but w/ tapered fingers and hyper curvature of the nail
SLE: "wandering" dilated **glomeruloid** vessels (normal capillary density just tortuous/dilated)

Drug-induced DM: "Big DISH" (As the mnemonic progresses, the drug is more common)

- BCG vaccine, D-penicillamine, Infliximab (TNF-α inh), Statins, **Hydroxyurea (MC)**
    - Hydroxyurea-induced: Huge (60 month) latency, no myositis, usually ANA (-)
    - Non-hydroxyurea-induced (Statins MC): Shorter latency (occurs within 60 days of drug intake), myositis present and has positive Serologies (usually ANA+)

Sjögren's syndrome (RST: "around S in the alphabet)

- Pathogenesis: a/w anti-Ro/SS-A > anti-La/SS-B antibodies which are Ribonucleoproteins
    - Lymphocytic infiltration of Secretory glands (lacrimal and Salivary glands)
    - Mutation in **TNFAIP3** → ↑ risk of MALT-type non-Hodgkin **lymphoma** (like DH)
- p/w triad of xerophthalmia, xerostomia, arthritis (can't see can't spit can't climb up shit)
    - May also see skin xerosis/pruritis (MC skin finding), dyspareunia, neuropathy, **vasculitis**, glomerulonephritis, lymphoma, ↑ risk of baby with neonatal lupus
- Dx (RST): Rose Bengal test and Schirmer Test to detect impaired lacrimal gland function
    - Serologies: **Anti-α-fodrin** (most sensitive and specific), anti-Ro/SS-A, anti-La/SS-B
    - ANA (+) with a Speckled nuclear pattern on IIF

- Rx (RST + SLE Rx): **Suck on candy**, Sialagogue therapy (pilocarpine), artificial Saliva/Tears
  - SLE treatments (SHaQ CRAB): Steroids, HCQ +/- Quinacrine and CRAB for recalcitrant cases (Cyclophosphamide, Rituximab Anifrolumab, Belimumab)

Relapsing Polychondritis (RP)

- Pathogenesis: Th1 mediated inflammatory disease (IFN, IL-2, IL-12) that is a/w HLA-DR4
- Autoantibodies against type II Collagen ("car-two-lage") and matrillin-1
- Need 3 of 6 diagnostic criteria: inflammation of joints (non-erosive inflammatory Polyarthritis), Cochlear/vestibular damage and Chondritis of 4 facial orifices:
  - Chondritis of both auricles **sparing earlobes** (MC criteria) → "Cauliflower" ears
    - **Ddx: auricular hematoma in a child (child abuse)**, lobomycosis (keloidal ear lesions), juvenile hyaline fibromatosis (fibrotic nodules on ears)
  - Chondritis of nasal cartilage: ↓ smell and saddle nose deformity (see below)
  - Chondritis of ocular structures: conjunctivitis, corneal ulceration, uveitis, iritis
  - Chondritis of respiratory tract: airway collapse → **Pneumonia (MCC of death)**
- Other features: vasculitis (2nd MCC of death), cardiac valvopathy, aphthous ulcers, EN
- Path: breakdown of normal lacunar cartilage structure with a neutrophilic infiltrate
- Rx: Prednisone is 1st line (MTX is best steroid sparing agent); dapsone for ppx
- MAGIC syndrome (Mouth And Genital Ulcers w/ Inflamed Cartilage) = RP + Behçet's
- Ddx for Nasal destruction (LMNOPqRST): Leishmaniasis (Mucocutaneous), NK T-cell lymphoma, Noma, Olfactory tumors, PLAID, Paracoccidoidomycosis, RP, Rhinoscleroma, Rhinosporidiosis, Syphilis (late congenital or tertiary), Sarcoid, SAVI, TB (lupus vulgaris)

Conditions with saddle nose deformity: "Saddle GIRL"
Granulomatosis with polyangiitis
Idiopathic
Relapsing polychondritis, Rothmund-Thompson
Late congenital syphilis, Lepromatous Leprosy

Mixed Connective tissue disease (MCTD)

- Mixed features of at least 2 of the following (PqRS): PM/DM, RA, SLE, Scleroderma
- Pathogenesis: a/w HLA-DR4 and autoantibodies against Anti-U1RNP
  - "The 4th DR has a fast car which goes VVRRUM": Vitiligo, pemphigus Vulgaris, Relapsing polychondritis, RA, Urticaria (chronic idiopathic) and Mixed CTD
- Abs to EBV and CMV cross-react w/ U1RNP, suggesting that Molecular Mimicry related to prior exposure to these viruses may be responsible for anti-U1RNP autoimmunity
- p/w Mixed features of PM/DM (myopathy, DM-rash), RA, SLE (arthritis), scleroderma (Raynaud's is the MC/earliest sign; pulmonary HTN is the most severe sign of MCTD)
- **DIF: characteristic immune deposits (IgG is MC) within epidermal cell nuclei**
- Dx: ANA (+) w/ a speckled pattern (fluorescent areas Mixed in w/ less fluorescent ones)
- Rx: prednisone (MTX is 1st line for severe arthritis but watch out for pulmonary fibrosis)

Systemic-onset juvenile idiopathic arthritis (Stills disease; “stIL-l disease”)

- Still’s disease is only 1 of 6 forms of JIA but is the most relevant to dermatologists
- Autoinflammatory syndrome (disorder of innate immune system) w/ **↑IL-1 cytokines**
- p/w **daily high fevers** (evenings) and an associated **transient salmon-pink exanthem** w/ a predilection for the axilla/waist; linear lesions due to köebnerization
  - Other features include symmetric arthritis, uveitis, LAD/HSM and serositis
- Dx: ↑ESR/CRP, **↑ ferritin** (RF and ANA are negative)
- Rx: NSAIDS, prednisone, MTX, TNF-α inhibitors depending on severity
  - IL-1 (anakinra, **canakinumab**, rilocept) and IL-6 inh (tocilizumab) work well
- Refer to ophthalmology for uveitis screening
- Prognosis: resolves in 50%, exhibits chronic course in 50% and progresses to life-threatening macrophage activation syndrome in 5% (pancytopenia, neurologic issues)

Adult onset Still disease

- Presents in young adults (but not < 16 years old by definition)
- Presents similar to Still’s with spiking afternoon fevers w/ an associated transient salmon-pink exanthem +/- persistent papules/plaques, arthralgias and LAD/HSM but is a/w a **flu-like prodrome w/ a sore throat, carpal ankylosis** (limited ROM in the carpal joints) and a more common progression to macrophage activation syndrome (15%)
- Criteria (needs 2 major and 3 minor)
  - Major: fever > 39°C x 1 week, arthralgias x 2 weeks, typical rash, WBC > 10,000
  - Minor: pharyngitis, LAD, ↑ LFTs, negative ANA and RF, ↑ ferritin
- Dx: labs are like Juvenile Still’s
- Path: neutrophil-predominant PV inflammation w/ necrotic keratinocytes in epidermis
- Rx: same treatment as for Juvenile Still’s

Rheumatoid arthritis

- a/w HLA-DRB1 (↑ propensity to form anti-CCP antibodies due to molecular mimicry to proteins produced by EBV and E. coli), HLA-DR1 and HLA-DR4
  - BP and DH have epitope spreading; RA and MCTD have molecular mimicry
- Pathophysiology: initiating event is binding of ECM components to TLR → ↑ inflammatory cytokines (TNF-α, IL-1, IL-6) → cartilage break down and B cell activation
  - RANKL binds RANK on osteoclasts (bone erosion)
  - RF and anti-CCP antibodies form immune complexes inside joints → activation of compliment → activation of neutrophils which cause cutaneous findings
- PTPN22 gene confers susceptibility to RA and other autoimmune conditions
- Skin findings (associated with ↑ RF antibody titers)
  - **Rheumatoid nodulosis p/w painless nodules over bony prominences**
  - Accelerated rheumatoid nodulosis p/w sudden appearance of multiple **painful** rheumatoid nodules following initiation of **methotrexate** or TNF-α inhibitors
  - Rheumatoid vasculitis (40% mortality) p/w systemic vasculitis (i.e. neuropathies, carditis) in patients with a long history of severe erosive RA
  - Bywater’s lesions p/w nailfold thromboses and **purpuric macules on distal digits**

        - Demonstrates LCV on path but no signs of systemic vasculitis
    - Felty syndrome p/w Low WBCs, Large spleen, Leg ulcers, Leukemias/Lymphomas
        - Rx: G-CSF or splenectomy
    - Superficial ulcerating necrobiosis (rheumatoid necrobiosis) resembles NLD
    - Rheumatoid neutrophilic dermatosis resembles Sweets syndrome
        - Rx: prednisone or antineutrophilic agents (dapsone, colchicine)
        - Other neutrophilic conditions a/w RA include EED, PG and IGDA/PNGD
- Path: early lesions resemble IGDA/PNGD (granulomatous or neutrophilic infiltrate) while later lesions show **palisading granulomas surrounding bright Red eosinophilic fibrin**
- Dx: RF (most sensitive) and anti-CCP (most specific)
- Rx (arthritis): NSAIDS, prednisone, DMARDS (MTX, hydroxychloroquine, sulfasalazine leflunomide) or biologics (TNF-α inhibitors, abatacept, tocilizumab)
    - Rheumatoid nodules do not respond to arthritis treatment (consider IL steroids)
- Prognosis: course waxes and wanes; MCC of death is ischemic heart disease

Morphea (localized scleroderma; "Go BLAND and wear hard HATS, SAM")

- Most cases present in childhood w/ F > M (except linear morphea in which M = F)
- Plaque type is MC variant overall and in adults but linear is MC variant in children
- Pathogenesis: genetic predisposition + environmental trigger (**Borrelia afzelii/garinii**) → vascular injury → inflammation → ↑ profibrotic Th2 cytokines (**↑ TGF-β**) → ↑ collagen
- Plaque morphea: p/w hairless and indurated hyperpigmented plaques w/ a **lilac** (violet tone on top of pink)**-colored inflammatory border** on the **trunk** or proximal extremities
    - This lilac (pink-purple) color is also the color seen in lupus and dermatomyositis
    - **Ddx: extragenital LS&A favors the breast/chest, back, posterior neck, butt and wrists and has a more porcelain-white and wrinkled center (vs hyperpigmented in plaque morphea) w/ a violaceous border +/- follicular plugging**
- Clinical subtypes of plaque morphea ("Go BLAND")
    - Generalized morphea: p/w widespread indurated plaques → muscle atrophy and difficulty breathing (constrictive effect of taut skin on chest); a/w fatigue/malaise
        - Pansclerotic variant: generalized morphea involving deep structures
    - Guttate morphea: p/w multiple, small flat or slightly depressed white macules
        - Appears similar to guttate LS&A but lacks follicular plugging
    - Bullous morphea: usually only seen in the context of generalized morphea
        - Diffuse sclerosis of skin → lymphatic obstruction → bullae formation
    - Linear morphea: p/w linear lesions of sclerosis on the **Legs**
        - a/w serious morbidity (permanent limb undergrowth/joint contractures)
        - Involves deeper (Lower) structures such as muscle, fascia and bone
        - "Lower age, lower extremities, lower structures involved, lower QOL"
        - Melorheostosis (roughened long bones seen underlying linear morphea and in Buschke-Ollendorff): resembles wax dripping off a candle on X-ray
        - Head/neck subtypes of linear morphea
            - En coup de sabre: variant of linear morphea of the scalp
            - Parry-Romberg syndrome: variant of linear morphea of the face

            - a/w epilepsy and eye involvement (**next step is eye exam**)
    - Atrophoderma of Pasini and Pierini: superficial variant of plaque morphea
        - p/w large hyperpigmented atrophic plaques w/ sharp "cliff-drop" borders
        - MC location is back and pre-sacral area ("Above the Ass")
    - Nodular or keloidal morphea: keloid-like nodules/streaks
    - Deep morphea (morphea profunda): p/w pseudo-cellulite look (like eosinophilic fasciitis) due to sclerosis of the deep structures (i.e. fascia, muscle)
        - Difficult to treat: poor response to steroids (vs eosinophilic fasciitis)
- MC associated symptom: **arthralgias** (esp w/ deep, generalized and linear morphea)
- Path: square punch, eccrine glands "trapped" and compressed by sclerotic collagen
    - Early lesions show a lymphocytic infiltrate w/ plasma cells at the dermal-SQ junction w/ loss of $CD34^+$ dendritic cells (vs ↑ in NSF and scleromyxedema)
- Dx/antibodies ("hard HATS"): all forms lack anti-Scl70/topoisomerase 1 (vs SSc)
    - Anti-Histone: linear and generalized forms (correlates w/ disease activity)
    - ANA: linear and generalized forms; correlates w/ disease severity/activity
    - Anti-Topoisomerase 2 (Two): all forms
    - Anti-ssDNA: linear form only (correlates w/ disease activity)
- Rx ("SAM"; Steroids, UVA phototherapy, Acitretin, Methotrexate)
    - Superficial localized lesions: topical or IL CS + UVA phototherapy
    - Superficial generalized lesions: UVA phototherapy
    - Deep lesions (linear or deep morphea): **pulsed IV Steroids + MTX**
    - Isotretinoin/Acitretin are affective for morphea, LS&A and sclerodermoid GVHD
- Prognosis: superficial plaque morphea is self-limited (resolves in 3 years); spontaneous resolution for deep and generalized morphea is uncommon
- High-yield points: treat deep/severe Morphea w/ MTX (1st line) or Mycophenolate mofetil (2nd line); see Melorheostosis in linear Morphea
    - Rx MCTD w/ MTX also (careful MTX and MCTD both cause pulmonary fibrosis)

Scleroderma (systemic sclerosis; SSc)

- Mainly affects F > M in 30-50s but older age, AA race, male sex, truncal distribution, anemia, internal organ involvement (**pulmonary**) and ↑ ESR confers a worse prognosis
    - "Old black males with truncal sclerosis, pulmonary disease and ↑ ESR do worse"
- Pathogenesis: key features include vascular dysfunction (earliest feature), immune activation (↑ Th2 profibrotic cytokines i.e. **TGF-β**, IL-1, -4, -6, -8, IGF) w/ autoantibody production (anti-centromere, anti-Scl70) and deposition of collagen in skin and organs
- SSc subtypes
    - Limited SSc: cutaneous sclerosis limited to face and distal extremities
        - CREST syndrome is a variant: Calcinosis, Raynaud's, Esophageal dysmotility, Sclerodactyly, **flat matted Telangiectasias on face/palms**
            - Ddx: HHT also has matted telangiectasias (but are in the mouth)
        - Lacks severe renal involvement or ILD (**MCC of death is PAH**)
    - Diffuse SSc: cutaneous sclerosis is generalized; a/w poor prognosis
        - a/w severe, rapidly progressive visceral disease (**MCC of death is ILD**)

- Cutaneous features (check the fingers)
  - **Skin thickening of fingers on both hands extending proximal to MCP**
    - **Solely having this feature is sufficient to meet the diagnostic criteria**
    - 3 stages: edema (1st year), induration (1-4 years), atrophy (forever)
  - Sclerodactyly: sclerosis/thickening of fingers distal to MCPs but proximal to PIPs
  - Fingertip lesions: **digital tip ulcers (due to recurrent vasospasm)** → pitting scars
  - Interphalangeal joint ulcers: due to trauma
  - Nail folds capillaries w/ dilated capillary loops alternating w/ areas of vessel loss
    - November 2022 JAAD CME 1 has a great picture of this
  - Pterygium inversum unguis (ventral pterygium) vs dorsal pterygium (seen in LP)
  - Raynaud's phenomenon: MC initial presenting sign of SSc (hand edema 2nd MC)
    - Whereas CCBs and α-blockers help Raynaud's, **β-blockers exacerbate it**
  - Other cutaneous features: mat telangiectasias, calcinosis, **"salt and pepper" pigmentation** (leukoderma w/ perifollicular sparing), microstomia, beaked nose, hyperpigmentation increased at areas of pressure (belt), pruritis and ↓ sweating
- Systemic features
  - Pulmonary disease is MCC of death: (**ILD in diffuse** and **PAH in limited**) and 2nd MC site of visceral involvement (after GI/dysphagia)
  - GI (dysphagia): MC site of visceral involvement (↑ morbidity but no ↑ mortality)
  - Renal: scleroderma renal crisis affects 20% of diffuse SSc (Rx: ACE inhibitors)
  - Other systemic features: restrictive cardiomyopathy, sicca syndrome, tendon friction rubs, arthralgias, proximal muscle weakness and esophageal dysmotility
- Path: same as morphea (differentiate w/ physical exam and autoantibodies)
- Dx: antibodies in systemic sclerosis ("Hard fibrotic CARS")
  - Fibrillarin (U3RNP): a/w diffuse skin disease and pulmonary HTN; poor prognosis
  - Centromere: a/w limited SSc/CREST (a/w pulmonary HTN and digital ulcers) > SSc
  - ANA (nucleolar staining pattern seen on IIF)
  - **RNA polymerase III**: a/w SSc (**Renal**; Rapidly progressive); ↑ risk of cancer
  - Scl-70 (DNA topoisomerase I): a/w SSc (lung disease) > CREST; poor prognosis
- Main ddx ("hard EGGS"): Eosinophilic fasciitis, Eosinophilia-Myalgia syndrome (ingesting L-tryptophan), Generalized morphea, GVHD (chronic), Scleredema, Scleromyxedema
- Rx (SSc/CREST): **avoid Cold exposure**, Cigarette Cessation and Corticosteroids +/- MTX in SSc (careful because **steroids can exacerbate renal crisis** and **MTX can exacerbate ILD**)
  - Skin sclerosis: **Celcept is 1st line**, Cyclophosphamide, MTX, phototherapy (PUVA/UVA1), CS, rituximab, tocilizumab (IL-6) help; HSCT if severe
  - ILD: MMF (**Celcept**) and Cyclophosphamide are first line; rituximab, nintedanib
  - Pruritis: Cannabinoid receptor modulator (lenabasum), antihistamines
  - Renal disease/HTN: ACE inhibitors (Captopril)
  - PAH: PDE5i (sildenafil), ERA (bosentan), guanylate cyclase stimulator (riociguat)
  - Raynaud's: non-cardioselective **CCBs** (nifedipine) are 1st line, PDE5i (sildenafil) and prostanoids (IV iloprost if severe) are 2nd line; botox, SSRIs, ARBS/ACEi help
  - Digital ulcers: same as Raynaud's but PDE5i alone or in combo w/ a CCB is 1st line

    - GI: PPIs for GERD, prokinetic agents (metoclopramide) for GI dysmotility, nutrient supplementation for malabsorption, antibiotics for bacterial overgrowth
    - Calcinosis cutis (poor evidence): Coumadin, CCBs, Cut it out, sodium thiosulfate
    - Mat telangiectasias: vascular lasers (**PDL**, KTP IPL), green-tinted camouflage
    - Microstomia: IPL, IL hyaluronidase, CO2 laser
- Screening in SSc (**CXCL4** is a new biomarker for PAH and skin/lung fibrosis)
    - ILD: **PFTs** at baseline and every 4-6 months for 3 years → high-resolution CT
    - PAH (DETECT algorithm): telangiectasias, anti-centromere Ab, BNP, serum urate or EKG w/ right axis deviation → echo-Cardiogram → right heart Catheterization
    - Renal: Cr, urea, electrolytes and UA at baseline and annually)

DNA Topoisomerases
Topoisomerase 1 (anti-Scl-70): auto-antibody in diffuse SSc
Topoisomerase 2 (DNA gyrase): autoantibody in morphea ("1 more topoisomerase than SSc")
Topoisomerase 2 and 4: fluoroquinolones inhibits topo 2 (G- coverage) +/- 4 (G+ coverage)

Eosinophilic fasciitis

- Pathogenesis: unknown but **TGF-β is markedly Elevated**
- a/w **strenuous Exercise** (precedes clinical findings by days to weeks in 30% of cases)
- p/w rapid onset (Eruptive), symmetric (Equal), Edema in Extremities (**spares hands/feet**)
    - Develops a peau d'orange/pseudo-cellulite look (like deep morphea and NSF) w/ a **"dry riverbed/groove sign" (linear depressions of veins within indurated skin)**
    - Lacks Raynaud's as it does not involve the hands/feet (vs SSc)
- a/w joint contractures, arthritis and loss of mobility > lymphoma/leukemia, MGUS, MM
    - "Can't Exercise anymore and they get leuk-E-mias"
- Ddx: SSc (but spares hands), Eosinophilic-myalgia syndrome (but lacks fever/myalgias), and deep morphea (but has excellent response to systemic steroids)
- Path (must obtain deep biopsy to fascia): fibrosis of deep fascia +/- eosinophils
- Dx: **CBC (peripheral Eosinophilia)**, ↑ ESR, ↑ TGF-β and normal ANA
    - **CBC may show pancytopenia due to an associated myelodysplastic syndrome**
        - "If veins are clinically low ("groove sign"), CBC may show low cell counts"
    - MRI or CT may show fascial thickening and eliminate the need for a biopsy
    - **Metalloproteinase inhibitor-1 (TIMP-1)** serves as a marker for disease activity
- Rx: Excellent response to **systemic steroids** (vs deep morphea) + **physical therapy**

Nephrogenic systemic fibrosis (NSF)

- Nephrogenic disorder (results from gadolinium contrast exposure in setting of AKI/CKD)
    - Pathogenesis: gadolinium leaks into tissues and triggers profibrotic cytokines → circulating fibroblasts ($CD34^+$, Pro-collagen I^+) recruited to skin → ↑ collagen
- p/w pruritis and edema → Stiffening of the skin w/ **Symmetric "ameboid" (Swirvy) plaques** w/ pSeudo-cellulite look MC on the extremities (**always Spares the face**)
    - Puckering/rippling/linear banding is seen due to focal areas of bound-down skin
- Extracutaneous involvement: **Scleral plaques (white-yellow plaques)** > organ fibrosis

- Path: identical to Scleromyxedema (↑ $CD34^+$ Spindled fibrocytes, ↑ collagen, ↑ mucin) except is generally deeper and may see "lollipop bodies" (elastic fibers stick like a lollipop stem out from the center of round osseous metaplasia)
  - vs morphea/scleroderma which has loss of $CD34^+$ cells in dermis (and no mucin)
- Rx: Failure of treatment; Fast death (try to treat kidney disease; physical therapy)

Mucinoses

Scleredema

- p/w erythema and woody induration of skin of face and trunk (distal extremities spared)
- 3 types ("It's probably hard to SKI with scleredema")
  - Type 1 (55%; a/w **Strep** infection): p/w induration of the cervicofacial region with extension to the trunk and arms; difficulty Swallowing; Self-resolves
  - Type 2 (25%; a/w **IgGκ** monoclonal gammopathy); a/w a "kronic course"
  - Type 3 (20%; scleredema diabeticorum; a/w **Insulin dependent diabetes**): p/w erythema/induration of the posterior neck/back; Indolent onset, chronic course
- Path: **wide spaces b/w deep dermal collagen bundles filled w/ mucin** (Ddx: back skin)
- Rx: type 1 is self-limited; treat Type 2 and 3 with phototherapy (UVA1 and PUVA)
- **Workup ("SKI"): next step is SPEP/Immunofixation to look for paraproteinemia**

Scleromyxedema

- p/w widespread **Symmetric Closely aligned papules in a Linear array** on the upper body
  - Surrounding skin is Shiny and Stiff (Sclerodermoid) w/ **Sclerodactyly of digits**
  - **Scalp is Spared** (face is spared in NSF; hands are spared in eosinophilic fasciitis)
  - **Shar-Pei Sign**: deep furrowing of back skin (named after shar-pei breed of dog)
  - **Donut Sign**: induration of skin w/ **Central Low-lying depression over PIP joint**
- Systemic symptoms: dysphagia, myositis, renal/lung disease, dermato-neuro syndrome (p/w high fever, flu-like symptoms, seizures and coma; this patient will be in the ICU)
- CNS involvement → Coma and Carpel tunnel
- **Leonine faces** (deep furrows of the glabella; "it is like the Shar-Pei sign on the face")
- a/w **IgGλ (Lamda)** paraproteinemia in 90% of patients
- Path: looks Like NSF (↑mucin, ↑collagen, ↑ $CD34^+$ fibroblasts) but not as deep into SQ
- Rx: **IVIG** (1st line; "I looks like a lower-case L"), Systemic Steroids + thalidomide (2nd line)
  - Refractory disease: autologous Stem Cell transplant, bortezomib, melphalan
  - Dermato-neuro syndrome: IVIG + systemic steroids +/- melphalan, bortezomib
- Workup: next step is **SPEP/immunofixation** to look for paraproteinemia
- **Lichen myxedematosus** is a localized variant w/ only the Symmetric Closely aligned Linear papules but w/o the surrounding sclerodermoid changes and systemic symptoms
  - a/w immune Suppression (**HIV**), Hepatitis C or exposure to L-tryptophan/toxic oil

PreTibial myxedema

- p/w waxy indurated Purple-brown Plaques w/ a Peau d'orange look on the Pretibia

- a/w Thyroid excess (**Graves** disease)
- **Path: Tuns of mucin in the dermis w/ attenuation/fragmentation of collagen fibers**
- Rx: Triamcinolone 10-40 mg/cc or steroids applied under occlusive dressings
- Generalized myxedema is seen in hypothyroid states and typically effects the face

4.11 NEUROCUTANEOUS SYNDROMES

- RASopathies were formerly called neurocutaneous syndromes but no longer as there are RASopathies that are not neurocutaneous syndromes (i.e. CM-AVM; good chart in section 4.6) and there are neurocutaneous syndromes that are not RASopathies
- RASopathies ("PLAN C") have mutations in the RAS-MAPK pathway and have overlapping features of craniofacial, cardiac (esp pulmonary valve) and MSK anomalies
  - Parkes-Weber, Legius, LAD type 3, CM-AVM, NF-1, Noonan and Noonan w/ multiple lentigines (LEOPARD), Cardio-facio-cutaneous syndrome, Costello
  - Noonan and the "Cs" in "PLAN C" commonly have lymphatic malformations like the entities in the PIK3CA-related overgrowth spectrum (i.e. CLOVES)

Neurocutaneous syndromes that are RASopathies

Neurofibromatosis type 1 (NF1; von Recklinghausen's disease; "LMNOPqRST")

- AD mutation in the tumor suppressor gene **Neurofibromin** (NF1), **a GTPase-activating protein which negatively regulates the RAS pathway** (turns active GTP to inactive GDP)
  - Loss of NF1 increases RAS activity → various cancers (**breast**, melanoma etc.)
  - Loss of heterozygosity is a phenomenon seen in many conditions (esp those w/ loss of function mutations in tumor suppressor genes) in which the pt has a germline mutation in one copy of the allele and every neurofibroma or tumor you see in NF1 is due to an acquired loss of heterozygosity of the second allele
- **Diagnostic criteria** (must have **2 or more** of the following: LMNOPqRST)
  - **Lisch nodules** (2 or more): tan hamartomas of the iris (90% by 10 years of age)
  - **Café-au-lait Macules** (**6 or more** > 0.5 cm prepubertal or > 1.5 cm postpubertal)
  - **Neurofibromas** (**2 or more**) or 1 plexiform neurofibroma (bag of worms feel)
  - **Optic nerve glioma** which can cause blindness (vs NF2 which can cause deafness)
  - **Pigment in the axilla or inguinal region (Crowe's sign)**; must be bilateral
  - Relative with NF1 (new criteria changed this to parent w/ NF1)
  - Sphenoid wing dysplasia: p/w pulsating exophthalmos but often asymptomatic
  - Tibial dysplasia/pseudarthrosis
- Other important features (LMNOPqRST)
  - Learning disability, Leukemias (**CML** is MC)
  - Macrocephaly (infancy), Meningiomas (prebubertal), Malignant peripheral nerve sheath tumors (rapid enlargement/pain of a plexiform neurofibroma in an adult)
  - Neurofibrosarcoma, Nevus anemicus
  - Ocular manifestations (Lisch nodules, optic glioma, glaucoma); Ovarian cancer
  - **Pheochromocytomas** (HTN); Prostate cancer
  - Pseudoatrophic macules (slightly depressed oval lesions) and blue-red macules are recently reported cutaneous findings (MC on trunk) seen in 10% of NF1 pts
    - Path: Perivascular neural tissue
  - **Pulmonic stenosis + NF1 features = Watson syndrome**
  - p53 is the MC 2nd mutation causing malignant transformation of a neurofibroma

    - Rhabdomyosarcoma
    - Seizures, Short stature, Scoliosis and other Skeletal dysplasias
- **1st sign = congenital skeletal dysplasia (tibial/sphenoid wing); 1st skin sign = CALMs**
- 25% of NF1 pts have a plexiform neurofibroma (PNF) which undergo malignant transformation to MPNST (neurofibrosarcoma) in 3-15% of pts which may p/w rapid growth, increased firmness, persistent pain within an established PNF or the development of new neurologic deficits (any of these symptoms in a PNF → **PET-CT**)
- Rx: MEK inhibitor Selumetinib is now FDA approved for associated tumors
- Workup: screen for hypertension (Renal angiography and 24-hour urine collection for catecholamines in those w/ HTN), NCCN recommends annual breast MRI starting at age 25 to 40, Ophtho exam before age 8, brain MRI for neuro symptoms and PET-CT (above)
- Consult: ortho, ophtho, endocrine and neuro

Legius syndrome
- AD mutation in SPRED1 (interacts with Ras)
- Has cutaneous features of NF1 except the neurofibromas (CALMs, Crowe's sign)
    - "Spred your legs so I can see your intertriginous freckles"
- Patients may also have learning disabilities, but all other NF features are absent

NF2 (bilateral acoustic Schwannomas)
- AD mutation in tumor Suppressor gene SCH (encodes schwannomin/**merlin**)
- Skin findings usually absent but can get CALMs and SQ neurofibromas (vs dermal in NF1)
    - **SQ neurofibromas have overlying pigment or hair**
- Bilateral vestibular Schwannomas → deafness/tinnitus/dizziness (DO NOT Swim alone)
- Ocular finding: juvenile posterior Subcapsular lenticular cataract
- MCC of death: CNS tumors (meningiomas, astrocytomas, ependymomas)

Genoderms with CALMs: "When a MAN FARTs he stays CALM"
MEN-1, McCune-Albright, Ataxia-telangiectasia, Noonan, NF1 and legius > NF2 Fanconi, Ataxia-telangiectasia again, Russel-Silver, Tuberous sclerosis

Costello syndrome (facio-cutaneo-skeletal syndrome)
- AR mutation in **HRAS** (RAS-MAPK pathway)
- p/w thick **loose skin on hands** w/ **deep palmoplantar Creases** and ulnar deviation of the wrist, Coarse facies w/ **nasolabial warts,** periorificial papillomas, AN, curly hair, MR
    - "Costellow" has low skin hanging on hands, low/deep hand creases and low IQ
- Cardiac: **pulmonic stenosis**, cardiomyopathy, lymphatic malformations (lymphedema)
- Cancer: rhabdomyosarcoma, transitional cell (bladder) cancer
- Phenotyping overlap w/ Cardio-facio-cutaneous syndrome (KRAS mutation)
- All RASopathies ('PLAN C' i.e. CFC, NF1) present similar; need genetic tests to distinguish

Neurocutaneous syndromes that are not RASopathies

Tuberous sclerosis complex (TSC)

- AD mutation in the tumor suppressor genes hamartin (TSC1) or tuberin (TSC2) which are GTPases that negatively regulate the **mTOR** (mammalian target of rapamycin) pathway
- Diagnostic criteria: "BOARDS CRAP"
    - **Brain stuff (MCC death)**: subependymal giant cell astrocytoma, subependymal nodules (often calcified), cortical tubers (often calcified)
        - **Seizures/infantile spasms is the MC presentation** (a/w worse prognosis)
    - Ocular hamartomas: retinal phakomas
    - Ash-leaf macules (> 2 hypopigmented macules > 5mm each): **1st cutaneous sign**
    - Renal stuff (renal failure is 2nd MCC of death): angiomyolipomas, cysts
    - Dental enamel pits (>2)
    - Shagreen patch: collagenoma MC on the back
    - CALMs (only entity in this mnemonic that is NOT in the diagnostic criteria)
    - Rhabdomyoma (cardiac) → Wolf-Parkinson-White arrythmia
        - **First sign and can be detected on prenatal US**
    - Angiofibromas on the face (>2; adenoma sebaceum) or fibrous cephalic plaque
        - Rx: topical rapamycin
    - Pulmonary lymphangioleiomyomatosis (p/w pneumothorax in females)
        - Dx: high resolution CT scan shows thin-walled cysts; VEGF-D blood test
    - Periungual fibromas (>1; Koenen tumors)
- All entities in the above mnemonic are major criteria except for renal cysts (minor), dental pits (minor) and CALMS (not a criteria); other minor criteria include confetti-like macules, nonrenal hamartomas, retinal achromic patch and oral fibromas
- Dx: **2 major** or **1 major and 2 minor** is a definitive diagnosis (order **MRI brain/abdomen**)
- Rx: topical rapamycin for facial angiofibromas; systemic mTOR inhibitors (sirolimus) for renal/hepatic angiomyolipomas and subependymal giant cell astrocytoma

Incontinentia pigmenti (IP)

- **XLD loss of function mutation in nuclear factor-kB (NF-kB)** essential modulator (NEMO; IKBKG) → protein product normally **confers protection against TNF-mediated apoptosis**
    - X-inactivation results in only cutaneous findings (blaschkoid pattern)
- Cutaneous manifestations follow a blaschkoid pattern and have 4 distinct stages
    - Vesicular (weeks): inflammatory vesicles and pustules
    - Verrucous (months): warty papules
    - Hyperpigmented (years): blue to brown streaks
    - Hypopigmented (persists throughout life): atrophic hypopigmented streaks
- Other skin findings: scarring alopecia, **subungual tumors (resemble SCC)**, nail dystrophy
- Other findings: pegged teeth/anodontia (50%; **MC extracutaneous finding)**, scarring alopecia (30%), seizures/MR (30%), **eye issues** (30%), breast/nipple abnormalities (20%)
- Path
    - **Vesicular: eosinophilic spongiosis w/ dyskeratotic keratinocytes**
    - Verrucous: hyperkeratosis and papillomatosis w/ dyskeratosis in whorls/clusters
    - **Hyperpigmented: pigment incontinence (dermal melanophages)**

    - Hypopigmented: normal to decreased melanocytes +/- dermal fibrosis
- Consult: dental, neuro, long term ophtho f/u for retinal vascular changes/optic atrophy
- Female carriers with mild IP phenotype can bear children with hypohidrotic ectodermal dysplasia w/ immunodeficiency (both HEP and IP have teeth and breast abnormalities)

5.4 FUNGAL DISEASES

Superficial mycoses

Dermatophytes

- Pathogenesis: hydrolases and keratinases are virulence factors which permit penetration into the stratum corneum inducing a Th1 response in the host
- Species: Microsporum, Epidermophyton, Trichophyton ("He MET dermatophyte fungi")
- T. rubrum is the MCC of many tinea infections and they can be remembered by the T. rubrum song: "Rub my yocki, my body, hands/feet/nails and secret spotty"
    - T. rubrum #1: Majocchi granuloma, tinea corporis/manus/pedis/unguium/cruris
    - Exceptions to this rule are vesicular variant of T. pedis and white superficial variant of T. unguium in adult patients in which MCC is T. interdigitale
        - Tinea pedis variants with MC associated organisms
            - Mocassin: T. rubrum > E. floccosum
            - Interdigital: T. rubrum > T. interdigitale
            - Bullous: **T. interdigitale**
        - Onychomycosis (T. unguium)
            - Distal lateral subungual (MCC): T. rubrum
            - Proximal white subungual (HIV pts.): T. rubrum (refer to section 5.2)
            - White superficial: T. mentogrophytes or T. rubrum (kids/HIV pts)
            - Other species include C. albicans (massive hyperkeratosis of nails w/ chronic paronychia), Fusarium and Scopulariopsis (p/w large quantities of cheesy debri; Dx: grows in chains: "cops put people in chains"), Scytalidium (black onychomycosis; also T. rubrum)
- Tinea capitis: **T. tonsurans is MC** ("Tinea on top of your head"), M. canis #2 (pt has pets)
    - Endothrix (**black dot**; arthroconidia within hair shaft so it never fluoresces)
        - T. schoenleinii, T. soudanense, T. gourvilli, T. violaceum, T. yaounde, T. tonsurans (These have O in the 1st syllable and are 'Twithin' the hairshaft)
    - Ectothrix (KOH shows arthroconidia on the outside of the hair shaft)
        - Flourescent: fluoresce greenish-yellow under Woods lamp from **pteridine**
            - M. canis, M. audouinii, M. distortum, M. ferrugineum, M. gypseum, T. schoenleinii (Just know that all M's fluoresce except M. nanum and that T. schoenleinii is both endothrix and fluorescent ectothrix)
        - Nonflourescent: M. nanum, T. gypseum, T. rubrum, T. mentagrophytes, T. verrucosum, T. megninii ("Just remember that non-fluorescent ectothrix are the T's (except M. nanum) that don't have an 'o' in the first syllable")
    - Favus: yellow crusting on the scalp w/ hair loss; a/w **T. schoenleinii** > M. gypseum, T. violaceum ("My Fav show is a gypsy violinist who has yellow hair")
        - Hair shafts infected with T. schoenleinii demonstrates air-filled spaces
    - Kerion: inflamed tender pus-filled area on scalp a/w T. mentagrophytes, M. canis, T. verrucosum, T. tonsurans → "Caring (kerion) Men Can be Very Tolerant"

- Tinea barbae: same species as seen in kerion ("Bearded Men Can be Very Tolerant")
- Tinea faceii: T. mentagrophytes, T. rubrum, T. tonsurans, M. canis
- Tinea imbricatum: T. concentricum ("fascinating concentric polycyclic rings of scales")
- Geography: zoophilic (primarily affect animals; massive inflammation), geophilic (found in soil) and anthropophilic (restricted to humans; mild inflammation)
    - Zoophilic: T. Mentagrophytes (rodents esp Mice), M. canis (cats), T. verrucosum (cows; "verru-cow-sum"), T. nanum (pigs) → "Men Can be Very Nasty animals"
    - Geophilic: M. Gypseum
    - Anthropophilic: all other species
- **p/w annular, erythematous scaling plaques w/ active border and central clearing**
    - Majocchi granuloma: fluctuant red dermal nodules w/ hair follicle invasion
    - Tinea incognito: patches w/ less erythema and more subtle scale
    - Tinea capitus: numerous scaly patches w/ broken hairs ("black dots") and LAD
        - M. canis is more inflammatory w/ papules and pustules
        - Kerion: severely inflamed w/ boggy, suppurating and crusted plaques
    - Tinea cruris: pruritic red scaly annular rash on inner thighs that spares scrotum
        - Ddx: candida burns more than itches, is redder and involves scrotum
    - Tinea manuum: non-inflammatory and often unilateral ("2 feet and 1 hand syndrome") w/ **scaly accentuation of palmar creases** +/- onychomycosis
        - Variants: interdigital, hyperkeratotic, vesicular, exfoliative, papular
    - Tinea pedis: presentation varies but usually p/w interdigital maceration (esp. lateral toe webs), red-scaly moccasin-like plaque on the sole (T. rubrum) or bullous lesions (T. interdigitale formerly T. mentogrophytes var interdigitale)
        - T. pedis commonly results in an id reaction that presents differently than normal → **p/w vesicles on lateral fingers (mimics dyshidrotic dermatitis)**
    - Tinea unguium
        - Distal lateral subungual (T. rubrum): p/w thick, yellow nails
        - Proximal subungual: p/w **beige-white opacity under proximal nail plate**
        - Superficial: entire nail is white (T. mentogrophytes) > black (T. rubrum)
- Dx: biopsy (horizontal septate hyphae in stratum corneum), KOH or culture (Wood's lamp is of no help in tinea skin infections but it will fluoresce in ectothrix T. capitis)
    - KOH: scrape scale onto a slide, apply 10% KOH and then heat up
        - Dimethyl sulfoxide can be added which does not require heating
        - Staining with chlorazol/black E or Parker blue-black can identify tinea
    - Dermatophyte Test Media (DTM) and Sabouraud media: culture is (+) if after 3 weeks a white fluffy colony grows AND the media has turned from yellow to red
    - Trichophyton produces elongated, cigar-shaped macroconidia w/ parallel sides
    - Microsporum species have **spindled** or boat/football shaped macroconidia
        - M. nanum is unique in being round
    - Scopulariopsis brevis grows in chains (must culture without cycloheximide)
    - For onychomycosis, nail clipping w/ PAS stain or PCR testing can be performed
- Rx: **topical or systemic** (better for hairy sites) **terbinafine** > azoles or naftin

    - Terbinafine (250mg) is now the preferred oral agent due to ↑ resistance and prolonged treatment course w/ griseofulvin **(20 mg/kg/day; take w/ fatty meal)**
        - Tinea skin infection: 2 weeks w/ terbinafine or 4 weeks w/ griseofulvin
        - T. capitis: 6 weeks w/ terbinafine and 8 weeks w/ griseofulvin
            - 2/4/6/8 who do we appreciate, terbinafine & griseo for skin & scalp
        - T. pedis: is notoriously refractile to therapy (esp if nails involved)
            - Terbinafine for 4 weeks for tinea pedis, 6 weeks for onychomycosis of the fingernails and 12 weeks for onychomycosis of the toenails
- Tinea infections are the 2nd MC sport-related infection (s. aureus/bacteria is #1)
    - T. capitus: athletes must be treated for 2 weeks before returning to competition, but students can return to school as soon as they start therapy
    - T. corporis: athletes must be treated for 3 days and lesions must be covered

Tinea versicolor (pityriasis versicolor)
- Species: Malassezia globosa and M. furfur
    - Unicellular yeast (skin flora) turns to multicellular mold (hyphae) in disease state
- Pathogenesis: overgrowth of normal flora; hypopigmentation due to tyrosinase inhibition by azelaic acid (byproduct of Malassezia)
- p/w hyper or hypo-pigmented scaly macules in seborrheic areas (trunk/upper arms)
    - In skin of color, the hypopigmented spots are more obvious and it can present in an inverse pattern (**face**, knees, elbows, groin, armpits, hands and feet)
- Diagnosis: biopsy, KOH, culture (requires **olive oil** for growth; shows '**ziti and meatballs**')
- Rx: topical or systemic antifungals, selenium sulfide shampoo, topical ciclopirox

Piedra
- Species
    - Black Piedra: Piedra hortae ("honorable black people")
    - White Piedra: Trichosporon asahii can cause disseminated disease in the immunocompromised ("Tricky Ass white people are very opportunistic"), T. inkum, T. ovoides, T. cutaneum ("Incredibly Overprivileged Cute white people")
- Microscopy: soft mobile nodules (white) or hard non-mobile nodules (black) along hair ("white people are soft and easy to move while black people are hard and immovable")
    - Ddx for nodes on scalp hair: hair casts (freely mobile), lice/pediculosis capitis (nits usually live close to scalp), trichorrhexis nodosa, monilethrix
        - Trichomycosis axillaris: presents in the axilla (not scalp) and p/w broad Tan concretions that encircle the Total length of hair shaft (vs nodes)
- p/w asymptomatic hair breakage on the **scalp** (MC)
- Dx: trichoscopy (multiple oval nodules), KOH, culture
- Rx: hair shaving/cutting or anti-fungals

Tinea Nigra
- Species: Hortaea werneckii
- p/w expanding homogenous brown macule/patch on the palms/soles

    - Ddx: **acral lentiginous melanoma** is multicolored and **talon noir** (calcaneal petechiae) is a homogenous dark brown plantar macule that arises after exercise
- Microscopy/pathology: dark brown septate hyphae
- Rx: topical azoles, Whitfield ointment (keratolytic w/ benzoic and salicylic acid)

Subcutaneous mycoses

- SCLE: Sporotrichosis, Chromoblastomycosis, Lobomycosis, Eumycetoma
    - All are due to traumatic inoculation in tropical areas (i.e. South America)
    - Organisms can be highlighted w/ PAS or GMS stain on path
    - All should be treated with **itraconazole** except lobomycosis (Rx: excision)

Sporotrichosis
- Species: Sporothrix Schenckii (dimorphic fungus)
- Geography: Saprophyte (fungus found on dying organic matter); MC in South America
- Pathogenesis: traumatic inoculation from Soil, wood Splinters or Sphagnum moss
- p/w multiple ascending SQ abscess/nodules (Sporotrichoid Spread)
- Path: **organisms usually absent** but when present are cigar ("Sigar")-Shaped yeast
- Dx: culture on Sabouraud agar (shows black and white rings of mold/hyphal colonies)
- Rx: SSKI or itraconazole

Chromoblastomycosis: "Pedro, Phil and Claud collect Copper pennies"
- Species: **Fonsecaea pedrosoi** > Phialophora, Exophiala, Cladophialophora, Rhinocladiella
- Pathology: PEH + puss differential w/ Copper pennies (pigmented medlar bodies)
- p/w pruritic papules expanding into **verrucous plaques w/ black dots + Central scarring**
- Chronic lesions can progress into SCC

Lobomycosis
- Species: Lacazia (loboa loboi) infects freshwater **dolphins** in South American rivers
    - "Lobo means wolf in Spanish and dolphins are the wolves of the water"
- p/w **keloid-like nodules** ("Wolves can scar you up")
- Pathology: **yeast in chains which look like "pop-beads"** ("Wolves hunt in packs/chains")
- Rx: surgical excision of keloids ("If you get attacked by a wolf you will need surgery")

Eumycetoma
- Species: **Pseudallescheria boydii** > Madurella, Exophialia jeanselmei, Acremonium
- p/w sinus tracks w/ draining grains MC on the feet (or any site of traumatic inoculation)
- Grains are black or white: "You (Eu-mycetoma) look at everything in black or white"
    - Black grain eumycetoma organisms: "My Lousy Greasy Jeans are black" (Madurella mycetomatis, Leptosphaeria, M. grisea, Exophiala jeanselmei)
    - White (non-pigmented) grain organisms: P. boydi is MC ("white boy")
- Rx: surgical debridement and several months of azole antifungals

Actinomycetoma (this is a bacteria but it is listed here to compare to eumycetoma)

- Mycetoma caused by bacteria such as Nocardia brasiliensis (white grains: "white note cards") > Actinomadura pelletieri (red grains), A. madurae (cream-pink grains), Streptomyces Somaliensis (yellow-brown grains; "Yellow like the Simpsons")
    - These colors (besides white) are specific to actinomycetoma
- Same presentation as eumycetoma (white grains are MC in both; know the other colors)
- Rx: sulfonamides (TMP-SMX)

Systemic dimorphic mycoses

- Dimorphic mycoses: "multicellular mold in cold (in culture), unicellular yeast in beast"
    - Sporotrichosis is also a dimorphic fungus but it is SQ, not systemic
- Consists of blastomycoses, histoplasmosis, coccidioidomycoses, paracoccidioidomycoses
- Pathogenesis: **inhalation (lungs)** and **hematogenous spread** to organs including skin
    - Only in blastomycoses it is common (> 75% of cases) to see spread to the skin
    - Only in paracoccidioidomycosis it is common (70%) to see spread to the mouth
- **Ddx: deep fungal infections are exophytic whereas cutaneous leishmaniasis p/w ulcers**
- Rx: itraconazole or amphotericin B depending on severity
    - "Blast his cocc and paracocc away with itraconazole or amphotericin"

Blastomycosis ("North American blastomycoses")

- Species: Blastomyces dermatitidis
- Geography: Eastern US (i.e. **Mississippi River valleys, Great Lakes, Ohio**)
- Pathogenesis: inhalation and hematogenous spread to skin and Bones
- p/w Bumpy ("heaped up") verrucous nodules w/ Black crust and sharp/irregular Borders
- **Skin disease is MC extrapulmonary manifestation** (Lung is MC organ affected)
- Culture (mold in cold): filamentous moldlike structures w/ short conidia ("**Lollipops**")
    - We also saw "lollipops" on path of nephrogenic systemic fibrosis
- Path (yeast in beast): PEH + puss ddx w/ Broad-Based Budding yeasts ("Snowmen")
    - If no budding is present, yeast can still be recognized by its thick refractile wall

Histoplasmosis

- Species: Histoplasma capsulatum var. capsulatum (var. duboisii in Africa)
- Geography similar to Blastomycosis (Ohio and Mississippi River valley)
- Pathogenesis: inhalation of bird and bat feces ("Histo Hides in bat caves")
- Skin lesions are more common in HIV and p/w umbilicated molluscoid pox-like papules
    - "HIV Can Cause Pox-like lesions": Histo, Coccidio, Crypto, Penicillium
- Culture: mold w/ tuberculate macroconidia and smooth-walled spherical microconidia
- Path (yeast): parasitized histiocytes w/ 2-4 µm yeasts evenly distributed in the histiocyte
    - Ddx: leishmaniasis (parasites rim the border of the histiocyte)

Coccidioidomycosis

- Species: Coccidioides immitis (US) or C. posadasii (South/Central America)

- Geography: desert Southwest US (Central Valley/San Joaquin Valley, California), Mexico
- Pathogenesis: inhalation → spreads to CNS, bones and sometimes skin
- p/w blastomycosis-like verrucous nodules +/- **erythema nodosum (good prognosis)**
    - In HIV, papules may be umbilicated ("molluscoid")
- Pathology: PEH + pus differential w/ large (100 μm) spherules containing endospores

Paracoccidioidomycosis ("Mexican/South American blastomycoses")
- Species: Paracoccidioides brasiliensis
- Geography: Mexico/South America > Southern US
- p/w verrucous ulcerative lesions in and around the Mouth w/ Massive LAD in **Men**
- Path: yeast w/ the characteristic "Mariner's wheel" configuration may be found

Opportunistic systemic mycoses

- CAP: Candidiasis, Cryptococcus, Aspergillosis/fusarium, Penicilliosis, Pheohyphomycosis
- Rx: antifungals (also excision of pheohyphomycosis/phaeohyphomycotic cysts)

Candidiasis
- Species: **C. albicans > C. tropicalis** (systemic), C. parapsilosis (chronic paronychia), C. glabrata and C. krusei (fluconazole resistance), C. dubliniensis (oropharyngeal in HIV)
- Pathogenesis: normal skin/GI/GU flora w/ pathologic state in the immunosuppressed and in microbiome imbalance → chitin, mannoprotein and glucan allow for adhesion whereas secretory aspartyl proteinases and phospholipases allow for tissue invasion
- Clinical presentations: "I MET AN Opportunistic Candidiasis"
    - Intertrigo: beefy red color w/ satellite lesions
    - Median rhomboid glossitis: smooth (loss of papillae) red patch on the tongue
    - Erosio interdigitalis blastomycetica: red plaque in the 3rd web space of fingers
    - Thrush: non-adherent "cottage cheese-like" lesions in the mouth
    - Angular cheilitis: erythematous plaques at the oral commissures
    - Neutropenic disseminated candidiasis: recalcitrant candidiasis in neutropenic pts w/ scattered ecthyma gangrenosum-like nodules and internal organ involvement
    - Neonatal candidiasis: monomorphic paps in diaper area +/- thrush at 2 weeks
    - Onychomycosis: thickened discolored nail plates
    - Congenital candidiasis: widespread eruption of monomorphic papules at birth
    - Chronic paronychia: inflammation of the nail fold lasting > 6 weeks
- Risk factors: diabetes and immunosuppression
- Microscopy: KOH shows yeast and pseudohyphae
- Rx: azoles (C. glabrata, C. krusei >> C. albicans are becoming resistant to azoles)

Cryptococcus: "Crypto is a Massive PIG"
- Species: C. neoformans (ubiquitous) and C. Gattii (tropics)
- Pathogenesis: similar to systemic dimorphic mycoses (inhalation of **pigeon** feces → primary pulmonary infection → hematogenous spread to CNS, bones and skin)

    - "PIG": more common in Pregnancy and in the Immunocompromised (HIV/AIDS); **Glucuronoxylomannan polysaccharide capsule is the virulence factor**
- p/w (Massive): Mulluscoid papules around Mouth (face, nose) +/- Meningoencephalitis
- Microscopy/path: single yeast cells w/ a thick capsule; collections of organisms look like soap bubbles ("this massive pig needs a bubble bath")
    - Stains w/ ("Massive PIG"): **Mucicarmine**, **Fontana-Masson**, PAS, India ink, GMS
- Rx: oral fluconazole (amphotericin B + flucytosine if CNS involvement)

Aspergillosis and fusarium (hyalohyphomycosis)

- Aspergillosis species: **A. fumigatus** (disseminated disease in the immunocompromised from inhalation; 'fumigate makes you think of inhalation') > A. flavus (direct inoculation)
- Fusarium species: Fusarium solani (direct inoculation in toes of **burn patients**)
    - "Solani should make you think soles of the foot/toes"
- Geography: ubiquitous in soil (saprophytic)
- 2nd MC opportunistic fungal disease (candida #1) in leukemic/solid organ transplant pts.
- Pathogenesis: hematogenous spread → vascular invasion → thrombosis and necrosis
- p/w red edematous plaques w/ necrotic centers (**sites of IV cannulas are MC locations**)
- Microscopy/path: **non-pigmented septate hyphae in tissue branching in Acute Angles**
- Rx: Azole antifungals, Amphotericin B (fusarium is difficult to treat)

Penicilliosis

- Species: Penicillium marneffei (caused by bamboo rat exposure in SE Asia)
- Pathogenesis (inhalation), presentation (pulmonary symptoms w/ molluscum-like lesions), path (parasitized histiocytes), Rx (itraconazole) are identical to histoplasmosis

Phaeohyphomycosis: "I think of Lizzo"

- Group of dematiaceous (pigmented) fungal infections
    - **Exophiala jeanselmei:** MCC of phaeohyphomycotic cyst; immunocompetent pts
        - "Lizzo always wears jeans"
        - E. jeanselmei also causes black grain eumycetoma/chromoblastomycosis
    - **Bipolaris** spicifera: MCC of invasive phaeohyphomycosis; immunocompromised
        - "Lizzo is bipolar because she yells at her staff for being overweight"
- p/w subcutaneous inflammatory abscess/cysts that may drain (may mimic Baker cyst)
- Path: **pigmented hyphae** in tissue w/ bubbly cytoplasm ('Lizzo is pigmented and bubbly')
- Rx: excision of cyst and itraconazole

Uncommon fungal, algae and protozoal pathogens

Zygomycosis (Rhinocerebral mucormycosis)

- Species: Rhizopus (order Mucorales)
- Geography: ubiquitous in soil and decaying vegetation
- Pathogenesis: enters via Respiratory tract → vascular invasion → thrombosis/necrosis

- A known virulence factor is its ability to sequester iron from host so **iron ("Rhino") overload syndromes** (pt w/ multiple blood transfusions) is a **major RF for infection**
- MC subtype is Rhinocerebral (a/w diabetics w/ DKA) but primary cutaneous (from surgery, burns), GI, pulmonary and disseminated types also exist
  - All forms are Rapidly progressive and commonly fatal
- p/w **necrotic eschars on face** (nose, mouth) w/ **periorbital cellulitis** and **epistaxis**
- Path: broad Ribbon-like non-septate hyphae which branch at Right angles
- Rx: **Resection** of necrotic tissue, **amphotericin B (1st line), isavuconazole** (2nd line)

Protothecosis: "Protothecosis reminds me of hyperthecosis in a WOMAN"

- Due to contact w/ **Water** that is contaminated w/ the algae Prototheca Wickerhamii
- p/w **Olecranon bursitis**
- Path: Morula (soccer ball-like) vs non-murulating form (looks like an eyeball)
  - "O's in protothecosis remind me of the round soccer balls or eyeballs on path"
- Rx: we treat this **Algae** w/ Antifungals (**Azoles** or Amphotericin B)
- Other clinical presentations include Nodules, ulcers or plaques

Rhinosporidiosis

- Species: Rhinosporidium seeberi (a protozoa)
- Geography: India, Sri Lanka
- Pathogenesis: like protothecosis it is caused by contact w/ contaminated water
- p/w slow growing red-purple lobulated nasal polyps in young men
- Path: huge (300 µm; 3x the size of coccidioides) sporangia w/ many endospores
- Rx: excision

6.1 KERATINOCYTIC NEOPLASMS

Benign keratinocytic neoplasms

Seborrheic keratosis (SK)

- Very common benign neoplasm w/ AD inheritance but incomplete penetrance
- PIK3CA and FGFR3 activating mutations ("PIK 3 Full Grown FReckles")
  - Do not confuse w/ FGFR2 in Apert, Beare-Stevens, Crouzon, nevus comedonicus
- p/w well-demarcated waxy-brown "Stuck on" papules on hair-bearing sites
  - Hyperpigmentation due to endothelin-1 (primary keratinocyte melanogen)
- Dermoscopy: milia-like cysts, fissures/ridges and gyri (cerebriform surface)
- Path: acanthosis, hyperkeratosis (loose lamellar "Shredded wheat" keratin) w/ pseudo-horn cysts and a completely flat base ("String Sign") composed of bland keratinocytes
- Sign of Leser-Trelat: eruptive SKs due to underlying adenocarcinoma (GI is MC)

Seborrheic keratosis clinical variants ("Sebastian, would you Ker to do LSD in CALI In MARCH?")

- Lichenoid keratosis: inflamed/regressing lentigo or SK that clinically mimics BCC or SCCIS
  - Path: resembles LP (also known as lichen planus-like keratosis)
- Stucco Keratosis: white scaly variant of SK found on B/L lower legs of adults (M > F)
  - a/w HPV-23b ('S' backwards looks like a 2 and a 'K' backwards looks like a 3)
  - Path: "church Spire" pattern (hyperkeratosis and papillomatosis)
- Dermatosis papulose nigra: hyperpigmented keratosis papules on face of AA patients
  - Path: identical to SKs except lacks pseudo-horn cysts
- Clear cell acanthoma: solitary red papule on the lower **legs**
  - Dermoscopy: Coiled vessels (like in SCCIS) but in a string of pearls configuration
  - Path: resembles psoriasis (w/ sharply demarcated zone of pale keratinocytes)
    - **Phosphorylase deficiency → glycogen accumulation →** PAS+ pale cells
- Acrokeratosis verruciformis of Hopf: tan warty papules on the dorsal hands/feet (Acral)
  - a/w ATP2A2 mutations (thus seen commonly in Dariers disease)
  - Path: identical to stucco keratosis ("church-spire")
- Large cell acanthoma: represents a flat SK w/ large cells on path
  - Path: SK w/ large and slightly atypical keratinocytes
- Inverted follicular keratosis: endophytic SK variant that p/w a pink papule on the face
  - Path: endophytic SK with many **squamous eddies**

Seborrheic keratosis histologic variants (Sebastian, would you Ker to do LSD in CALI In MARCH)

- Inflamed: lymphocytes and spongiosis are present in the epidermis
- Irritated: whorls of pink keratinocytes (squamous eddies)
- Melanoacanthoma: heavily pigmented (mostly in dendritic melanocytes > keratinocytes) acanthoma w/ parakeratosis instead of the typical loose lamellar keratin
- Acanthotic: MC and most typical (acanthosis, horn cysts and lamellar hyperkeratosis)
- Reticulated: thin pigmented interlacing downward proliferation of epidermis; horn cysts

- **Clonal**: clonal islands of small keratinocytes with bland nuclei (vs clonal Bowens) and no ducts in the clones (vs hydroacanthoma simplex type of poroma)
- Hyperkeratotic: prominent hyperkeratosis and papillomatosis ("church-spire")

Porokeratosis ("BDSM Porn")

- All lesions on path display a cornoid lamella (angled column of parakeratosis overlying dyskeratotic cells) which correspond to a **slightly raised peripheral rim of scale clinically**
  - Blaschkoid/linear porokeratosis: linear lesions MC on the legs in infancy
    - Highest risk of SCC (since develops in kids cancers have a lifetime to form)
  - Disseminated Superficial Actinic porokeratosis: numerous brown-red scaly macules on sun exposed areas (MC on the legs) in middle-aged adults
    - 2nd lowest risk of SCC (punctate form has no risk)
    - Mutation in **SART3** gene
    - a/w SLE
  - Porokeratosis of Mibelli: classic large circinate plaque MC on the legs of children
    - "Porokeratosis (and angiokeratoma) of Mibelli is far from the belly"
  - Punctate porokeratosis: 1-2 mm "seed-like" papules on the palms and soles of adolescence; no risk of SCC
    - Ddx: punctate keratosis of palmar creases
  - Porokeratosis Palmaris Plantaris et disseminate (PPPD): seed-like keratotic papules start on the palms/soles but then disseminate; MC in adolescence
    - Ddx: arsenic keratosis but look for the raised rim of scale in porokeratosis and **age of pt is a good clue** (arsenic keratosis develops 20-30 years after exposure whereas PPPD presents in the first 20 years of life)
  - Porokeratosis Ptychotropica: scaly verrucous gluteal/peri-anal lesions in adults
    - Path: cornoid lamella are multiple w/ surrounding orthohyperkeratosis
  - Porokeratotic eccrine ostial and dermal duct nevus (PEODD): type of epidermal nevus w/ **keratotic**/comedo-like papules/plaques (MC on an extremity) of infants
    - Represents a **mosaic form of KID syndrome** (GJB2 encoding connexin 26)
      - "The KID PEED (PEODD) in the sandbox"
    - **Path: cornoid lamella centered exclusively over acrosyringia**
- Dermoscopy: diffuse vessels as dots and well-defined peripheral thickened rim
- Rx: cholesterol/lovastatin cream; monitor for malignant transformation (rare)

Epidermal nevus

- Hamartoma of the epidermis or papillary dermis which occurs in the first year of life
- Same activating mutations seen in SKs (**PIK3CA and FGFR3**); also **HRAS** > KRAS, NRAS
- **p/w hyperkeratotic papillomatous linear plaques in Blaschko's lines**
- Variants
  - Nevus comedonicus: linear comedones on face (MC); path shows horn cysts
  - Ichthyosis hystrix: hyperkeratotic spiny scales mainly over extensor limbs w/ PPK
  - Nevus Sebaceous: in infants it usually presents as a **yellow-tan plaque on the scalp** (subtle, may just look like alopecia) which gets thick and bumpy at puberty

      - a/w **HRAS** > KRAS mutations and CNS abnormalities (seizures, MR)
  - Inflammatory linear verrucous epidermal nevus (ILVEN): p/w itchy, red, warty plaques in in Blaschkoid pattern that presents at birth or shortly after
      - Ddx: red/itchy (vs epidermal nevus), does not respond to topical steroids (vs linear psoriasis) and does not regress (vs lichen striatus)
  - Epidermal nevus syndrome (Shimmelpenning): a/w CNS, eye, MSK abnormalities
      - a/w vitamin D-resistant rickets and postzygotic mosaic HRAS mutations
- Path: "church-spire" (epidermal papillomatosis and orthohyperkeratosis)
  - May see epidermolytic hyperkeratosis (EHK) due to mosaicism (**offspring have a risk of full blown bullous congenital ichthyosiform erythroderma**)
- Workup: eye and developmental assessment; biopsy after childhood to rule out EHK

Flegel disease (hyperkeratosis lenticularis perstans)

- AD inheritance with adult-onset hyperkeratotic papules on the distal extremities
- Path: thick orthohyperkeratosis overlying thin epidermis; lichenoid dermal infiltrate
- Electron microscopy: absent/altered Odland bodies (lamellar granules)

Warty dyskeratoma

- p/w solitary verrucous papule w/ central keratotic plug on the head/neck in adults
- Path: acantholytic dyskeratosis (corp ronds/grains) in a cup-like configuration

Borderline or malignant keratinocytic neoplasms

Actinic keratosis (AK)

- RFs: Caucasian, immune-Compromise, old males (more Time in sun), UVR/Tanning (#1)
- Pathogenesis: **UVB** damages DNA by formation of photoproducts such as **Cyclobutene pyrimidine dimers (MC)** which forms Thymine dimers (covalent bonds between two thymines) which are mutagenic (**C → T is MC mutation**; "aC→Tinic keratosis")
- Mutations in the tumor suppressor gene **p53** are highly a/w actinic keratosis and SCCs
  - "p53 puts a freeze on the cell cycle"
- p/w red scaly plaques on sun-exposed sites
- Dermoscopy: strawberry pattern (red pseudo-network around follicular openings)
- Path: keratinocyte atypia in the lower 1/3rd of epidermis w/ overlying parakeratosis (pink) which alternates w/ orthokeratosis (blue) → "Flag sign"
- Rx: Cryo, ED&C, CO_2 ablation, 5-FU, imiquimod, PDT, diclofenac, Tirbanibulin, TCA peel
- Prophylaxis: oral niacinamide/nicotinamide (vitamin B-Three), Tretinoin (retinoids)
- Erosive pustular dermatosis occurs on the scalp of elderly patients and presents w/ sterile pustules and chronic crusted erosions on a severely actinically damaged scalp
  - Rx: potent topical corticosteroids

Bowen's disease (Squamous cell carcinoma in situ)

- Risk factors: similar to AK but also include HPV and arsenic exposure
- Pathogenesis: AK, Bowen's and SCC have the same pathogenesis

- p/w hyperkeratotic erythematous plaque in sun-exposed site (but may occur anywhere)
- Dermoscopy: glomerular vessels (most specific) or vessels as red dots
- Path: full-thickness keratinocyte atypia w/ a disorganized "wind-blown" architecture
- Variants
    - Bowenoid papulosis: Brown penile papules (a/w HPV); low risk of invasive SCC
    - Erythroplasia of Queyrat: red erosive plaques on glans penis: high risk of invasion
- Rx: ED&C, excision or MOHs (consider AK treatments in non-surgical candidates)
- Prognosis: 3-5% Bowen's disease progress to invasive SCC

Squamous cell carcinoma ("ABC's of SCCs")

- Risk factors: similar to AK and Bowen's but also include
    - Genetic syndromes: helicase family mutations (XP, Werner, Bloom, Rothmund-Thompson), albinism, DEB, EDV, dyskeratosis congenita, KID syndrome)
    - Chronic inflammation/ulcers (**MC RF in AAs**): leg ulcers, hypertrophic LE/LP
    - Drugs (ABC): Azoles (voriconazole), Azathioprine, Biologics (TNF inhibitors), BRAF inhibitors (vemurafenib), Corticosteroids, Calcineurin inhibitors, Cyclosporine
    - Chemical exposures (ABC): **Arsenic**, Bromide, Cigarettes (tobacco)
- Mutations: p53 > CDKN2A, NOTCH1/2, deregulation of EGFR pathway
- MC malignant tumor of **nail** unit (a/w HPV 16, 31 and 73; melanoma is #2)
    - BCC and aggressive digital papillary adenocarcinoma are less common
- MC skin cancer in AAs, Asian Indians and in organ transplant recipients
    - Transplant recipients: **SCC (65x risk)**, Merkel (25x), BCC (10x), melanoma (3.4x)
        - SCCs are more aggressive in transplant pts (vs BCCs → usually superficial)
    - Treatment of multiple SCC's in organ transplant recipients: add **oral acitretin** (most effective); change CNI to the mTOR inhibitor sirolimus (2nd most effective)
- p/w red scaly papulonodule MC on the head/neck or dorsal extremities
- Dermoscopy: same vessels as in Bowens (glomerular or dots) which may be in a linear distribution, brown circles in a linear distribution (pigmented SCC), scale, tan-orange structures (keratin), white scale, blood spots and crystalline (white reflective) structures
- Path: dermal invasion of atypical keratinocytes which are often more differentiated (more eosinophilic/keratinizing) than surrounding keratinocytes
    - Variants: acantholytic, spindled, pseudoglandular, KA and verrucous carcinoma
- IHC: **AE1/AE3** cytokeratin stains, p63, MN116, S100 can highlight PNI of SCC
- Staging systems
    - Brigham and Women's Hospital: the 4 risk factors are the 4 D's (Diameter **≥ 2 cm**, perineural invasion in a nerve ≥ 0.1 mm in Diameter, poor Differentiation, Deep invasion (beyond fat but excluding bone which automatically upgrades to T3)
        - **T1 = 0 risk factors (0.1% 10-year nodal metastasis)**
        - **T2a = 1 RFs (3% 10-year risk of nodal metastasis)**
        - **T2b = 2-3 RFs** (20% 10-year risk of nodal metastasis → **get CT head/neck**)
            - "T2B, 2 to 3 (RFs), twenty, get CT" (it all rhymes)
        - **T3 = 4 RFs or any bony invasion (67% 10-year risk of nodal metastasis)**

- AJCC8: stage 0 (in situ), stage 1 (T1), stage 2 (T2), stage 3 (T3 even w/o nodal disease or N_{any}), stage 4 (T4 w/ N_{any}, T3 w/ ≥ 2 nodes of ≥ 3 cm, or M_{any})
    - T1: ≤ 2 cm in greatest diameter
    - T2: 2.01 cm to 4 cm in greatest diameter
    - T3 (ABCD): tumor invasion After Adipose (or > 6 mm deep), minor Bone erosion, PNI of ≥ 0.1mm Caliber (or subdermal) nerve, Diameter > 4 cm
    - T4: extensive bone invasion (T4a) or invasion of the base of the cranium or invasion through the skull foramen of the base of the cranium (T4b)
        - "4A is A-xtensive bone invasion, 4B is to Base or through 4men"
    - BWH is a better predictor of SCC metastasis than AJCC8 (perhaps because **AJCC8 does not include poor differentiation** as a risk factor for staging)
    - Hint: AJCC8 staging always has the "point-zero" round down (i.e. 2.0 cm SCC is AJCC8 T1; for BWH however, a 2.0 counts as a risk factor)
- **NCCN management guidelines (HIGH YIELD)**
    - ED&C should only be used to treat local low risk SCCs and w/ 3 caveats…
        - Do not use in areas w/ terminal hairs (scalp, axilla, pubis, beard in males)
        - If SQ later is reached during ED&C then excision must be performed next
        - If high risk path features are identified then perform additional treatment
    - WLE: usually 0.4 cm margins; 0.6 – 1 cm margins if SCC is high risk (see below)
        - Positive margins → consider MOHs or re-excise
        - Negative margins but extensive PNI (> 0.1 mm) → consider adjuvant RT
    - MOHs: most effective treatment option for high-risk local disease (see below)
        - Positive margins or extensive PNI → consider adjuvant RT
    - RT: pts w/ unresectable, inoperable or incompletely resected disease (combine w/ cisplatin) or combine w/ PD-1 for pts w/ regional recurrence or distant mets
        - **CI: genetic syndromes like Gorlin's or XP;** (relative CI w/ scleroderma)
    - Guidelines for SLNB for SCC is not established, but publications (Tejera et al. Derm surg 2020) endorse SLNB for AJCC8 T3 and BWH T2b and greater SCCs
    - Bone invasion is suspected → order CT w/ contrast ("B next to C in alphabet")
    - Perineural invasion is suspected → order MRI w/ contrast ("P next to M")
    - Local recurrence is suspected → biopsy (MRI can assess extent of local disease)
    - PET-CT is best for assessing metastases
    - Pembrolizumab and cemiplimab are TOC for locally advanced/metastatic disease
        - 2^{nd} line is chemo (cisplastin) or cetuximab (EGFR inh that is FDA approved for mucosal laSCC of head/neck but used off label for cutaneous SCC)
    - LN dissection for regional disease (ipsilateral dissection +/- RT for solitary or multiple ipsilateral LNs; B/L neck dissection + RT for B/L nodal involvement)
    - **Stepwise process of evaluating aggressive SCCs of ≥ T2 ("ABC")**
        - **Assess LNs** (clinically or with imaging) → if nodal disease then…
        - **Biopsy (US guided FNA** or **core needle**) involved LNs, and if positive get…
        - **CT scan w/ IV Contrast of involved nodal basin** and get a **PET-CT** or **CT Chest/abdomen/pelvis** to rule out distant metastases
    - Ppx ("I'm a FAN of SCC prophylaxis"): 5-FU cream, Acitretin, Nicotinamide 500mg
- Follow-up

- Local disease: at least annually but more frequent if pt/lesion are high risk
    - Node negative: 3-12 months x 2 years and 6-12 months x 3 years
    - Node negative but high risk: 3-6 months x 2 yrs then 6-12 months for life
  - Regional disease: q 2-3 months x 1 year; then q 2-4 months x 1 year; then q 4-6 months x 3 year; then q 6-12 months for life
- ↑ risk of mets/death (ABCDE): Breslow depth > 2mm, arising in Burns/scars, immune-Compromised, **Diameter > 2cm** (#1 risk), poorly Differentiated/Desmoplastic, Ear/lip
- Management for nail SCC: currently no guidelines exist for proper use of imaging but since many nail SCCs involve the bone which may change management (from MOHs to distal digital amputation), imaging may be warranted (CT w/ contrast > X-ray)
- In a recent JAAD article, **hydrochlorothiazide** was significantly a/w NMSC, especially SCC

VerrucouS carcinoma
- Low grade but locally destructive SCC a/w low risk HPV types (6 and 11)
- Large verrucous nodule w/ three clinical variants: epithelioma caniculatum (plantar foot), Buschke-Lowenstein tumor (anogenital) and oral florid papillomatosis (mouth)
  - MC location is in **mouth** and also a/w chewing tobacco ("Snuff dipper's cancer")
- **Risk factors: HPV 6 (Six) > 11, Sexual activity, history of STDs, Smoking, immuno-Suppression, male gender (boys have more Sex), age > 60 (Sixty)**
- Path: cytologically bland (no atypia) but is massive with a bulbous/pushing border
- Rx: **Surgical excision** (also diagnostic because must see architecture); **do NOT irradiate!!**

Keratoacanthoma
- Unique variant of SCC w/ initial rapid growth (weeks) → resolves/involutes (months)
  - Subungual KAs never involute
- RFs: UV, immunosuppression, HPV, drugs (vemurafenib, vismodegib, sorafenib)
- Genetic syndromes with KAs
  - Ferguson-Smith: AD mutation in **TGFBR1** ("Fergie parties, TGIF") which p/w rapid-onset of multiple KAs in adolescents → resolves → may reappear later
  - Gryzbowski: sporadic inheritance w/ numerous milia-like KAs in later adulthood which affect the airway ("Old Grizzlies Growl") and cause scarring → ectropion
    - Rx: surgery and/or oral retinoids
  - Muir-Torre ("more SICK"): Sebaceous tumors, Internal Cancers (Colon MC), KAs
- Path: crateriform nodule composed of well-differentiated keratinocytes, central keratin plug, neutrophilic micro-abscesses, eosinophils and elastic trapping peripherally
- Subungual and mucosal KAs are unlikely to regress spontaneously
- Rx: IL MTX/5-FU, excision or MOHs (may observe if it is already involuting)
  - 66/69 KAs were successfully treated with IL MTX (25 mg/mL) in one study

Basal cell carcinoma
- RFs (similar to SCC): intermittent/intense UV exposure (vs cumulative for SCC), sunburns at any age (vs sunburns in childhood for SCC), arsenic exposure (produces superficial BCCs), immunosuppression, radiation, fair skin, old age

- Mutations in the tumor suppressor genes PTCH (MC; 66%) and p53 (#2; 50% of BCCs)
    - Patched normally inhibits smoothened (sonic hedgehog molecule is in the unbound state) but when mutated, it activates GL1/2 transcription factors to promote transcription of genes involved in cell growth
- Essentially no metastatic potential (depends on stroma for growth)
- BCC variants
    - Nodular: MC variant; pearly pink papule which favors the head/neck
        - Path: large dermal nodules w/ peripheral palisade and stromal retraction
    - Superficial: MC type in younger patients; red scaly plaque on trunk or extremities
        - MC subtype in those who have received an organ transplant (in this setting even superficial BCCs should be considered high risk)
        - Path: multiple epidermal buds in the papillary dermis
    - Morpheaform: scar-like pink-white plaque
        - Path: small angulated nests in a **sclerotic stroma** w/ minimal retraction
    - Infiltrative: path term referring to an aggressive BCC with jagged angulated nests
    - Micronodular: best thought of as an intermediate tumor between nodular (not aggressive) and infiltrative (aggressive) composed of small round nests
    - Fibroepithelioma of Pinkus: pedunculated lesion MC on the lower back
        - Path: thin anastomosing pink epithelial strands w/ embedded blue basaloid buds and **eccrine ducts**
    - Infundibulocystic: resembles benign follicular tumors; not aggressive
        - SUFU mutations are commonly seen
        - Path: pink epithelial strands w/ blue buds and horn cysts
    - Adenoid: histologic variant w/ blue islands in an adenoid pattern (clear spaces)
    - Basosquamous: aggressive tumor w/ features of BCC and SCC (MC perinasal)
- Genoderms with BCCs ("Bad X-Rays Synthesize Basals"): Brooke-Spiegler, XP, Rombo, Schöpf-Schulz-Passarge, Basal cell nevus syndrome, Basex-Dupré-Christol
- Dermoscopy: arborizing vessels, leaf-like structures, large blue-gray ovoid nests
- Path: basaloid nests w/ ↑N:C ratio, peripheral palisading w/ disorganized pattern centrally, retraction between stroma and tumor, myxoid stroma, ↑ mitoses
    - IHC: **Ber-EP4, strong diffuse BCL-2 staining, AR**, AE1/AE3, CD10, MN116
- Rx: excision, MOHs, ED&C, radiation (NCCN: adjuvant therapy if PNI ≥ 1mm or invasion of nerve below dermis) or vismodegib/sondegib/cemiplimab (inoperable)
    - "The Vegetarian PAs bind and inhibit the smooth doctor (inhibit smoothened)"
        - Erivedge, Patched, cyclopamine (itraconazole also treats BCCs but not by binding to smoothened, rather it inhibits smoothened accumulation)
    - SUFU mutations are downstream of PATCH (would not respond to vismodegib)
- NCCN management guidelines
    - ED&C should only be used to treat local low risk BCCs and w/ 3 caveats…
        - Do not use in areas w/ terminal hairs (scalp, axilla, pubis, beard in males)
        - If SQ later is reached during ED&C then excision must be performed next
        - If high risk path features are identified then perform additional treatment
    - WLE: usually 0.4 cm margins; 0.6 – 1 cm margins if BCC is high risk (see below)

- MOHs: most effective treatment option for BCCs that are high risk (see below)
- Topical therapies can be considered for superficial BCC if surgery or RT is CI
  - 5-year tumor-free survival is 80% w/ imiquimod and 70% w/ 5-FU
  - Cryo (poor cosmesis) and PDT can be considered but have poorer efficacy
- Radiotherapy (RT) can be considered as a primary therapy in those w/ limited metastatic BCC (mBCC), locally advanced BCC (laBCC) that have recurred following surgery and those patients who cannot tolerate surgery
  - RT is recommended for BCCs w/ positive margins or significant PNI
  - RT as a primary treatment for BCC had higher recurrence (7.5% vs < 1%), poorer cosmetic outcomes and more complications than excision
  - Consider post-op RT for T3/T4 lesions, recurrent lesions, desmoplastic lesions in setting of immunosuppression or for close/positive margins
  - RT is CI in genetic syndromes (Gorlins) and relative CI in scleroderma
- Hedgehog inhibitors (HHI) are approved in those w/ metastatic, locally advanced and inoperable BCCs and those w/ genoderms such as Gorlin's syndrome
  - Sonidegib is only approved for locally advanced BCC (not metastatic)
- Cemiplimab is approved for mBCC or laBCC in those previously treated w/ HHIs or in whom HHI are not appropriate (cannot tolerate their SEs)
- Bone invasion is suspected → order CT w/ contrast ("B next to C in alphabet")
- Perineural invasion is suspected → order MRI w/ contrast ("P next to M")
- Local recurrence is suspected → biopsy (MRI can assess extent of local disease)
- Nodal metastasis is suspected → histologic analysis (i.e. FNA or core biopsy)
- Metastasis is suspected → CT or PET-CT can be used to gauge extent of disease
- Ppx: 20% decrease in BCCs after 1-year w/ nicotinamide (acitretin, celecoxib, acitretin, capecitabine and tazarotene are also effective prophylactic agents)

- **Less than 0.1%** of BCCs develop metastasis (MC to **lymph nodes and lungs**)
- Follow-up (NCCN): complete skin exam every 6-12 months x 5 years and then annually
- Boards factoid: 2018 JAAD article showed higher risk of BCC's in men who take statins

NCCN guidelines high risk factors of NMSC: "HIGH RISK NMSC"

- **High risk locations (H or M; esp ears/lips for SCC)**; Huge tumor (> 2 cm)
  - > 1 cm on M area is high risk (any size in H is high risk; SCC-KA on central face)
  - > 4 cm wide and > 6 mm deep or invasion beyond fat is very high risk in SCC
    - Any of these would automatically result in SCC being AJCC8 stage T3
- Immunosuppressed patient (transplant/lymphoma/iatrogenic/HIV)
- Growth is fast (this feature is specific for SCC)
- **High risk Histo**
  - BCC: infiltrative, sclerosing, morpheaform, micronodular, metatypical
  - SCC: spindle cell/desmoplastic or adenoid subtype or poorly differentiated tumor
    - Notice desmoplasia is higher risk in NMSC but lower risk in melanoma
- Radiation previously done at site of cancer; Recurrence
- Ill-defined borders; Invasion of nerves > 0.1mm, lymphatics or vessels on path
- Scars (Marjolin's ulcers; this high-risk feature is specific for SCCs)

- Key patient characteristics (genetic syndromes or immunocompromised)
- Neuro symptoms (SCC only): pain/paresthesia/anesthesia/diplopia (suggests PNI)

Recommended WLE margins for malignant neoplasms (“SLAM’D Angi Extra”)

- SCC/Sebaceous carcinoma: 0.4-1 cm (melanoma in-Situ is 0.6-1 cm)
- Leiomyosarcoma/AFX/Merkel cell carcinoma/Melanoma: 1-2 cm
- DFSP: 2-5 cm (2 cm → 40% recurrence rate; 3 cm → 15%; 5 cm → 5%)
- Angiosarcoma/Extramammary Paget’s disease: 5 cm

8.1 SURGICAL ANATOMY

Head and neck anatomy (and other high-yield dermatology locations)

- Arterial system: face receives blood from the External and Internal Carotid A. (ECA/ICA)
  - Branches of the ECA from superior to inferior are "STOMP on his face"
    - Superficial temporal artery: supplies temple, scalp and lateral forehead
    - Transverse facial artery: supplies the parotid gland and lateral face
    - Occipital artery: supplies occipital scalp and posterior neck
    - Maxillary artery ("Max has a high BMI"): branches into Buccal (supplies cheek), Mental (chin) and Infraorbital (midface/nasal dorsum) artery
    - Posterior auricular artery: supplies the scalp behind the auricle
    - Facial artery: branches into inf./sup. labial arteries (supplies lips) before becoming angular artery (arterial injection w/ filler in NL fold may occur)
      - Angular artery anastomoses w/ branches of ICA (**Dorsal Nasal A.**)
        - "DNA is a molecule with many anastomoses"
  - The branch of the ICA that supplies the midforehead/midnose is the **ophthalmic artery** which travels through the optic canal to branch into "SLIDER" → **Supraorbital, Supratrochlear,** Lacrimal, **Infratrochlear** (BUT NOT INFRAORBITAL), Dorsal nasal, External nasal, Ethmoidal (anterior and posterior), Retinal arteries
    - Fast pushes of filler if injected into a nasal or glabellar artery can cause retrograde flow of product into the retinal artery → blindness
- Venous system: whereas most veins follow their associated arteries (ex. angular vein → facial vein → internal jugular vein), many veins corresponding to arteries supplied by the ICA (ex. **supratrochlear and infratrochlear veins**) drain directly through the orbit and into the cavernous sinus (infections in this area around the nose can cause cavernous sinus thrombosis, meningitis and brain abscesses)
- Lymphatic system: generally, drains from superior to inferior and medial to lateral
  - Lymph nodes (LN) from superior to inferior ("PASS that lymph")
    - Posterior cervical LNs: receive lymph from scalp, neck and upper ear
    - Preauricular superficial parotid LNs: cheek, forehead, temple, lateral ear
    - Anterior cervical LNs: oropharynx, tongue, anterior neck, lower ear
    - Submandibular LNs: upper lip/mandibular area
    - Submental LNs: chin, lower lip
- Superficial musculoaponeurotic system (SMAS): fibrous network of lower 2/3rds of face (platysma inferior and zygomatic arch superior) that connects facial muscles to dermis
  - Allows muscle forces to be evenly distributed and contain infections superficially
  - Sensory nerves (CN V) are Superficial and motor nerves (CN 7) are deep to SMAS
- Facial sensory nerves: face supplied entirely by CN V ("LISA IZZ A Bloody Little Mess")
  - V1 (ophthalmic; LISA)
    - Lacrimal nerve: lateral eyelid, conjunctiva, lacrimal gland
    - Infratrochlear nerve: **nasal root/bridge, medial canthus**
    - Supratrochlear n: medial upper eyelid/forehead/frontal scalp/conjunctiva
    - Supraorbital n: lateral upper eyelid/forehead/frontal scalp/conjunctiva

        - Border b/w supratrochlear and supraorbital sensory innervation on forehead skin is an imaginary line drawn up at the medial iris
    - Anterior ethmoidal nerve: inferior central nose (dorsum, tip, columella)
  - V2 (maxillary; IZZ)
    - Infraorbital nerve: nasal sidewall, nasal ala, **upper lip**, medial cheek, **lower eyelid**, upper teeth and maxillary gingiva
    - Zygomaticofacial nerve: malar eminence
    - Zygomaticotemporal nerve: medial temple, supratemporal scalp
  - V3 (mandibular; A Bloody Little Mess)
    - Auriculotemporal nerve: superior anterior ear, auditory canal, tympanic membrane, lateral temple, temporoparietal scalp, TMJ
      - Also has **parasympathetic innervation to parotid**
      - Frequently injured during TMJ surgery or parotidectomy → Frey syndrome (ear or lateral temple sweats while eating spicy food)
    - Buccal nerve (Boca = mouth in Spanish): angle of mouth, buccal mucosa
    - Lingual nerve (Lengua = tongue in Spanish): anterior 2/3rds of tongue
    - Mental nerve (mentón = chin in Spanish): chin and lower lip
    - **V3 also provides motor innervation to the muscles of mastication**
      - Muscles of Mastication arise from 1st branchial arch and are innervated by V3, muscles of facial expression arise from 2nd branchial arch and are innervated by CN7
- Salivary glands: exocrine glands that produce saliva (listed from largest to smallest)
  - Parotid gland: located on the lateral face and commonly encountered during MOHs (path shows triad of **basophilic cells + eosinophilic ducts + adipocytes;** adipocytes predominate in the elderly, obese and those w/ Sjögrens or diabetes)
    - Parasympathetic innervation by auriculotemporal nerve (branch of V3)
    - "Vessels trapped in parotid gland are not FREE" (from superficial to deep)
      - Facial nerve: 2 branches enter the gland and 5 branches exit
      - Retromandibular vein: enters gland as maxillary and superficial temporal vein and exits as facial and posterior auricular vein
      - External carotid artery: exits gland superiorly as the maxillary and the superficial temporal artery
  - Submandibular gland: located deep to the body of the mandible (path shows predominance of clear foamy mucus-rich glands)
    - **Duct crossed laterally by the lingual nerve** (parasympathetic innervation)
    - Gland is supplied by the facial and lingual artery
    - Common site of calculus formation
  - Sublingual gland: located below the floor of the mouth
    - Ducts open onto the floor of the mouth
    - Like submandibular gland, it receives parasympathetic innervation from the lingual nerve (chorda tympani) and is supplied by the facial artery
  - Labial glands: circular (**size of a small pea**) glands located between submucosa and orbicularis oris muscle (labial artery also runs here but is not circular)

        - Labial glands and artery are encountered during MOHs (recognize them!)
- Cervical plexus: come from Erb's point along w/ accessory nerve ("Erb has Great LATS")
    - Great auricular nerve (C2, C3): **angle of jaw, mastoid process**, ear, lateral neck
    - Greater occipital nerve (C2): majority of occipital scalp
    - Lesser occipital nerve (C2): posterior neck, **postauricular scalp**
    - Accessory nerve (CN XI): motor innervation to SCM/trapezius → damage causes wing scapula, poor shoulder abduction, shoulder/neck pain, trapezius atrophy
    - Transverse cervical nerve (C2, C3): anterior neck ("where the Thyroid is")
    - Supraclavicular nerve (C3, C4): chest, clavicle, Shoulder ("in-between Shoulders")
- Nose sensory nerves: V2 innervates the B/L nasal dorsum and alae (V1 supplies the rest)
    - Smell is by olfactory nerve (CN1) and the nasal muscles by facial nerve (CN7)
- Ear sensory nerves: great auricular nerve (C2, C3) innervates entire ear except…
    - **Anterior helix (including crux and tragus) → auriculotemporal nerve (V3)**
    - Conchal bowl → CN7, **CN10 (vagus nerve)** ("I always bowl a 7-10 split")
    - Superior posterior ear: lesser occipital nerve (C2; also supplies scalp behind ear)
    - Posterior portion of external auditory canal: chorda typani (branch of CN7)
- Ring block around ear anesthetizes everything **except conchal bowl** +/- EAM
    - To block the great auricular nerve inject behind inferior aspect of earlobe
    - To block the auriculotemporal nerve inject superior and anterior to tragus
- Hand sensory nerves: Radial, Ulnar and Median nerves ("I drink RUM with my hands")
    - Radial nerve: first 3 and a half digits on **dorsal side** from wrist to PIP joint
        - However, they may show you dorsal side of 2nd/3rd fingertip → median n.
    - Ulnar nerve: **dorsal and ventral** 5th digit and ulnar half of 4th digit
        - Block: inject radial to flexor carpi ulnaris tendon at proximal wrist crease
    - Median nerve: first 3 and half digits on **plantar side** and **dorsal tip of 2nd/3rd digit**
        - Median is Major nerve in hand as it has Most of palm and Most of fingers
        - Block: inject radial to palmaris longis tendon at proximal wrist crease
- Foot sensory nerves (all 5 are branches of sciatic nerve except the saphenous nerve)
    - Posterior tibial nerve: heel and **sole of foot** and plantar toes ("Plantar Only")
        - Block: inject Posterior to medial malleolus
    - Sural nerve: **lateral** foot, lateral heel and lateral 5th toe
        - Block: inject posterior to lateral malleolus
    - Deep peroneal nerve: in-between 1st and 2nd toes
        - Block: lateral to extensor hallucis longus tendon at foot/ankle junction
    - Superficial peroneal nerve: dorsal toes and dorsal foot ("per-toe-neal")
        - Block: inject along dorsum of foot from malleolus to malleolus
    - Saphenous nerve (branch of the femoral nerve): medial (middle) ankle/foot/calf ("poor sap in the middleclass can't support his femily")
        - Block: inject medial (middle) and superior to medial (middle) malleolus
        - All other foot nerves are branches of the sciatic nerve → sciatic nerve block will provide complete anesthesia below the knee w/ the exception of the area innervated by the saphenous nerve (branch of femoral nerve)

- The common peroneal nerve (as it wraps around the head of the fibula) is susceptible to injury in cutaneous surgeries involving the lateral upper leg
- Sensory nerves for digits: to perform digit block inject perpendicular to the digit at Wallace's line at least 1 cm distal to the web space using 1-2 cc on each side (max volume of lidocaine that can be used is 4 mL per digit according to Napier et al.)
    - "If you use > 4 cc of anesthetic in a digit the patient will be left with only 4 digits"
- Tongue nerves
    - Sensory innervation to the anterior 2/3rds of tongue is w/ the lingual nerve of V3
    - Taste of anterior 2/3rds of tongue is w/ the chorda tympani branch of CN 7
        - "Taste the Tympani"
    - Taste and sensory innervation to posterior 1/3rd of tongue is w/ CN IX
    - Motor innervation to tongue is w/ CN 12
- Face motor innervation: **facial nerve** (CN7) emerges from skull base at the stylomastoid foramen, travels in parotid gland, splits into 5 branches ('Ten Zebras Bit Mandy's Cervix')
    - Temporal branch: supplies frontalis (eyebrow elevation; **has no bony attachments**), corrugator supercilia (pulls eyebrows inferiomedially), upper orbicularis occuli (tight closure of eyelids)
        - Injury: can't make shocked face (**can't fully open eyes; no forehead lines**)
        - Botox: frontalis (forehead lines), orbicularis oculi (crow's feet), corrugator supercilli (vertical glabellar lines)
    - Zygomatic branch: supplies alar portion of nasalis (flares nostrils), procerus (horizontal glabellar lines; shortens nose), lower portion of orbicularis occuli (**tight closure of eyelids**), zygomaticus major (elevates upper lip)
        - Injury: **can't make a mean face** ("I call it zygo-mean-icus")
        - Botox: procerus (horizontal glabellar lines), alar nasalis (flared nostrils)
        - CN III (levator palpebrae superioris) opens eyelids ("III looks like three columns keeping eyelids open) whereas CN 7 (orbicularis oculi) closes eyelids ("7 looks like a hook which shuts eyelids")
        - Extensive rami exist so injury to this nerve usually is not permanent
    - Buccal branch (boca is mouth in Spanish): supplies Buccinator muscle of mastication (along w/ V3), transverse portion of nasalis (Bunny lines), depressor septi nasi (pulls columella toward lip) and upper lip muscles (orbicularis oris, zygomaticus major/minor, risorius, levator anguli oris, levator labii superioris)
        - Injury: can't talk (muffled speech), can't kiss (pucker), can't smile (**uneven smile even at rest**), can't eat (food accumulates b/w cheek and teeth)
        - Botox: levator labii superioris (gummy smile), trans. nasalis (bunny lines)
        - Extensive rami exist so injury to this nerve usually is not permanent
        - These muscles converge and are held together by a fibrous tissue known as the modiolus (just superior and lateral to the oral commissure)
    - Marginal mandibular branch: supplies lower lip muscles (orbicularis oris, depressor anguli oris, depressor labii inferioris, mentalis, upper platysma)
        - Muscles of facial expression in the lower face have no bony attachments
        - **Most superficial nerve; highest risk of permanent motor deficits**
        - Injury: can't evert lower lip, **uneven smile visible when smiling**

- Cervical branch: supplies platysma (depresses lower jaw and tenses neck skin)
  - Injury: can't depress lower jaw to make a grimace
  - Botox: platysmal bands
- Posterior auricular branch of CN7 (not in mnemonic as it branches before entering the parotid close to the stylomastoid foramen; "Ten Zebras Bit Mandy's Cervix Auright"): supplies occipitalis muscle and muscles of the ear
- Facial nerve also provides sensory innervation to the conchal bowl ("bowl a 7, 9 or 10"), taste to ant 2/3rds of tongue, parasympathetic innervation to submandibular/sublingual glands and motor innervation to muscles of mastication (buccinator supplied on superficial side, all others on deep side)
- CN7 is prone to injury w/ retroauricular surgery in kids (immature mastoid)

- Cutaneous danger zones
  - Labial/angular artery: skin necrosis if injected near the base of ala w/ filler
  - Supratrochlear artery: skin necrosis or blindness if injected in glabella w/ filler
    - Approximated by most prominent glabellar frown line by medial eyebrow
  - Temporal branch of CN7: eyebrow ptosis if severed over zygomatic arch
  - Marginal mandibular branch of CN: facial asymmetry upon smiling if severed 2-3 cm inferolateral to oral commissure as it passes over the mandible
  - Spinal accessory nerve (CN XI): **shoulder pain** (MC presenting symptom), **inability to abduct arm** (MC presenting sign), **internal rotation of humoral head**, **winged scapula** and **trapezius atrophy** if severed at Erb's point (6.5 cm below EAM at posterior border of SCM)
  - Parotid duct: sialocele requiring surgery if injured as it courses over masseter

9.1 LASERS

Speaking the language of lasers

- Laser terminology
  - Energy (Joules): unit of work
  - Fluence (Joule/cm^2): energy per cm (↑ fluence → ↑ energy per unit area)
    - Changing the spot size does not alter fluence
  - Power (Watts = Joule/s): energy as a rate
  - Irradiance (Watts/cm^2): power delivered per cm^2
    - If given irradiance and fluence (dose) and asked how long to keep them in light box, use formula: Fluence (J/cm^2) = Irradiance (J/s cm^2) x Time (s)
      - "People who love UV light are generally more FIT"
  - Pulse width aka pulse duration (seconds): duration of laser exposure
    - ↑ pulse duration → ↑ energy or heat delivered to tissue
    - Pulse duration should be ≤ TRT to prevent collateral damage to tissue
  - Spot size (mm): diameter of the laser beam hitting skin
    - **↑Spot size** → ↓ Scatter → **↑ Span (depth) of penetration**
  - Wavelength (nm): **longer wavelengths (shorter frequency) penetrate deeper**
    - UV (10-400 nm): excimer laser (308 nm)
    - Visible (400-700): argon (488-514), KTP (532), PDL (585-600), Ruby (694)
    - Infrared (700 nm-1 mm): alexandrite (755), diode (800), Nd: YAG (1064), Erbium: Glass (1540), thulium (1927), Erbium: YAG (2940), CO_2 (10,600)
    - Radiofrequency/microwaves (> 1 mm)
  - Chromophore: absorptive target of laser (i.e. melanin, hemoglobin and water)
  - Thermal relaxation time (TRT): time for heated tissue to dissipate 50% of its heat
    - TRT (seconds) is proportional to the **square of the target's Radius**
    - Pulse duration should be ≤ TRT ("P is before/less than T or R in alphabet")
    - Since **tattoo** ink and melanosomes are the chromophores with the smallest diameters (0.1 and 0.5 micrometers respectively), they have the smallest TRTs (**10** and 250 **nanoseconds** respectively) and thus require Q-switched lasers with pulse durations in the nanoseconds
    - Standard pulse duration for PDL on **telangiectasias is 10 milliseconds**
      - "Tattoo (ns) and Telangiectasias (ms) are both Ten"
  - Photomechanical effect: sudden heating produces cavitation (Pits/Pockets)
    - **Cavitation causes vessel rupture with PDL** and skin whitening with QS
- Lasers are characterized by the 3 Cs
  - Coherence: light waves travel in phase spatially and temporally
  - Collimation: light waves travel together in a parallel fashion (minimal divergence)
  - Mono-Chromatic: light waves are all same wavelength (which determines color)
- Three different media exist
  - Gas ("You don't gas a check you cash A CHECK"): Argon, CO_2, Helium-neon, Excimer laser (xenon chloride), Copper vapor, Krypton
  - Liquid ("liquid dye"): pulse dye light (rhodamine dye)

- Solid: two classes
    - Crystal ("Krystal"): KTP, ruby, Er-YAG, Nd-YAG, alexandrite
    - Semiconductor ("solid semicon-diode"): diode
- Properties of selective photothermolysis of target tissue (light kills the Pesky Wuhan Flu)
  - **Pulse duration should be ≤ TRT** (thermal relaxation time) → minimizes the Propagation of heat and resultant "collateral damage" to tissue
  - **Wavelength** must reach appropriate depth to target the desired chromophore
  - **Fluence** must be high enough to damage target tissue but no higher
- Four types of laser waveforms
  - Continuous: light is emitted continuously; low power (i.e. CO_2, argon)
  - Quasicontinuous: light is emitted in multiple small rapid bursts; low power (i.e. copper vapor, potassium titanyl phosphate (KTP)): "When cops pat you down, they do it in a quasicontinuous way (multiple rapid low power pats)"
  - Pulsed: light is emitted periodically with short pulse durations (milliseconds) and high power (i.e. PDL, ruby, alexandrite, diode, Erbium:glass, Erbium:YAG)
    - Perhaps coincidentally, but these are the lasers from 585-2940 nm
  - Quality switched (Q-switched): variant of pulsed lasers with extremely short pulse durations (nanoseconds); extremely high power (i.e. ruby, alexandrite, Nd:YAG) → laser of choice for pigmented lesions, tattoos and drug deposits
- Treated skin will have one or more of the following four interactions: "STAR light":
  - Scattering: light waves bounce off fibers within dermis/SQ → **limits the Span of penetration** (↑ Spot size → ↓ Scatter → ↑ Span/depth of penetration)
  - Transmission: light waves Transmit Through Tissue without having any effect
  - Absorption (desired effect): light it absorbed by its target tissue
  - Reflection: **4-7% of light it reflected** by skin surface
- Epidermal damage can be minimized by cooling (three methods)
  - Precooling with cryogen (tetraflouroethane) spray; side effect is dyspigmentation (i.e blue skin); most aggressive and effective method
    - "When you are in preschool (precool) you cry on the flour until you turn blue. Crying is an aggressive but effective way to get what you want."
  - Parallel cooling is done during the laser treatment
  - Postcooling is used to ↓ pain, erythema and edema afterwards (i.e. ice packs)
- The various lasers can be remembered by "Excimer APP ReADY glass TEC"
  - Excimer (308 nm; UVB range): UV light (used for psoriasis and vitiligo)
  - Argon (488-514nm): Blue visible light ("A next to B in alphabet")
  - Potassium titanyl phosphate (KTP; 532 nm): green light ("Green KaTerPillar")
  - PDL (585-600 nm): yellow visible light ("PD-yellow")
  - Ruby (694 nm): Red visible light ("69 is a # that makes you blush/turn red")
  - Alexandrite (755 nm): infrared light ("Alexander the Great was born in 755 BC")
  - Diode (800 nm): infrared light ("I call it Di-ocho because ocho is 8 in Spanish")
  - Nd: YAG (1064 nm): infrared light ("YAG lasers are close to 1000 and 3000")
    - **Penetrates deepest, bypassing melanocytes so is safest in skin of color**
  - Erbium glass (1540 nm): infrared light ("glass was discovered in 1540")
  - Thulium (1927 nm): infrared light ("thulium was discovered in 1927")

- Erbium YAG (2940 nm): infrared light ("YAG lasers are close to 1000 and 3000")
- CO_2 (10,600 nm): infrared light
- Nonlaser energy sources are much less selective and less powerful
  - Intense pulsed light (IPL): **xenon flashlamp** emits light that do not follow any of the 3 Cs → noncollimated, noncoherent, polychromatic
    - Broad wavelength (range 500-1200 nm) requires filters to narrow it down
    - **IPL has been shown to Increase expression of p53**
    - **IPL is "No HELP": Non-ablative resurfacing, Hair removal, Erythematotelangiectatic rosacea, Lentigos, Poikiloderma of Civatte**
  - Radiofrequency (RF): alternating electric current **locally heats tissue** ('think Fire')
    - Has specificity for Fat (used to treat cellulitis); potential SE is lipoatrophy
    - Also affective for scars and skin tightening and is **safe in all skin types**

Laser safety

- Lasers are divided into 4 safety classes (1-4) where class 4 is the most hazardous and powerful laser class and also the MC class used in dermatology
- The four main concerns are **ocular damage**, inhalation of biohazardous plume (**HPV**), cutaneous burns and fire hazard ("blindly inhaling burning fire")
  - Various ocular complications may arise depending on which laser is used
    - UV laser (excimer) damages the **lens → cataracts** ("Lens filter UV")
    - Visible light and near infrared lasers ("APP ReADY"; 400-1400 nm) target **melanin/hemoglobin** → damages pigment containing structures **(retina)**
      - **p/w sudden painless Vision loss** which improves over weeks
        - Painless because the retina does not have pain receptors
      - Near infrared/Q-switched **("ReADY") has highest risk of blindness**
    - Mid and far infrared lasers ("glass TEC"; 1400-100,000 nm) target **water** → damages water containing structures such as the cornea and sclera
      - **CO_2 laser → Corneal ulceration → pain**, tearing and blurry vision
  - Inhalation of biohazardous plume specifically **HPV** particles
    - Concern for oral SCC; prevent w/ smoke evacuator held 1 cm away + N95
  - Cutaneous burns may occur with any laser as a result of operator error
  - Risk of fire hazard is greatest with Erbium:YAG and CO_2 lasers ("TEC") and when oxygen is being administered (reduce intraoperative O_2 concentration < 40%)
    - Other fire hazards ("a fire CAME"): CO_2 lasers, Aluminum chloride (drysol), Alcohol, Malathion, Erbium:YAG lasers

Types of lasers

Vascular lasers (poster child is PDL)

- Utilizes selective Photothermolysis to damage blood vessels
  - Uses: telangiectasias, PWS, superficial hemangiomas, poikiloderma of Civatte, scars, psoriasis, PG, angiofibromas and warts
- Targets hemoglobin: Oxyhemoglobin (O next to P in alphabet) > deoxyhemoglobin

- Absorption peaks are 418, 542 and 577 nm ("I just remember 543 and 567")

- SEs: **Purpura (#1)**, **Pigmentary changes/PIH (#2)** and blisters; **prevent side effects by Precooling, decreasing the fluence and using longer Pulse durations (ms not ns)**
  - In darker skin types: warn patients of transient PIH, increase skin cooling, **lengthen pulse width/duration** and decrease energy/fluence
  - Site of eye damage is retina
- To make vascular lasers more effective you can ↑ fluence or use shorter pulse durations
- Desired treatment endpoint for PDL is Purpura (a result of cavitation and vessel rupture)
  - With pulse durations of 10 ms or more, there is little to no immediate purpura as cavitation and vessel rupture is avoided **(short pulse durations cause purpura)**
  - "Pulse Stacking" w/ PDL changes Shape of RBCs, is more effective and causes less purpura (because you can ↓ pulse duration) but it causes more Swelling/edema
- Desired treatment endpoint for KTP and Nd:YAG is immediate disappearance of vessels
- PDL is TOC for most vascular lesions except
  - Telangiectasia: **KTP** (short wavelength good for superficial lesions), IPL, PDL
  - Poikiloderma of Civatte: IPL (PDL is next best)
  - Vascular ectasias on the Lower legs: Long-pulsed Nd:YAG (diode is next best)
    - Whereas long-pulse Nd:YAG is a vascular/depilatory laser, QS Nd:YAG is a hyperpigmentation/tattoo/nevus of ito laser

Hair reduction lasers and light sources

- Utilizes selective photothermolysis to destroy the **bulge and bulbar stem cells** (ideally)
- Targets melanin (absorption spectra is 300-1000 nm w/ no peaks; linear; "me-line-in")
  - White hair is impossible to remove because it lacks melanin chromophore
- SEs are PIH (↑ risk in skin of color; **Nd:YAG safest in POC**), leukotrichia, blisters/scars
  - May see paradoxical hypertrichosis as low fluence induces hair growth (happens in darker skin types because we use lower fluences in these patients)
  - Site of eye damage is retina
- Can shave before treatment but do not fully remove hair shaft (will lose chromophore)
- Desired treatment endpoint is **transient perifollicular edema**
- Diode is the most efficacious Depilatory laser; Nd: YAG is the safest in skin of color

Resurfacing lasers

- Uses selective photothermolysis to target water (absorptions peaks: 1450, 1950, 3000)
  - Uses: skin tightening, wrinkles, age spots, sun damage, **hypertrophic scars**, warts
- May be ablative (vaporization of target tissue) or nonablative (thermal effect on dermis)
  - Ablative lasers (more downtime but requires less treatments)
    - Erbium:YAG (2790 nm) and Erbium:YSGG (2790 nm) have less thermal injury → ↓ coagulation, **↑bleeding**, ↓ collagen retraction, ↓ down time
    - CO_2 (10,600 nm) has more thermal injury → ↑ Coagulation, ↓ bleeding, ↑ Collagen retraction, ↑ recovery time/PIH/erythema
  - Nonablative lasers (less downtime but requires more treatments)
    - Vascular lasers (PDL), infrared lasers (Nd: YAG, diode, Er:glass), IPL and RF

- May be fractionated (↓ downtime, ↓ erythema but less effective) or nonfractionated
  - Fractionated ablative lasers create columns of thermally damaged skin next to islands of normal skin (these keratinocyte reservoirs lead to more rapid healing)
  - Unlike fully ablative lasers, fractionated lasers can be used on the neck and chest
- SEs are erythema, PIH and infections (↑ risk of HSV in 1st week); consider prophylaxis
  - Site of eye damage is cornea and sclera → corneal ulceration/pain/blurry vision
  - Prophylaxis 1 day before to 10 days after resurfacing laser if history of HSV
  - MC short-term SEs are erythema (100% of patients) and **post-inflammatory hyperpigmentation** (40%); MC long-term SE is **hypopigmentation** (16%)
  - Infrared non-ablative lasers penetrate deeper into the skin (bypass melanocytes) than the ablative lasers and thus cause less PIH in darker skin types
  - Ectropion can occur after periorbital resurfacing (↑ risk if pt had blepharoplasty)
- CI to Ablative lasers: Active infections/inflammation in treatment zone (herpes labialis, impetigo), history of keloid formation, history of AI-CTD (lupus or scleroderma)
- Most sources say to wait between 1 and 6 months after isotretinoin to use fully ablative lasers (and medium/deep peels) but there is no need to wait for other types of lasers such as fractional ablative, non-ablative, PDL and laser hair removal

Hyperpigmentation/tattoo removal lasers

- Targets tattoo pigment: small radius → small TRTs (nanoseconds) → **QS-lasers required**
- For black, blue or brown pigment use ReADY lasers (except diode → hair removal)
  - Alikhan refers to these as the RAN lasers, I call them RAY or RAN lasers
  - These QS-RAY lasers are also best for pigmented lesions (lentigines, ephelides, nevus of ota, minocycline pigmentation, post-sclerotherapy hyperpigmentation)
    - QS Ruby 694 ("RAY") works best; **QS-1064 Nd-YAG ("RAY") safest in POC**
- For green pigment use "QS-RA" (QS **ruby 694** is best but QS Alex 755 is safer in POC)
- For turquoise the 755 QS-alexandrite is best
- For Yellow, Red, Violet, White or Orange use the frequency doubled **Nd:YAG (532 nm)**
  - "Your Return Visit Will be On May 32" (532 nm frequency doubled Nd:YAG)
- Red tattoos w/ mercuric sulfide (cinnabar) are most likely to cause allergic reactions
- A Granulomatous reaction to red tattoo is a type 4 (Four) delayed hypersensitivity or a Foreign body reaction ("F next to G in alphabet")
- If patient is allergic to tattoo die, laser treatment may cause anaphylaxis
- **Desired endpoint is immediate tattoo whitening (a result of cavitation)**
  - White tattoos may undergo paradoxical darkening 2° to **reduction** of Ti^{4+} → Ti^{3+}
    - We can treat this complication w/ further Q-switch laser but it may be permanent (a test spot is recommended to avoid this complication)
  - Flesh toned/light red tattoos (permanent lip liner) may undergo paradoxical darkening 2° to reduction of Ferric oxide (Fe^{3+}) → Ferrous oxide (Fe^{2+})
    - "Think iron in steak which goes from red when rare to black when burnt"
    - Opposite of methemoglobinemia ("ferrous to ferric, will make you sick")
- Tattoo pigment origin: black, blue (cobalt; "cops are blue"), brown (ochre; "brown oak tree"), green (chromium; chlorophyll is green), yellow (cadmium; photoallergic tattoo

reactions; “yellow cab”), white (titanium), **red (mercuric sulfide; MCC of allergic tattoo reactions**; “mercury is a red planet”), violet (manganese; “non-ripe mangos are violet”)

- Pigment alteration/erythema is the MC SE with Q switch lasers (all lasers really)
- Laser-induced chrysiasis (p/w blue-gray skin) is a SE specific to QS and picosecond lasers that occur in those w/ a history of gold ingestion (old treatment for RA)

10.3 DERMPATH DIFFERENTIALS

Neutrophils

Neutrophils in the horn/corneum ("SPASTIC")

- Seb derm: psoriasiform spongiotic dermatitis w/ perifollicular neutrophilic scale/crust
- Psoriasis: discussed under reaction patterns
- Amicrobial pustulosis of the folds
- Syphilis: long Sexy Skinny rete, Swelling of vessel lumen, plasma cells
- Tinea: neuts and horizontal hyphae in stratum corneum which is commonly basket-weave w/ a narrow zone of compact corneum just above the granular layer
- Impetigo: neutrophils and **clusters of gram-positive cocci** in the stratum corneum
- Candida: neutrophils and pseudohyphae (often vertically oriented) in stratum corneum
    - "Tinea swims and candida dives (candi-dives)"

Subcorneal neutrophilic pustule ("A CAT PISS")

- IgA pemphigus: intrepidermal neutrophilic infiltrate with minimal acantholysis (IEN type) or subcorneal pustule (subcorneal pustular dermatosis type); both have **(+) DIF**
    - DIF: intercellular IgA in upper epidermis (IEN) or lower/entire epidermis (SPD)
- AGEP: in contrast to IgA pemphigus and Sneddon-Wilkinson, AGEP has **eosinophils** in the dermal infiltrate, papillary dermal edema and necrotic keratinocytes
- Candida: neutrophils and pseudohyphae (often vertically oriented) in stratum corneum
- Acropustulosis of infancy: intraepidermal neutrophils and eos → subcorneal pustule
- Transient neonatal pustular melanosis: subcorneal (superficial) pustule w/ (-) gram stain
- Tinea: may show a subcorneal pustule but would see hyphae on GMS or PAS
- Pustular psoriasis: subcorneal pustule w/ other psoriasis features (hypogranulosis etc.)
- Impetigo: neutrophils and clusters of gram-positive cocci in the stratum corneum
- SSSS: subcorneal split w/ neuts but no bacterial colonies (vs bullous impetigo)
- Sneddon-Wilkinson (subcorneal pustular dermatosis): subcorneal pustule w/ **(-) DIF**

If there are many neuts in the dermis, next question to ask yourself is if there's a vasculitis

- Pink fibrinoid degeneration of vessel wall + neutrophils extravasating the vessels walls + leukocystoclasia/nuclear dust ("ant particles")

Medium vessel vasculitis (polyarteritis nodosa)

- LCV with necrotizing arteritis of medium sized arteries in the superficial SQ

Mix of small and medium vessel leukocytoclastic vasculitis ("Big 5 pattern of vasculitis")

- Involves both small vessels (post-capillary venules) and large ones in the reticular dermis
- Endothelium is frequently necrotic with fibrin deposition → RBCs extravasation
- Karyorrhexis (neutrophilic debris)
- MARS: Mixed cryoglobulinemia, ANCA vasculitis, Rheumatoid vasculitis, Septic vasculitis

- Granulomatosis w/ polyangiitis (Wegener's granulomatosis): LCV evolves into stellate abscesses (palisaded granuloma w/ central neutrophils) and giant cells
    - "Blue granuloma"
- Churg-Strauss: LCV evolves into stellate abscesses (palisaded granuloma w/ central eosinophils); granuloma composed of epithelioid cells w/o giant cells
    - "Red granuloma"
- Microscopic polyangiitis: no granuloma so only non-distinct ANCA+ vasculitis
- ANCA-positive vasculitis secondary to drug (eosinophils) or levamisole (thrombi)
- Septic vasculitis: CSSV + thrombi + dense dermal inflammation

Small vessel leukocytoclastic vasculitis ("Little 5 pattern")

- Involves only small vessels (post-capillary venules)
- Endothelial necrosis is rare but will still see fibrin deposition within the vessel wall
- Karyorrhexis (neutrophilic debris) w/ RBC extravasation
- "Small vessel vasculitis is A HUGE Sickness": Acute hemorrhagic edema of infancy, **Henoch-Schönlein purpura**, Urticarial vasculitis, Granuloma faciale, EED, Serum Sickness, Secondary CSVV (ABCD; Autoimmune, Bug, Cancer, Drug)
    - Granuloma faciale: Grenze zone w/ mixed infiltrate; facial skin (no granuloma)
    - EED: LCV early w/ onion skin fibrosis in later lesions; not on facial skin

Neutrophils in the dermis but no obvious vasculitis or stellate abscess ("NEUTS")

- Neutrophilic dermatosis of the dorsal hand: diffuse neutrophilic dermal infiltrate
- Erythema marginatum: superficial perivascular neutrophilic infiltrate
- Urticaria: superficial dermal vessels filled w/ neutrophils (early) → perivascular and interstitial neutrophilic infiltrate and superficial dermal vessels filled w/ neutrophils
- TRAPS and other autoinflammatory periodic fever syndromes: neutrophilic infiltrate
- Sweet's syndrome: nodular and diffuse infiltrate of neutrophils (karyorrhexis and focal LCV common) w/ marked papillary dermal edema
    - PG, Behçet's and Bowel-Bypass syndrome can all look like Sweets

Stellate abscesses (palisaded granuloma w/ central neutrophils): "No CLAST"

- Nocardia: stellate abscess w/ acid fast and gram-positive filaments present
- Cat scratch: stellate abscess that stains w/ Warthin-Starry or B. henslae IHC
- Lymphogranuloma venereum: stellate abscess w/ plasma cell infiltration and sinus tracts
- Atypical mycobacteria: stellate abscess w/ acid fast-positive organisms within histiocytes
- Sporotrichosis: stellate abscesses +/- cigar-shaped yeasts
- Tularemia: stellate abscess
- Other conditions that show stellate abscesses include Churg-Straus and GPA (would also show vasculitis) and mycetomas (would also show Splendore-Hoeppli phenomenon)

Lymphocytes

Perivascular lymphoid infiltrate ("LGBT Plus PRIDe" where PRIDe represents a true lymphocytic vasculitis → damage/edema to vessel walls w/ lymphs extravasating vessels)

- Lupus: **vacuolar interface** w/ dermal **mucin**, **dense PV/PA lymphocytes**, BMZ thickening
- Gyrate erythema: dense "coat-sleeve" perivascular lymphoid infiltrate
    - Includes erythema marginatum (neuts and lymphs), erythema migrans, EGR, EAC
- Bullous lupus: subepidermal blisters w/ neuts, dermal PV/PA lymph infiltrate, ↑ mucin
- Tumid lupus: PV/perieccrine lymphocytic infiltrate, ↑ mucin, minimal vacuolar interface
- PMLE: **superficial** > deep **PV lymph infiltrate w/ papillary dermal edema; not acral skin**
- PPD: **purely lymphoid inflammation which surrounds capillaries (above level of the post-capillary venule), RBC extravasation** and hemosiderin deposits over time
- Pityriasis lichenoides (PLEVA): **compact hyperkeratosis w/ Parakeratosis** (+/- neuts in the horn), Lichenoid to vacuolar infiltrate but w/ lymph in every whole, Extravasated RBCs, V (wedge)-shaped dermal lymph infiltrate w/ neuts but never any eosinophils, Apoptotic/dyskeratotic keratinocytes in a spongiotic epidermis w/ exocytosis of lymphs
- Perniosis: **superficial and deep PV lymphoid infiltrate w/ papillary dermal edema**, edema around vessels and expansion of vessel walls without fibrin deposition; **acral skin**
- Rickettsial disease: lymphohistiocytic perivascular infiltrate w/ RBC extravasation edema, vacuolar interface +/- fibrin thrombi, capillary wall necrosis and karyorrhexis
- Insect bites: **dense wedge (V)-shaped perivascular lymphoid infiltrate w/ eosinophils and papillary dermal edema** +/- focal LCV
- Degos disease: V-shaped (superficial and deep) PV lymphoid infiltrate w/ vessel damage (chronic lesions display a central epidermal depression w/ necrosis of dermal structures)

Lupus ddx (dense nodular Lymphiod infiltrate in the dermis)

- Lupus: vacuolar interface w/ mucin and **PV/PA lymphocytes** and BMZ thickening
- Light (PMLE): papillary dermal edema and a dense **perivascular lymphocytic** infiltrate
- Lymphoma: lymphocytes are diffuse and atypical
- Lymphocytoma cutis: nodular lymphocyte-predominant infiltrate in the dermis
- **Lichen striatus: LP-like interface at the DEJ and peri-eccrine lymphocytic inflammation**
- Lues (syphilis): psoriasiform hyperplasia with long "sexy" rete ridges, lichenoid interface, abundant lymphocytes and plasma cells and endothelial cell swelling

Eosinophils

Eosinophilic spongiosis ("HAAPIED")

- Herpes gestationis: histologically indistinguishable from BP (subcorneal split w/ eos)
- Arthropod bite: moderately dense superficial and deep infiltrate of lymphs and eos
- Allergic contact dermatitis: spongiotic dermatitis w/ many eosinophils
- Pemphigus (early stage): prior to an intrepidermal split you see eosinophilic spongiosis
- Pemphigoid (early urticarial stage): perivascular and interstitial infiltrate of lymphocytes and many eosinophils that tend to line up at the DEJ
- Incontinentia pigmenti (early vesicular stage): eosinophilic spongiosis w/ **dyskeratosis**
    - Verrucous stage: hyperkeratosis w/ papillomatosis w/ whorls of dyskeratosis

- Pigmented stage: pigment incontinence
- Hypopigmented stage: decreased melanocytes

- Erythema toxicum neonatorum: spongiosis adjacent to follicle + eosinophils
- Drug reaction: variable but commonly includes interface dermatitis w/ necrotic keratinocytes, spongiosis and superficial and deep mixed infiltrate w/ eosinophils
  - This pattern is not distinct and can be seen in viral exanthems also

Perivascular lymphocytic infiltrate with eosinophils (“BAD eos hang around blood vessels”)

- BP (urticarial phase): presence of eos going into the epidermis or lining up at the DEJ
- Arthropod/Well’s: infiltrate is deeper (even down into the SQ) and more dense
  - May also see papillary dermal edema (discussed above in “PASTE” mnemonic)
- Drug: this is favored if the eos are not superficial (epi → BP) or deep (near fat → bug)
  - Path is variable but commonly see a vacuolar interface and a polymorphous infiltrate w/ eosinophils which is less exuberant than in arthropod assault

Plasma cells: “KRLASMA”

- Kaposi’s, Rhinoscleroma, **Rosai-Dorfman**, Leishmaniasis, Lymphomas, nodular Amyloid, Syphilis, Scleroderma, SCC, SPAP, Melanoma, Morphea, AK)
- So if you are ruling out one of these conditions, try not to take a biopsy from an area of the body that also commonly recruits PLASMA Cells (Posterior neck, face esp near the Lips, Anogenital area, Shins, **Mouth**, Axilla, Chest/breasts)

Granulomas

- All CD68+ and HAM56+ except Langerhan cell histiocytosis
- Always perform special stains on all these to rule out fungal and mycobacterial infections and examine under polarized light to exclude foreign material
  - Fungal infections are discussed below
  - Cutaneous tuberculosis (i.e. lupus vulgaris): lymphocytic and suppurative (neuts) granuloma w/ epithelioid histiocytes and Langhan (horse-shoe shaped) giant cells palisading around **central caseous necrosis**

“GRANulomas are XLR”: GRAN are palisading and XLR are a sea of histiocytes in dermis

- Granuloma annulare (palisade around **mucin/degenerated collagen**): eosinophils in 40%
  - Interstitial pattern (MC): interstitial histiocytes + lymphs (busy dermis) + mucin
    - Ddx: PNGD/IGD (neuts +/- LCV), breast cancer/leukemia cutis (atypia)
  - **Palisading pattern: histiocytes surround altered dermal collagen and mucin**
    - Ddx: NLD (deeper, more fibrosis, layered cake pattern), actinic granuloma (on actinically damaged skin, elastophagocytosis, less mucin)
  - Deep (SQ): histiocytes palisading around degenerated collagen in deep dermis
    - Ddx: looks similar to rheumatoid nodule but color is not as bright pink
- Gout: feathery crystals of uric acid w/ surrounding granulomatous inflammation (looks amorphous from low power but it actually has a feathery/needle-shaped appearance)

    - Ddx: similar to IL ken which shows palisade around amorphous bubbly material
- Rheumatoid nodule: large palisading granuloma surrounding **pink fibrin (no mucin)**
- Actinic granuloma (annular elastolytic giant cell granuloma): looks like palisading GA except it **lacks mucin and there is solar elastosis**; VVG shows loss of elastic tissue
- Acne agminata (lupus miliaris disseminates faciei): palisaded granuloma w/ central light pink caseous necrosis
    - Ddx: TB (i.e. military TB/lupus vulgarus) also has a palisading around caseous necrosis (lighter pink than the fibrin in rheumatoid nodule) but TB should have horsehoe-shaped Langhan Giant cells and would stain for acid-fast bacilli
- Necrobiosis lipoidica: palisade of histiocytes w/ pale degenerated collagen in a sclerotic dermis forms a **"layered cake"** appearance; square punch biopsy w/ plasma cells
- Necrobiotic xanthogranuloma (NXG): neutrophils, histiocytes and **large Touton giant cells palisading around large X-shaped pink zones of necrosis** w/ **cholesterol clefts**
- Xanthogranuloma: sea of histiocytes w/ **many Touton giant cells** (no hemosiderin vs DF)
- Langerhans cell histiocytosis (LCH): superficial sea of histiocytes w/ reniform nuclei that commonly shows epidermotropism (Langerhan cells are meant to be in the epidermis)
    - IHC: S100+, CD1a+, peanut agglutinin+, langerin (**CD207**)+; CD68-
    - Ddx: mastocytoma looks similar on low power but mast cells are "fried egg" or "polka dot" in appearance and do not cross the DEJ into the epidermis
- Reticulohistiocytoma: "sea" of mononuclear histiocytes in the dermal w/ a two-toned (one lighter and one darker area) cytoplasm ("reticulohistiocy-two-ma")
- Rosai-Dorfman disease: sheets of **lymphocytes** (dark-staining), **plasma cells** and histiocytes (light-staining) w/ emperiopolesis (lymphocytes passing through histiocytes)
    - IHC: CD68+ and **S100+**

Sarcoidal ("naked") granulomas: "Get naked and talc about Milking Sir Crohns tattooed cactus and Sarcing Ross' silly berries" (look for $CD68^+$ foreign body giant cells in foreign body ones)

- Talc granulomas: sarcoidal granulomas w/ white birefringent crystals on polarized light
- Milkersson-Rosenthal: granulomatous (especially sarcoidal) inflammation of lip stroma
- Zirconium: non-polarizable foreign material
- Cutaneous Crohn's: sarcoidal granulomas and crypt irregularities
- Tattoo granuloma: **black** material in the dermis (melanin and hemosiderin is brown)
- Cactus granuloma: sarcoidal or foreign body granulomas w/ PAS+ spines
- Sarcoidosis: epithelioid histiocytes forming naked granulomas (no lymphs); **no necrosis**
- Granulomatous rosacea (lupus miliarius disseminates facei is a variant): folliculocentric sarcoidal granulomas **(caseating necrosis in LMDF); dilated vessels**
- Silica granuloma: sarcoidal granulomas w/ polarizable birefringent crystalline particles
- Beryllium granuloma: non-polarizable foreign material
- Tuberculoid leprosy: sarcoidal granulomas that track horizontally along nerves

Foreign body granulomas: "foreign languages don't STICK AZ well as A-B-Z"

- Silica (Silicon dioxide): Sarcoidal granulomas containing birefringent crystalline particles
- Starch: foreign body granulomas w/ ovoid basophilic **PAS+** starch granules

- Splinters (wood): PAS+ birefringent material made of regularly sized cells
- Spines of Jellyfish/coral/sea urchin: lichenoid dermatitis w/ birefringent calcite crystals
- Suture: foreign body granulomas w/ birefringent suture material within a scar
- Tattoo ink: black material in the dermis (melanin and hemosiderin is brown)
- Talc (hydrous magnesium silicate): sarcoidal granulomas w/ white birefringent crystals on polarized light ("Don't talc to white cops")
- IL steroids: foreign body granulomas w/ central pale bluish material (resembles mucin)
- Cactus (Opuntia): sarcoidal or foreign body granulomas w/ PAS+ spines
- Keratin: foreign body granulomas w/ birefringent keratin flakes on polarized light
- Arthropod parts: PAS+ birefringent material on polarized light
- Zinc: dense neutrophilic infiltrate w/ birefringent **rhomboidal** crystals on polarized light
- Aluminum: granulomas w/ no polarizable particles seen
- Beryllium: caseating granulomas w/ no polarizable particles seen
- Zirconium: sarcoidal granulomas; no (Zero) polarizable particles seen
    - A-B-Z (words also end in "ium") are the ones that are non-birefringent
    - Arthropod parts, Aluminum, Starch, cactus Spines and wood Splinters are PAS+

Foreign body reactions to filler material

- Hyaluronic acid: amorphous basophilic material that stains w/ mucin stains
- Paraffin: **cystic spaces of varying sizes (Swiss-cheese)** which stain with Oil red O
- Silicone: **cystic spaces of varying sizes (Swiss-cheese)** which do not stain with Oil red O
- PMMA w/ bovine collagen (Artefil): round non-birefringent bodies in cystic spaces
- PMMA w/ hyaluronic acid (DermaLive): irregular polygonal, pink, non-birefringent particles that resemble broken glass in cystic spaces
- Poly-L-lactic acid (Sculptra): irregular fusiform, oval and Spikey ("Sculptra and Spikey sound alike") birefringent particles in cystic spaces that resemble suture material
- Calcium hydroxylapatite (Radiesse): blue-gray round or oval microspheres
    - "Looks kind of like the saponified calcium seen in pancreatic panniculitis"

Xanthomas ("Plane VET")

- Planar xanthomas (DIXX): **foam cells** (lipidized histiocytes) located **superficially**
    - Diffuse plane xanthoma: a/w IgG paraproteinemia but no lipid abnormalities
    - Intertriginous xanthoma: a/w type 2 hyperlipoproteinemia (2 antecubital fossae)
    - Xanthoma striatum palmaris: a/w type 3 hyperlipoproteinemia (3 palmar lines)
    - Xanthelasma: lipid levels are normal in around half of patients
        - Path also shows many vellus follicles and **striated muscle** (eyelid skin)
- **Verruciform xanthoma**: papillomatosis and parakeratosis ("wart on fire") w/ foam cells in dermal papillae; a/w CHILD but not a/w lipid abnormalities; MC in mouth (esp tongue)
- Eruptive xanthoma: foam cells + extracellular lipid (looks like mucin so confused w/ GA)
- **Tuberous xanthoma**/Tendinous xanthoma: foam cells + fibrotic nodule

Reaction patterns

Lichenoid: sawtooth rete, band-like lymphoid infiltrate, Civatte bodies and hyperkeratosis

- Lichen planus: only one that never has parakeratosis or eosinophils
- Benign lichenoid keratosis: usually can see a solar lentigo or SK towards the periphery
- Lichenoid drug: lichenoid interface dermatitis often w/ **eosinophils** +/- parakeratosis
    - Civatte bodies can be seen up higher in the epidermis than in LP
- Lichenoid GVHD: lichenoid interface that may have parakeratosis but rarely has eos
- Lichenoid regression of a melanoma: lichenoid interface on heavily sun-damaged skin
- Hypertrophic lupus: lichenoid interface (rarely has parakeratosis or eos) w/ superficial and deep infiltrate that is often peri-adnexal (↑ $CD123^+$ plasmacytoid dendritic cells)
- Syphilis can also have a lichenoid interface but do not have the other features of LP
    - "Looks like LP and psoriasis had a messy baby"; **plasma cells are your main clue**

Vacuolar interface w/ a lymph in every hole ("Most Bangable Player puts it in every hole")

- Mycosis fungoides: large angular hyperchromatic lymphocytes in the epidermis (sometimes in Pautrier's microabscesses) and at the DEJ w/ papillary dermal fibrosis
- Benign lichenoid keratosis (early): vacuolar interface > lichenoid interface
- PLEVA: **compact hyperkeratosis w/ Parakeratosis** (+/- neuts in the horn), Lichenoid to vacuolar infiltrate but w/ lymph in every whole, **Extravasated RBCs**, V (wedge)-shaped dermal lymph infiltrate w/ neuts but never eos or plasma cells, Apoptotic/dyskeratotic keratinocytes in a spongiotic epidermis w/ exocytosis of lymphs
    - PLC is a variant with less pronounced histologic features (less interface, less inflammation, less RBCs, less apoptotic keratinocytes)
    - Ddx: syphilis is the great imitator and can look similar (look for plasma cells!!)

Vacuolar interface ("when you see holes at the DEJ, you DELVe G")

- DM: similar to lupus but generally has more dermal Mucin and epidermal atrophy
- Drug including fixed drug eruption: many eosinophils (w/ pigment incontinence in FDE)
- EM: basket weave stratum corneum, vacuolar interface w/ spongiosis and dyskeratosis
- Lupus: vacuolar interface w/ thin epidermis, PV/PA lymph infiltrate and BM thickening
    - From ACLE to SCLE to DLE, the vacuolar interface becomes more obvious, lymphoid infiltrate becomes deeper, epidermis becomes thinner w/ more follicular plugging
    - Tumid lupus is not in this ddx because while it has ↑ dermal mucin and PV/PA lymph infiltrate, it lacks epidermal changes (vacuolar interface, BM thickening)
    - DIF: continuous **granular** band of IgG/A/M and C3 (full house) at the BMZ
- Viral exanthem: mild vacuolar interface w/ perivascular lymphocytes (non-specific)
- GVHD: discussed below

Vacuolar interface with Dyskeratotic cells in the epidermis (abcDEFG)

- EM: **more dyskeratosis** (can be full thickness in areas) than the others, basket weave stratum corneum, superficial PV lymphocytic infiltrate +/- eosinophils
    - Early (macules/papules): mild lymph vacuolar interface, dyskeratosis, pap dermal edema, spongiosis (intercellular edema) and ballooning (intracellular edema)

    - Fully developed (vesicles): more lymph vacuolar interface which progresses to subepidermal vesicle, confluent necrotic keratinocytes, more papillary dermal edema and more spongiosis/ballooning (intraepidermal vesicle)
- FDE: mixed infiltrate that extends Deeper into the dermis than the others with **Eosinophils**, **pigment incontinence** and dermal fibrosis
- acute GVHD
    - Vacuolar interface w/ lymphs next to apoptotic cells (satellite cell necrosis)
    - **Hair follicles** are also affected w/ dead keratinocytes and interface change
    - Dysmaturation of epidermis (resembles Bowenoid AK)
- Paraneoplastic pemphigus can also be in this ddx (discussed below)
- PLEVA can also be in this ddx but you would see **compact hyperkeratosis w/ Parakeratosis** (+/- neuts in the horn), Lichenoid to vacuolar infiltrate but w/ lymph in every whole (**never eos**), Extravasated RBCs, V (wedge)-shaped dermal lymph infiltrate, Apoptotic/dyskeratotic keratinocytes in a spongiotic epidermis w/ exocytosis of lymphs
- Ddx: phototoxic drug will see the necrotic keratinocytes but no vacuolar changes

Psoriasis

- Neutrophils alternating w/ confluent parakeratosis (sandwich sign) in stratum corneum
- Collections of neutrophils in stratum corneum (Munro microabscesses)
- Collections of neutrophils in spinous layer (spongiform pustules of Kogoj)
- Regular acanthosis w/ skinny, club-shaped rete that are "test-tube like" (vs irregular and thick rete in LSC) w/ thin suprapapillary plates (vs thick in PRP) and hypogranulosis (vs hypergranulosis in LSC)
- Dilated capillaries in dermal papillae

Psoriasisiform ddx ("My BIG Necrolytic PSoRIATIC Plaque")

- Mycosis Fungoides: large angular hyperchromatic lymphocytes in the epidermis (no neutrophils) and at the DEJ w/ papillary dermal fibrosis
- Bowen's disease: atypical keratinocytes at all levels of the epidermis
- ILVEN: **horizontally alternating ortho and parakeratosis** with ↓ granular layer under areas of para (which is typical when we see parakeratosis)
- Granular parakeratosis: stratum corneum w/ granules and parakeratosis and a preserved granular layer (granular layer is usually ↓ under areas of parakeratosis)
- Necrolytic acral erythema: psoriasiform hyperplasia **w/ necrotic keratinocytes**
- PRP: no neuts in the stratum corneum, Parakeratosis (but no neuts) and **acantholysis** at follicular ostia, horizontally and vertically alternating **"checkerboard" ortho- and parakeratosis** and acanthosis w/ **thick** suprapapillary plates
- Seb derm: Spongiosis w/ Superficial PV/PF lymph infiltrate; neuts next to follicular ostia
- Syphilis: Sexy Skinny rete, Swelling of vessel lumen and plasma cells (looks like LP and psoriasis had a messy baby; PLEVA is also in the ddx but this does not have plasma cells)
- Reiter's disease: looks identical to psoriasis
- ILVEN: see above description

- Acrodermatitis enteropathica (zinc deficiency) is MC but same pattern is seen in other nutritional deficiencies like pellagra or necrolyic acral erythema: psoriasiform hyperplasia but instead of neuts in the horn you see **pallor** of corneum/upper epidermis
- Geographic Tongue: looks like psoriasis but in mouth (no granular or corneal layer)
- ILVEN: see above description
- **Clear cell acanthoma**: sharp demarcation of clear (glycogenated) psoriasiform epidermis
- **Porokeratosis**: look for **cornoid lamella** (angled column of parakeratosis w/ dyskeratotic cells below); zone between cornoid lamella may appear lichenoid or psoriasiform

Mounds (patchy/non-confluence) of parakeratosis ("PEGS")
- PR: mounded Parakeratosis, RBC extravasation, PV lymph infiltrate, mild spongiosis
- EAC: mounds of parakeratosis, **dense "Coat sleave"** PV lymph infiltrate, mild spongiosis
- Guttate psoriasis: **neuts above mounded para**, RBC extravasation, mild spongiosis
- Small plaque parapsoriasis: mounds of parakeratosis, mild spongiosis

Spongiotic dermatitis (edema between keratinocytes:): "SAND PITZ"
- Stasis dermatitis: cannonball-like angioplasia in the superficial dermis with hemosiderin
  - **Acroangiodermatitis (pseudo-kaposi) is clinically like Kaposi but on path can show features of both stasis and Kaposi's sarcoma**
- Seb derm: psoriasiform spongiotic dermatitis w/ perifollicular neutrophilic scale/crust
- Allergic contact dermatitis: eosinophilic spongiosis (HAPPIED mnemonic below)
- Nummular dermatitis: spongiosis of the epidermis
- Dyshidrotic dermatitis: spongiosis of the epidermis
- Pityriasis Rosea: spongiosis w/ Parakeratotis focally, RBC extravasation
- ID reaction: spongiosis of the epidermis
- Tinea: hyphae in a stratum corneum with a basket weave layer above a compact layer
- **Zoon's balanitis: pale flattened, diamond-shaped keratinocytes in a thin epidermis, w/ dilated dermal capillaries and a dermal infiltrate composed entirely of plasma cells**
- Candida: neutrophils and vertically oriented pseudohyphae in stratum corneum
- Note that chronic dermatitis does not have spongiosis rather it has irregular acanthosis and compact stratum corneum

Stages of spongiotic dermatitis
- Early (papules): spong w/ superficial perivascular infiltrate and basket-weave corneum
- Early (vesicles): spongiotic microvesiculation, parakeratosis/scale-crust, psoriasiform hyperplasia and superficial perivascular infiltrate
- Late (psoriasiform plaques): LSC changes (compact orthokeratosis, hypergranulosis w/ stratum lucidum, psoriasiform hyperplasia and course collagen fibers in vertical streaks in the papillary dermis) with no spongiosis
  - Only time you will see a stratum lucidum on hair bearing skin ("hairy palm" sign)

Blisters
- Get a DIF

Subcorneal split (all attack desmoglein 1 except friction blister; "I have to take a Frickin' PISS")

- Friction blisters: superficial split just below the granular layer **w/ epidermal necrosis**
- Pemphigus foliaceus: subcorneal acantholysis w/ scattered **eosinophils**; **DIF (+)**
    - DIF: netlike pattern of IgG (upper epidermis > lower epidermis)
        - Pemphigus erythematosus is a variant which also has linear MBZ staining
- bullous Impetigo: **neutrophils** within the subcorneal bullae +/- **staph colonies**; DIF (-)
- SSSS: clean subcorneal split w/ no inflammation or staph colonies; DIF (-)

Intra-epidermal acantholysis ("Pem-PHHAAA-gus")

- Pemphigus vulgaris: **acantholysis above basal layer ("tombstoning")** that **tracks down hair follicles**; **+ eosinophils**
    - DIF: net-like deposition of IgG and C3 (lower epidermis > upper epidermis)
- IgA Pemphigus (IEN type): intrepidermal **neutrophilic** infiltrate w/ minimal acantholysis
    - DIF: intercellular IgA in upper epidermis (SPD type) or lower/entire epi (IEN type)
- Paraneoplastic pemphigus: **lichenoid or vacuolar infiltrate** w/ necrotic keratinocytes
    - DIF: net-like IgG and C3 (like PV) **but also linear deposition at the DEJ** (this DIF pattern is most reminiscent of pemphigus erythematosus)
- Hailey-Hailey: **acantholysis at all levels of an acanthotic epidermis (resembles a dilapidated brick wall)** but **does not track down follicle**; minimal dyskeratosis; DIF (-)
- **HSV 1-3 (M)**: intraepidermal blister w/ solitary keratinocytes w/ nucleic changes of viral infection→ Multinucleated keratinocytes, **Margination of chromatin**, nuclear Molding
    - Early lesions (papulovesicles): ballooning of epidermal > follicular keratinocytes w/ resultant **multinucleated keratinocytes** (nuclear molding) with pale nucleus (center) and steel-gray chromatin (periphery)
    - Fully developed (vesicle): same features but with intraepidermal vesicles +/- LCV
    - Late lesions (crusted vesicle): "ghosts" of necrotic multinucleated keratinocytes
- Acantholytic acanthoma: acantholysis of acanthotic epidermis like in Hailey-Hailey but does not have any dyskeratosis
- **Acantholytic AK/SCC**: large atypical acantholytic keratinocytes (usually a/w hair follicle)
    - Mechanism: desmoglein-3 and E-cadherin defect
- Bullous Arthropod assault: intraepidermal acantholysis w/ dense superficial PV lymphocytic infiltrate w/ numerous eosinophils

Intra-epidermal acantholysis w/ prominent dyskeratosis (corps ronds and grains; "DAWG")

- Dariers (keratosis follicularis): intraepidermal acantholysis can be seen at all levels of the epidermis w/ numerous dyskeratotic cells including corps Ronds (cells w/ pyknotic nucleus, clear perinuclear halo and brightly Red/eosinophilic cytoplasm) and corps grains (oval cells w/ elongated cigar-shaped nuclei and abundant keratohyalin granules mostly seen in the stratum corneum); orthokeratosis; lymphs (but no eos) in the dermis
- Acanthoyic dyskeratoma: solitary lesion w/ acantholytic dyskeratosis
- Warty D: cup shaped/**endophytic** solitary lesion w/ acantholytic dyskeratosis
- Grovers: smaller involved foci than in Darier's +/- **eosinophils** (absent in Darier's)

  - In reality, these are not able to be differentiated from each other on H&E

Subepidermal autoimmune (positive DIF) blistering diseases: "ABCDE"

- linear IgA bullous dermatosis: sup-epidermal split w/ **neutrophils** lined up at the DEJ
  - DIF: linear IgA at DEJ
- Bullous lupus: sub-epidermal blister w/ **neutrophils**
  - DIF: full house (IgG, C3, IgA, IgM) in a granular pattern along the DEJ
- BP: sub-epidermal split with **eosinophils** and neutrophils in vessel cavity ("BAD eos": BP, Arthropod and bullous Drug can all do this so you will need DIF to differentiate)
  - DIF SSS: stains roof of blister cavity (vs stains floor in EBA)
  - DIF: linear C3 and IgG along DEJ in an n-serrated pattern ("bullous pen-phigoid")
- Cicatricial pemphigoid: subepi split w/ lymphs > plasma cells and eos; **dermal fibrosis**
  - DIF: linear IgA, IgG and C3 along the DEJ and along appendageal structures
- DH: sup-epidermal split w/ **neutrophilic micro-abscesses in tips of dermal papillae**
  - DIF: granular IgA within dermal papillae
- EBA: noninflammatory subepi split (mechanobullous) vs BP-like (inflammatory form)
  - DIF SSS: stains floor of blister cavity (vs stains roof in BP)
  - DIF: linear IgG > C3 at DEJ in u-serrated pattern ("epidermoysis bulossa aquisita")

Non-autoimmune subepidermal blisters (all are pauci-inflammatory except bullous CSVV)

- **PCT**: subepidermal split on acral skin w/ minimal inflammation, protuberance of rigid dermal papillae into blister cavity ("festooning"), amphophilic BMZ (type 4 collagen) material in overlying epidermis above blister cavity ("caterpillar bodies") and thickened hyalinized PAS+ material around BVs under blister cavity w/ surrounding solar elastosis
  - DIF (typically negative): **IgG, IgM and C3 in vessels** >> linear at DEJ
- SJS/TEN: subepidermal bulla w/ **eosinophils** and **full-thickness epidermal necrosis**
- Bullous dermatitis artifacta (self-inflicted): subepidermal blister w/ epidermal necrosis
- Bullous CSVV: LCV with massive subepidermal edema and epidermal necrosis
- Coma blisters: cell poor subepidermal blister with **necrosis of eccrine glands**
- Edema blisters: marked epidermal spongiosis w/ dilated dermal BVs and mild infiltrate
- Bullous drug eruption: variable presentation but will contain a blister w/ eosinophils
- Bullous diabeticorum, delayed postburn/postgraft blisters, fracture blisters and suckling blisters also fit into the category of pauci-inflammatory subepidermal split

Neoplastic

Squamous eddies ("Eddy is Tricky IF Very Irritated")

- Trichilemmoma: smooth lobules of glycogenated cells hanging down from the epidermis w/ a peripheral palisade and outlined by a thick eosinophilic basement membrane
  - IHC: $CD34^+$; PAS^+ (becomes negative if diastase is added due to ↑ glycogen)
- IFK (inverted follicular keratosis): endophytic lesion resembling an expanded hair follicle

- Verruca vulgaris: exophytic, papillomatosis, rounded parakeratosis, coarse hypergranulosis, koilocytes (vacuolated cells w/ hyperchromatic shriveled nuclei)
    - Myrmecia (palmoplantar wart; HPV-1): endophytic w/ **red** cytoplasmic inclusions
    - Verruca plana (flat wart): bird's eye cells (like koilocytes but lack shriveled nuclei)
    - Epidermodysplasia verruciformis: widespread flat warts w/ **blue cytoplasm**
    - Condyloma acuminata: SK-like (horn cysts/squamous eddies) but in genitals
- Irritated SK: type of SK (loose keratin, acanthosis, horn cysts) but w/ squamous eddies
    - Other types of SKs include pigmented (pigmented keratinocytes), hyperkeratotic (papillomatosis), inflamed (lymphocytes), **clonal (islands of small keratinocytes)**, Dowling-Degos disease (SK coming off hair follicles); refer to 6.1 for more on SKs

Clonal ("Clone Chriss BoSH")
- Clonal Bowens: clonal islands of atypical keratinocytes w/ pagetoid scatter
- Clonal SK: clonal islands of small keratinocytes w/ bland uniform nuclei
    - Regular SKs are too easy so this is the type of SK most likely to be tested
- Hidroacanthoma simplex (intraepidermal poroma): clonal islands of small eosinophilic acrosyringial keratinocytes but w/ small ducts in the center

Malignant tumors in the epidermis/Pagetoid spread ("Bowens SPITS")
- Bowens: atypical keratinocytes w/ clear cytoplasm which crush basal layer ("eye-liner sign"); abnormal keratinocyte maturation ("wind-blown appearance"); **parakeratosis**
    - AK only has atypical keratinocytes in the lower epidermis (not full thickness)
    - Keratin stains (p63, HMWK), p16 (- in AK), EMA and PAS (only clear cell Bowens)
- SebAceous carcinoma (ocular type): malignant epidermal sebaceous appearing as cells w/ scalloped nuceli and clear cytoplasm containing microvesicles; no parakeratosis
    - Adipophilin, Androgen receptor, EMA
- Paget's/EMPD: **large cells (some are signet ring) w/ ample bluish mucinous cytoplasm (glandular differentiation apparent)** spitting into the epidermis and crushing basal layer
    - Clue: apocrine glands, dermal blood vessels (gives the red "strawberry and cream" color clinically); no parakeratosis or solar elastosis (vs Bowens disease)
    - IHC: EMA, PAS (diastase resistant), CEA, CK7, gross cystic protein (GCDFP-15)
        - "EMA passed by and resisted a sea of 7 cocks, gross"
        - Disease extending from internal cancer is **CK20^{+}**, CK7^{-} and **GCDFP-15^{-}**
            - "But internally EMA wanted 20 cocks and did not find them gross"
- Spitz nevus: well-circumscribed lesion w/ large junctional nests that "rain down" and exhibit clefting +/- Kamino bodies (pink clumps of collagen 4) in the epidermis; Pagetoid spread of melanocytes and superficial mitoses can be seen
    - IHC: p16^{+}, Melan-A^{+}, S100^{+}, S100A6^{+} ("when a girl PMS'S she spitz at you")
        - p16 is positive in Spitz but lost in atypical Spitz/spitzoid melanomas
- Intraepidermal porocarcinoma: like poroma but w/ cytologic atypia and atypical mitoses
    - IHC: CEA, EMA, and PAS highlight ducts and intracytoplasmic lumina
- Thin melanomas (in-situ): **basal layer involved (so it is not crushed)**; look for pigment and **nests of atypical melanocytes** at all levels of epi (usually spits into the s. corneum)

    - Lentigo maligna: confluence of atypical melanocytes at the DEJ (less pagetoid)
    - IHC: melan-A, S100, SOX10

Stages of Paget's/extramammary Paget's (EMPD)

- Neoplastic cells in Paget's migrate to the epidermis from **lactiferous ducts** whereas in **EMPD they begin in the epi ("adenocarcinoma in situ")** and migrate down into adnexa
    - EMPD due to mets from GI/GU/cervix adenocarcinoma makes up < 20% of cases
- Early lesions (patch): single Paget cells in epidermis, especially lower half
- Fully developed (plaque): Paget cells nested in aggregates throughout entire epidermis
- Late lesions of EMPD due to GI/GU/cervix adenocarcinoma: aggregates of Paget cells in epidermis, dermis, superficial SQ and within vascular spaces

BCC look-a-likes ("Bacon Lettuce Tomato")

- BCC: blue islands w/ peripheral palisade and retraction artifact in a fibromyxoid stroma
    - Variants: superficial (blue buds come off epidermis), nodular (large round blue islands), micronodular (small round blue islands), infiltrative (spikey blue islands), morpheaform (thin infiltrative strands of BCC in sclerotic stroma), infundibulocystic (pink strands, blue buds, horn cysts), fibroepithelioma of pinkus (pink strands, blue buds, eccrine ducts), adenoid (clear spaces in islands)
    - IHC: **Ber-EP4; strong diffuse BCL-2 staining and androgen receptor**
- Lymphadenoma (adamantanoid trichoblastoma): tumor islands w/ 1-2 layers of basaloid cells peripherally and clear cells centrally
    - IHC: same as trichoblastoma
- Trichoblastoma: large blue islands in the dermis but there is no peripheral palisade, retraction is seen between collagen in stroma not between tumor and stroma, stroma is fibroblast-rich and papillary mesenchymal bodies may be seen in stroma
    - Trichoepithelioma is a variant which is smaller (trichoblastoma is bigger) and has cribriform (Swiss-cheese) basaloid nodules
    - IHC: CD10, CK20, CD34 (stroma), BCL-2 (stains periphery of islands)

SCC-like

- SCC: atypical keratinocytes invading the dermis w/ a jagged border
- Keratoacanthoma: invasive proliferation of glassy red keratinocytes, keratin-filled crater, neutrophilic microabscesses common w/ eosinophils common in surrounding infiltrate and trapping of elastic fibers common at the periphery of the tumor
- **Verrucous carcinoma**: well-differentiated glassy eosinophilic keratinocytes w/ blunt/rounded border that push into the underlying tissue in a bull-dozing fashion
    - Seen in anogenital region (Buschke-Lowenstein tumor), mouth (oral florid papillomatosis) and foot (epithelioma caniculatum); HPV-6, 11
- **Lymphoepithelioma-like carcinoma**: at scan it looks like a germinal center but on closer look there are keratin-positive atypical epithelial cells surrounded by inflammation

Papillomatosis ("An Erratic Church Spire Pattern): undulation of epi in a "church spire" pattern

- Acanthosis nigricans, Axillae/groin skin, Acroverruciformis of Hopf, Epidermal nevi, CARP, Stucco keratoses, Papillomatous/hyperkeratotic SK
- Differentiation between entities in this mnemonic based on pathology is not possible

Malignant spindle cells SLAM against the epidermis

- Spindled (sarcomatoid) SCC: biphasic tumor w/ thin epithelial elements surrounded by spindled cells; may have epidermal component
    - HMWK, CK5/6, p63 and p40 (newest and most specific)
- Leiomyosarcoma: no epidermal component; tumor is very pink; cells are cigar-shaped w/ perinuclear vacuoles ("glycogen-snack" for these muscle cells)
    - SMA, desmin
- Atypical fibroxanthoma: no epidermal component; cells are very pleomorphic, multinucleate and bizarre
    - Can be (+) for S100A6, **CD10** and SMA
- **Melanoma (desmoplastic)**: can have epidermal component (usually lentigo maligna pattern); **lymphoid aggregates is a great clue;** may or may not see melanin
    - **S100, Sox-10, p75**

Cysts: "A DUMB Cystic VEST"

- Apocrine/eccrine hydrocystoma: flat cuboidal epithelium w/ large empty cystic space
- Apocrine cystadenoma: has been used interchangeably w/ apocrine hydrocystoma but inside the cyst there are papillary projections ("mix between HPAP and hydrocystoma")
    - Ddx: tubular apocrine adenoma has many small cyst-like spaces w/ these papilla
- Auricular pseudocyst: cystic space in cartilage w/ no epithelial lining or chondritis
- Dermoid cyst: stratified squamous epithelium (SSE) w/ granular layer and adnexal structures (sebaceous/eccrine/apocrine glands and **hair follicles**); may contain red-tooth lining like steatocystomas; lamellar keratin is inside cyst cavity
- Digital mucus cyst/ganglion cyst: mucin inside pseudocyst (no epithelial lining)
- Urachal cyst: ectopic bladder epithelium (cuboidal or flat, atrophic epithelia)
    - Ddx: omphalomesenteric duct remnant (GI mucosa w/ goblet cells on path) vs umbilical granuloma (vascular proliferation on path)
- Median raphe cyst: dirty debris ("penis' are dirty") in cyst; genital skin features
- Mucocele: mucin inside pseudocyst (no epithelial lining) +/- adjacent salivary glands
- Branchial cleft: variable lining w/ surrounding **Lymphoid Aggregates (germinal centers)**
- Bronchogenic cyst: ciliated columnar epithelium; cartilage, smooth muscle, goblet cells
- Cutaneous Ciliated Cyst (lower legs) and Ciliated Cysts of vulva: Columnar Ciliated epithelium +/- dirty debris in cyst ("vulvas are dirty")
- Cutaneous endometriosis ("GIRL"): cyst-like Glandular spaces, Iron, RBCs, Loose concentric fibromyxoid stroma
- Verrucous cyst: looks like a pilar cyst but w/ papillomatosis, hypergranulosis, koilocytes
- Vellus hair cyst: looks like epidermoid cyst but w/ multiple vellus hairs in cyst cavity
- Ear pits: lined by stratified squamous epithelium (SSE) w/ a granular layer
- Epidermoid cyst: cystic cavity filled w/ **laminated keratin** lined by SSE w/ granular layer

- Steatocystoma: lined by SSE w/ no granular layer and thin bright pink corrugated ("Shark-tooth") cuticle w/ Sebaceous glands in wall (no hair follicles in cyst wall or lamellar keratin in cyst cavity like in dermoid cyst)
- Trichilemmal: SSE w/ no granular layer and dense pink homogenized keratin +/- calcium
- Thyroglossal duct cyst: variable lining w/ bright pink thyroid follicles in the wall

Melanocytic neoplasms

- Ephelides/CALMs: ↑ basilar keratinocyte hyperpigmentation w/o ↑ melanocyte density
- Solar lentigo: long rete ridges w/ basilar keratinocyte hyperpigmentation ("dirty socks")
- Lentigo simplex: ↑ basilar keratinocyte hyperpigmentation; mild ↑ melanocyte density
- Mucosal melanotic macule: acanthosis with mild basilar hyperpigmentation and mild ↑ in melanocyte density
- Becker's nevus: ↑ basal melanocytes w/ a smooth muscle hamartoma in the dermis
- Dermal melanocytosis: dendritic melanocytes oriented east to west in lower dermis
- Nevus of Ito/Ota: dendritic melanocytes oriented east to west scattered between collagen bundles in the upper 1/3rd of the dermis; no sclerotic stroma
- Common blue nevus: small: wedge-shaped lesion in upper 2/3rd of dermis composed entirely of long dendritic melanocytes embedded in a sclerotic stroma
- Cellular blue nevus: proliferation of both epithelioid (plump) and dendritic melanocytes in a sclerotic stroma that bulges into the SQ ("big black ball in the fat")
- Epithelioid blue nevus: proliferation composed entirely of epithelioid (plump) melanocytes but no sclerotic stroma (looks like animal-type melanoma w/ no atypia)
- Deep penetrating nevus (DPN): very similar to cellular blue nevus (both have epithelioid melanocytes diving deep into SQ) but **DPN'S Dive Down (more wedge-shape; classically tracking down adnexa/nerves), more Pigmented, more Nested and are more Spitzoid (cells have more abundant cytoplasm) compared to a CBN**
- Spitz nevus ("shpitz"): symmetric and circumscribed lesion w/ large junctional nests of epithelioid melanocytes Hanging down from a Hyperplastic epidermis; overlying Hyperkeratosis and Hypergranulation; **kamino bodies** within epidermis
- Pigmented spindle cell nevus of Reed: like spitz but more superficial (superficial dermis), more pigmented (more melanophages) and melanocytes are spindled (vs epithelioid)
- BAP-1 inactivated nevus: combined nevi composed of conventional nevus (w/ preserved BAP-1 expression) and zone of large spitzoid cells w/ amphophilic cytoplasm (BAP loss)
- Recurrent melanocytic nevus: atypical junctional melanocytic proliferation (resembling MIS) located above a dermal scar; benign dermal nevus seen below/next to dermal scar
- Balloon cell nevus: > 50% dermal melanocytes are "balloon cells" (large pale melanocytes w/ vacuolated cytoplasm); always can find conventional nevus somewhere
- Halo nevus: bland nevus w/ lymphocytes intertwined w/ melanocytes ("cocktail party")
- Congenital melanocytic nevus: symmetrical broad proliferation of melanocytes in the dermis w/ maturation, splaying between collagen bundles and permeation of erector pilli muscles, blood vessels, adnexal structures and nerves (this is not worrisome)

- Acquired melanocytic nevus: symmetrical and circumscribed junctional, compound or intradermal proliferations of nested epithelioid nevoid cells which show maturation w/ depth (cells are more spindled/neuroid and singly spaced in the deep dermis)
- Dysplastic (Clark's) nevus: asymmetry, atypical nests at the junction extend > 2 rete ridges beyond dermal component, "bridging" of junctional nests, papillary dermal fibrosis, cytologic atypia (graded based on degree of architectural and cytologic atypia)
- Melanoma: asymmetry, poor circumscription, irregular nests w/ discohesion, confluent melanocytes along DEJ, pagetoid growth, epidermal consumption w/ extension down adnexa, lack of maturation of dermal component (large nests at base), cytologic atypia w/ prominent red nucleoli (vs lilac-colored nucleoli in spitz nevi)

Melanoma variants

- Melanoma in situ (MIS): broad lesion, nests vary in size and shape and are found at all levels of the epidermis and extend down adnexa
- Lentigo maligna: form of MIS on sun-damaged skin; confluence of atypical melanocytes at the DEJ (junctional growth w/o Pagetoid spread)
- Lentigo maligna melanoma: lentigo maligna w/ a vertical growth phase
- Superficial spreading melanoma: broad lesion w/ buckshot scatter of atypical melanocytes in the epidermis, nests are not confined to tips and sides of rete (vs dysplastic nevus) and are not evenly spaced
- Nodular melanoma: may be symmetrical, no in situ component, dermal component consists of a cohesive nodule of tumor cells with pushing border, deep mitoses and necrosis; plasma cells common in center and lymphs common at base ("riot police")
- Regressing melanoma: effaced rete pattern, dermal fibrosis and lichenoid inflammation w/ abundant melanophages
- Desmoplastic melanoma: elongated spindle cells (long fascicles of parallel nuclei) and abundant fibrous stroma; foci of **lymphoid aggregates** is a useful clue
    - Tumors w/ ↑ cellularity and ↓ stroma best classified as spindle cell melanoma
    - Ddx: nodular fasciitis also has large pleomorphic fibroblasts and lymph aggregates but the stroma is loose and myxoid ("tissue culture-like") w/ RBCs

Benign adnexal neoplasms

- Eccrine poromas: endophytic proliferation of epidermal poroid cells (small red cuboidal cells) w/ variably spaced sweat ducts in the epidermis and a highly vascularized stroma
    - Poroma variants
        - Hidroacanthoma simplex: nests of small poroid cells only in the epidermis
        - Dermal duct tumor: dermal nodules of poroid cells; no connection to epi
- Hidradenoma ("SPoCK aka eyebrows are high"): Squamoid cells + Poroid cells + Clear cells (any of these 3 cell types may dominate) + Cystic degeneration + Keloidal collagen
- Syringoma: tadpole or comma-shaped sweat ducts (amorphous sweat within lumen) in a sclerotic stroma confined to the upper half of dermis (benign superficial tumor)
- Mixed tumor (chondroid syringoma): circumscribed tumor made of glandular structures in a Myxoid stroma; branching (apocrine) and small tubular (eccrine) types exist

- Spiradenoma: large "blue ball" in the dermis w/ cystically dilated ducts and PAS+ hyaline droplets (3 types of cells on high power: large pale + small blue + lymphocytes)
- Cylindroma: jigsaw pattern of tumor lobules encircled by thick hyalinized BMZ material
- Syringocystadenoma papilliferum: papillary glandular proliferation w/ decapitation secretions which opens to Skin Surface ("looks like you can Slide in"); many plasma cells
- Hidradenoma papilliferum: papillary glandular proliferation w/ decapitation secretions "hiding" in the dermis (lacks epidermal connection vs SPAP) w/ a "maze-like" architecture ("hide in the maze")
    - Ddx: nipple adenoma (looks like an HPAP but it is on the nipple), apocrine cystadenoma (mix b/w hidrocystoma and HPAP w/ papillary fronds budding into the lumen of the cyst), tubular apocrine adenoma ("small cystic mazes") and aggressive digital papillary adenocarcinoma (consider if HPAP-like lesion on digit)
- Papillary eccrine adenoma (PEA) and tubular apocrine adenoma (TAA): proliferation of small round sweat ducts w/ Papillary projections in lumen
    - "Whereas HPAP is one huge maze, PEA/TAA looks like many tiny mazes"
- Syringofibroadenoma: long strands of sweat-duct epithelium that hang down from epi
- Hidrocystoma: cyst lined by cuboidal or columnar cells +/- decapitation secretions

Aggressive and Malignant adnexal neoplasms

- Aggressive digital papillary adenocarcinoma: looks like SPAP/HPAP due to numerous papillary projections but it is deeply infiltrative w/ ↑ mitoses and atypia
- Adenoid cystic (S sound) carcinoma: cribriform ("Swiss-cheese") appearance of basaloid tumor nodules with mucin in the cribriform lumens extending deep into the SQ fat; PNI
- Mucinous carcinoma: blue tumor cells in sea of mucin ("blue islands in a sea of snot")
- Microcystic adnexal carcinoma (MAC): deeply infiltrative basaloid proliferation w/ Multi-lineage differentiation (horn cysts and sweat ducts), PNI and lymphoid aggregates

Follicular neoplasms ("There is hair on my Filthy Toilet Paper")

- Fibrofolliculoma: central follicle w/ numerous thin strands of immature (**no mature follicle/hair shafts**) follicular epithelium radiating from it; loose fibromyxoid stroma
- Trichofolliculoma: dilated central follicle w/ smaller but still **fully formed vellus follicles radiating from it** ("mother and her babies")
- Trichoepithelioma: well-Circumscribed tumor of basaloid Cribriform nodules, numerous horn Cysts, Calcification and papillary mesenchymal bodies in a pink Collagenous stroma
    - Desmoplastic TE: Cords of blue cells in Collagenous stroma, Calcium, horn Cysts
- Trichilemmoma: circumscribed proliferation of clear outer root sheath cells (contains glycogen) that hangs down from the epidermis (broad connection to epi) and has a peripheral palisade w/ hyalinized BMZ and a warty surface
    - Desmoplastic trichilemmoma: **often mistaken for invasive SCC** due to pseudo-infiltrative epithelial strands but you see a classic trichilemmoma embedded in a sclerotic stroma off to the periphery
- Trichoadenoma: dermal proliferation of keratinizing milia-like cysts in a sclerotic stroma (cysts are often in pairs resembling eyeglasses)

- Tumor of the follicular infundibulum: downward epidermal proliferation of clear cells that run parallel to the epidermis (clinically and histologically may mimic superficial BCC)
- Pilar sheath acanthoma: similar to trichofolliculoma but babies radiating from central dilated pore are acanthotic ("fat fingers radiating from dilated pore")
- Pilomatricoma: cystic architecture w/ "rolls and scrolls", blue matrical cells which have an abrupt transition to anucleate shadow cells
- Proliferating pilar tumor: same "rolls and scrolls" architecture as pilomatricoma but has dense keratin instead of shadow cells and lacks basaloid matrical cells

Sebaceous neoplasms

- Nevus sebaceous of Jadassohn: "broad, bald (no terminal hairs), bumpy (papillomatous) and bubbly (↑ sebaceous and apocrine glands)"
- Sebaceous hyperplasia: enlarged sebaceous glands w/ normal architecture (central mature white sebocytes w/ thin peripheral rim of immature blue seboblasts)
- Sebaceous adenoma: **superficial** proliferation w/ central mature sebocytes w/ thicker rim of blue seboblasts ("but still more white than blue")
  - "In sebaceous adenoma we add some blue seboblasts to the periphery"
- Sebaceous epithelioma (sebaceoma; "sa-blue-ceoma"): purely intradermal mass comprised of **more blue seboblasts than white sebocytes ("more blue than white")**
- Sebaceous carcinoma
  - Ocular: intraepithelial spread of malignant cells in a pagetoid pattern w/ minimal sebaceous differentiation (need stains to identify)
    - IHC (Seb-A-ceous): EMA, AR, Adipophilin, Factor XIIIa (AC-1A1)
  - Extraocular: infiltrative sebaceous basaloid proliferation ("may have more white than blue or more blue than white") w/ atypia, mitoses and tumor necrosis

Paisley tie: all show tadpole shaped islands in a red sclerotic stroma (**Tied** up think BDSM)

- BCC (morpheaform): clefts Between epithelium and stroma; no sweat ducts
  - IHC: (+) for BCL2 diffusely and Ber-EP4
- DTE (think Cs): Clefts within stroma, **Central dell (umbilication), horn Cysts (follicular differentiation)**, Calcifications on Cheek of Child (young female); **no sweat ducts**
  IHC: (+) for CK20, CD10 in stroma, CD34, PHLDA, BCL-2 only stains periphery
- Syringoma: Small benign tumor (never deep) of eccrine gland w/ Sweat in ducts
- MAC: deep infiltrative pattern w/ horn cysts, **perineural extension and lymphoid aggregates w/ bi-linear differentiation** (follicular horn cysts + sweat ducts; "MAC is bi")

Downward epidermal proliferations ("Peanut Butter STICKS")

- Poroma: epidermal proliferation of small monotonous keratinocytes w/ ducts
- BCC (superficial, fibroepithelioma of Pinkus and infundibulocystic): blue islands, peripheral palisade, retraction artifact and fibromyxoid stroma
  - Basaloid follicular hamartoma looks identical to infundibulocystic BCC on path
- Seborrheic keratosis: broad sheets of small polygonal keratinocytes w/ intervening horn cysts; loose lamellar "shredded wheat" keratin

- Tumor of the follicular infundibulum: acanthoma composed of pale pink cells forming anastomosing narrow strands that run parallel (just underneath) to the epidermis
- Inverted follicular keratosis: endophytic lesion w/ squamous eddies
- Clear cell acanthoma: distinct transition between normal epidermis and acanthoma made of clear glycogenated (PAS+ diastase sensitive) cells w/ overlying parakeratosis
- Syringofibroadenoma of Mascaro: Slender downward ductal proliferation
    - "#1 site is on legs and it kind of looks like long slender legs"

Deep dermal tumors: "SANS SANG with a deep voice

- Spiradenoma: big blue ball in dermis composed of dark cells, light cells, lymphocytes, hyaline droplets and widely ectatic blood vessels
- Angiolipoma: proliferation of capillary sized blood vessels and lipocytes
- Nodular hidradenoma ("SPoCK"): Squamoid cells + Poroid cells + Clear cells (any of these 3 cell types may dominate) + Cystic degeneration + Keloidal collagen
    - Clear cell hidradenoma mimics metastatic RCC but in **hidradenoma you see ductal structures** and in metastatic **RCC you see RBCs and vascular spaces**
- Spindle cell lipoma: "I call it a ropey myxoid spindle cell lipoma to remind me that you see ropey collagen in a myxoid stroma with surrounding spindle cells and lipocytes"
    - Pleomorphic lipoma looks like a Spindle cell lipoma but it has floret giant cells
- Schwannoma: deep encapsulated tumor w/ hypercellular Antoni A areas containing Verocay bodies (looks like leaves raked into piles) and hypocellular Antoni B areas
- Angioleiomyoma: deep nodule of smooth muscle with round slit-like vascular spaces
- NoduLAR fasciitis: Loose myxoid stroma, Large pleomorphic fibroblasts, Aggregates of inflammatory cells, RBC extravasation and Red thick collagen bundles in older lesions
- Giant cell tumor of the tendon sheath: deep tumor composed of foamy macrophages, plump fibroblasts, hemosiderin in a Fibrous pseudo-capsule; osteoclast-like Giant cells

Fibrohistiocytic neoplasms: "JACK F***ING DIES and the Police Department is PAGED Fast"

- Benign (JACK F***ING DIES)
    - Juvenile hyaline fibromatosis: pink amorphous hyaline material in dermis
    - Angiofibroma: dermal proliferation of stellate factor 13a+ fibroblasts w/ concentric perivascular fibrosis and ectatic thin-walled vessels
    - Multinucleate cell Angiohistiocytoma: looks like an angiofibroma but contains **bizarre, angulated Multinucleated giant cells**
    - Adult myofibroma: fibrovascular proliferation w/ myofibroblasts, staghorn vessels and large nodules w/ a blue hue ("Adult myofi-blue-ma")
    - Acquired digital fibrokeratoma: compact hyperkeratosis and acanthosis w/ vertical collagen bundles in the dermis; no nerves (vs supernumerary digit)
    - Calcifying aponeurotic fibroma: plump epithelioid fibroblasts palisading around foci of **cartilage or calcification** +/- osteoclast-like giant cells
    - Keloid/scar: discussed elsewhere
    - Fibromatosis: proliferation of long fascicles of corkscrew-shaped myofibroblasts
    - Fibrous hamartoma of infancy: "This FFibrous ham is SAVy" → 3 tissue types

        - Fat cells (mature): S100+
        - Fibroblast cells consisting of plump spindle cells in Fascicles: Actin+
        - Myxoid Mesenchymal cells (immature): Vimentin+
    - Pleomorphic Fibroma: skin tag w/ hyperchromatic, bizarre, multinucleated cells
    - Infantile myofibromatosis: same path as in fibromatosis (above)
    - Nodular fasciitis: described above in deep dermal tumor mnemonic
    - Giant cell tumor of the tendon sheath: above in deep dermal tumor mnemonic
    - Dermatofibroma ("DF'S"): Dermal spindle cell proliferation w/ collagen trapping, thickened epidermis w/ Folliculo-sebaceous induction; when present, ringed lipidized Siderophages are pathognomonic
        - IHC (DF'S): D2-40, Factor 13a ("13A you're A OK"), Stromelysin-3
    - Dermatomyofibroma: spindle cells w/ east-west orientation that respect adnexa
    - Infantile digital fibroma entire ("intire") dermis is filled w/ intersecting fascicles of plump spindle cells which contain pink-red inclusion bodies
    - Elastofibroma dorsi: large fibrous tumor w/ beaded elastic fibers visible on VVG
    - Epithelioid cell histiocytoma: well-circumscribed dermal proliferation of epithelioid cells (spitz nevus-like) w/ epidermal collarette and dermal sclerosis
    - Sclerotic fibroma: acellular collagen bundles w/ van Gogh's Starry night pattern
- Borderline (Police Department)
    - Plexiform fibrohistiocytic tumor: Poorly demarcated dermal or SQ mass w/ multinodular Plexiform fascicles of fibrohistiocytic spindle cells +/- giant cells
    - Desmoid tumor (aggressive fibromatosis; "ABCDEFG"): Aggressive fibromatosis, Bland wavy nuclei on path, β-Catenin mutation, Desmoid tumor, Deep infiltration, Fibroblastic proliferation, a/w Gardner's syndrome
- Malignant (PAGED Fast)
    - Pleomorphic undifferentiated sarcoma: similar to AFX but more deeply located
    - Pleomorphic dermal sarcoma: similar to AFX but more deeply located
    - AFX: fairly well-circumscribed but overtly malignant dermal proliferation of anaplastic, pleomorphic cells w/ high mitotic rate and wildly atypical mitoses which fills the dermis but does not invade into the fat (if it does then call it a PDS or superficial UPS/PUS); composed of 4 cell types ("call it atypical fibro-histiocytic xanthoma") → bizarre atypical multinucleated giant cells, spindle cells, histiocyte-like cells, xanthomatous cells
        - IHC (non-specific): **CD10**, CD68
    - Giant cell fibroblastoma: multinucleated giant cells lining pseudovascular spaces
    - Epithelioid sarcoma: transition from epithelioid to plump pleomorphic spindle cells **w/ central necrosis** and a surrounding lymphohistiocytic infiltrate
        - "Looks like a palisaded granuloma"
    - DFSP: Densely hypercellular Dermal proliferation infiltrating **Fat (creates a "honeycomb" pattern)** consisting of Spindled cells w/ a **Storiform Pattern**
        - IHC: **CD34+** (D2-40, Factor 13a and Stromelysin-3 negative)
        - 15% of DFSP's may undergo fibrosarcomatous degeneration → ↑ atypia, ↑ cellularity, ↑ mitoses w/ **"herringbone" pattern**, loss of CD34 staining
    - Fibrosarcoma: densely hypercellular tumor w/ herringbone ("Fir tree") pattern

Vascular neoplasms

- Infantile hemangioma: lobular proliferations of GLUT-1 (+) dermal capillaries lined by plump endothelium (transiently positive for LYVE-1 but lacks other lymphatic markers)
- Intravascular papillary endothelial hyperplasia (IPEH): Intraluminal Papillary Proliferation of Endothelial cells Entirely contained within a Hyperplastic thrombosed vessel (i.e. vein)
- Reactive Angioendotheliomatosis: proliferation of grouped capillaries in dermal BVs
- Angiokeratoma: superficial dilated vessels a/w an acanthotic epidermis ("**bloody SK**")
- Targetoid hemosiderotic lymphatic malformation: **superficial dilated BVs** with endothelial cells that protrude (hobnail) into the lumen in the upper dermis; the **lower dermis has hemosiderin and BVs get thin and slit like**; vessels stain w/ lymph markers
- Verrucous venulocapillary malformation (verrucous hemangioma): dilated capillaries and venules that stain with CD31 and CD34
- Pyogenic granuloma: circumscribed, exophytic capillary proliferation in a lobular pattern
- Tufted angioma ("tough LUCK"): tightly packed Lobules (Ultrastructural studies show Weibel-Palade bodies) of **Capillaries in a "Cannonball" pattern in Clusters** with an empty appearing Crescent surrounding each lobule (Clefting) which represent dilated lymphatic Channels; CD61 confirms platelet trapping in cases of Kasabach-merritt
- Glomeruloid hemangioma: RBCs + capillaries in dilated vessels (resembles glomerulus)
- Cherry angioma: dilated capillaries within the papillary dermis
- Microvenular hemangioma: small flat monomorphic blood vessels that fill the dermis
- Epithelioid hemangioma: Enlarged thickened vessels lined by Enlarged plump Epithelioid Endothelial cells which have intracytoplasmic vacuoles; **Eosinophilic inflammation**
- Congenital Cutaneous Angiomatosis w/ thrombocytopenia: thin-walled vessels form intraluminal papillary projections; Coexpresses BV (CD34) and lymph (LYVE-1) markers
- Sinusoidal hemangioma: dilated intersecting channels in a Sieve-like/Sinosoidal pattern
- Spindle cell hemangioma ("spindle MIKE"): Mixture of Cavernous ("Kavernous") spaces and more cellular areas containing spindle cells w/ aggregates of vacuolated Epithelioid cells; resembles "hemorrhagic lung" with alveolar spaces at scanning power
    - "Spindle MIKE vapes all day which gave him a hemorrhagic lung"
- Glomus tumor: rows of round dark nuclei w little cytoplasm ("String of black pearls") which Surround Subtle vascular Spaces (vs glomangioma which has prominent vessels)
- Acroangiodermatitis (pseudo-Kaposi sarcoma): proliferation of thick-walled capillaries in a lobular pattern, RBC extravasation, hemosiderin, dermal fibrosis, PV lymphs; plasma cells are also found and may resemble Kaposi's but there is a lack of atypical cells
- Kaposiform hemangioendothelioma: nodules of spindle cells w/ slit-like lumina containing RBCs (like nodular Kaposi's)
    - CD61 confirms platelet trapping in cases of Kasabach-merritt
    - Unlike infantile hemangiomas it is **GLUT-1 negative**
- Kaposi Sarcoma
    - Patch/Plaque stage: subtle or more pronounced (in plaque stage) infiltration of Lymphatic-like vessels w/ Pale appearing stroma (looks understained) + Hemosiderin + Plasma cells + Promontory sign + Erythrocyte Extravasation

- Nodular stage: nodules composed of fascicles of parallel spindle cells w/ slit-like lumina containing RBCs; cells are never as atypical as those seen in angiosarcoma
  - Expression of latency-associated nuclear antigen (**LANA-1**) of HHV-8
  - Mixed lymphatic/blood vascular phenotype (CD34, VEGFR-3, LYVE-1)
- Angiosarcoma ("SARC"): large, pleomorphic, crowded endothelial cells that **Stack on top of each other ("multilayered/piled-on")** and dissect between collagen bundles forming Anastomosing vascular networks + RBC extravasation + Cytoplasmic vacuoles
  - IHC ("Angi sarcs and FUCs MY External Genitalia"): FLI-1$^+$, Ulex europaeus lectin$^+$, CD31$^+$, CD34$^+$, c-MYC amplification (radiation-induced), ERG$^+$ (most sens/specific)

Neural neoplasms (M-T)

- **Merkel cell carcinoma**: tightly packed and atypical small blue cells w/ ↑N:C Ratio which exhibit nuclear molding, numerous mitoses and speckled salt and pepper chromatin
  - **IHC: perinuclear dot pattern w/ CK20**
- MPNST: densely cellular (hypercellularity is more characteristic than atypia) proliferation of atypical intersecting spindle cells ("Herringbone pattern") w/ large areas of necrosis; low power shows marbling due to alternating hypo- and hypercellular areas
- Neuroblastoma: small blue cells which form a dermal/SQ nodule
- Neurofibroma: loose arrangement of haphazard small wavy (S-shaped) spindle cells in pale myxoid stroma; numerous mast cells
- Neurothekeoma: multiple (plexiform) myxoid lobules containing bland spindle cells
- Cellular Neurothekeoma: multiple nests/lobules of epithelioid or spindled cells
- Nasal glioma: astrocytes w/ round vesicular nuclei in a loose neurofibrillary stroma
- Perineuroma: circumscribed tumor of spindle cells w/ onion skin-like whorls
- PEN: **superficial** dermal tumor w/ a **thin delicate capsule surrounding fascicles of spindle cells separated by clefts** (recapitulates normal nerve; Schwann cells:axons = 1:1)
  - May see **adjacent sebaceous glands and vellus hair follicles** typical of facial skin
- "Qutaneous" ganglioneuroma: localized collection of wavy Schwann cells admixed w/ large polygonal ganglion cells (big cells hanging on in the superficial dermis in "gangs")
- gRanular cell tumor: sheets of large polygonal cells w/ granular cytoplasm due to pink cytoplasmic inclusions (**Pustulo-Ovoid bodies)** w/ pseudoepitheliomatous hyperplasia
- Schwannoma (neurilemmoma): deep encapsulated proliferation of plump wavy Schwann cells w/ hypercellular (Antoni A) areas containing Verocay bodies (palisaded nuclei around acellular pink material) and hypocellular (Antoni B) myxoid areas
- Supernumerary digit: acral (thick stratum corneum) papule w/ nerve bundles
- Traumatic neuroma: like PEN, there is a haphazard proliferation of small nerve fascicles (resembles normal nerve in that Schwann cells and axons are in 1:1 ratio) separated by clefts however traumatic neuroma also has a background of scar and is not on the face

Neoplasms of fat ("fatty FLAPS")

- Fibrolipoma: large collagen bundles ("fibro") + mature fat ("lipoma")
- Lipoma: well-circumscribed tumor of mature lipocytes
- Angiolipoma: thrombosed capillary sized blood vessels ("angio") + mature fat ("lipoma")

- Angiomyolipoma: smooth muscle coming off vessel walls ("Angiomyo") + fat ("lipoma")
- Pleomorphic lipoma: **multinucleate floret cells** (overlapping nuclei arranged at the periphery like petals of a flower) + myxoid areas + ropey collagen bundles + mature fat
- Spindle cell lipoma (like pleomorphic lipoma w/o the floret cells): myxoid areas + ropey collagen bundles + mature fat ("I call it a ropey myxoid spindle cell lipoma to remind me of the ropey collagen in a myxoid stroma with surrounding spindle cells and lipocytes")

Smooth muscle neoplasms
- Pilar leiomyoma (piloleiomyoma): circumscribed but non-encapsulated (ill-defined) dermal nodule of intersecting bundles of smooth muscle cells (spindle cells w/ bright Pink cytoplasm and cigar-shaped nuclei w/ Perinuclear vacuole containing glycogen)
- Genital leiomyoma: similar to pilar leiomyoma but often larger and less circumscribed
- Angioleiomyoma: deep SQ nodule of smooth muscle containing slit-like vascular spaces
- Leiomyosarcoma: like leiomyoma but ↑ cellularity, atypia, pleomorphism and mitoses
- Smooth muscle hamartoma: numerous haphazard bundles of smooth muscle in the dermis (bundles are fewer and more discrete than those seen in leiomyoma)

CTCLs w/o a cytotoxic phenotype
- Mycosis fungoides: lichenoid band w/ epidermotropism of lymphs which may be in clusters (Pautrier's microabscesses) and lined up at the DEJ
- Lyp: discussed below
- Primary cutaneous anaplastic large cell lymphoma (ALCL): diffuse sheets of large, atypical $CD4^+$ > $CD30^+$ cells in the dermis +/- Hallmark (horseshoe) cells

Cytotoxic CTCLs: "EGS are fatty (located in fat) and toxic (cytotoxic: granzyme B+, TIA-1+)"
- Extranodal NK/T-cell lymphoma, nasal type: dense lymphoid infiltrate in SQ and dermis with angiodestruction; EBV+, CD2+, CD3epsilon+ and CD56+
- Gamma-delta T-cell lymphoma: **infiltrate in SQ, dermis and epidermis**; CD3+, CD56+
- Subcutaneous panniculitis-like T-cell lymphoma: **infiltrate is in SQ in a lace-like pattern with rimming of adipocytes (no dermis/epi involvement)**; CD8+
- Primary cutaneous aggressive epidermotropic CD8+ cytotoxic T-cell lymphoma has a cytotoxic phenotype but is superficial and epidermotropic (looks like MF)

Lymphomatoid papulosis subtypes ("My Ana is aggressive and a little Extra")
- Type A (typical LyP): wedge-shaped dermal infiltrate w/ lymphocytes, neutrophils, **eosinophils** and large $CD30^+$ (Ki-1) Reed-Sternberg-like cells
  - Resembles PLEVA but PLEVA never has large $CD30^+$ cells or eosinophils
- Type B (Mycosis fungoides-like): epidermotropic infiltrate of CD30 negative cells
- Type C (Anaplastic large cell lymphoma-like): diffuse sheets of large cells in the dermis
- Type D (primary cutaneous aggressive epidermotropic $CD8^+$ cytotoxic T-cell lymphoma-like): markedly epidermotropic $CD8^+/CD30^+/TIA\text{-}1^+/\text{granzyme } B^+$ cells
- Type E (extranodal NK/T-cell lymphoma-like): angioinvasive $CD30^+/\beta\ F\text{-}1^+$ cells

Primary cutaneous B-cell neoplasms ("Full MIL")

- As the mnemonic progresses, the entities go from more common to less common, the MC location goes from superior to inferior and the prognosis gets worse
- Follicle center cell lymphoma: irregular neoplastic follicles which lack follicle polarity, lacks a well-defined mantle zone, lacks tingible body macrophages and lacks t(14;18) translocation (vs systemic follicular lymphoma); **BCL-6^{+}**/BCL-2^{-}
- Marginal zone lymphoma: "inverse" pattern of central darker benign reactive lymphocytes w/ surrounding neoplastic Monocytoid B-cells (marginal zone cells) and numerous plasma cells, some w/ Dutcher bodies; BCL-6^{-}/**BCL-2^{+}**
- Intravascular B-cell lymphoma: large, atypical B-cells within vessels; CD20^{+}, CD79a^{+}
- Diffuse Large B-cell lymphoma: dense infiltrate of Large atypical cells; **MUM^{+}**/BCL2^{+}/BCL6^{+}/FOXP1^{+} ("Your MUM is quite large")

Connective tissue/depositional

Occlusive/thrombotic vascular disease ("SCLE")

- Stasis change: **cannon-ball tufting of vessels in the superficial dermis**, RBC extravasation, dermal hemosiderin +/- thrombi of small vessels
    - **Acroangiodermatitis (pseudo-kaposi) is clinically like Kaposi but stasis on path**
- Coagulopathy (APLS, factor V Leiden, purpura fulminans etc.): thrombi in vessels
- Cryoglobulinemia (type 1): **pink jelly** material **occluding vessel lumen**, RBC extravasation
- Livedoid vasculopathy (atrophie blanche): **hyalinized vessel walls ("pink crayon")** w/ thrombi and stasis change (cannon-ball tufting of superficial vessels, hemosiderin, RBCs)
- Cholesterol Embolization: arteriole containing cholesterol clefts

Square punch/dermal sclerosis ("Nice Sclerotic Cutis Can Make Good Square Biopsies")

- Necrobiosis lipoidica: horizontal acellular, pale degenerated collagen alternating w/ horizontal layers of granuloma ("layered cake"); plasma cells +/- cholesterol clefts
- Nephrogenic systemic fibrosis (NSF): ↑ CD34+ dermal fibroblasts, ↑ dermal mucin
    - Can be indistinguishable from from scleromyxedema but in NSF the process is **deeper**, denser and more cellular and less mucinous than in scleromyxedema
        - May extend into adipose, fascia and muscle (need deep incisional biopsy)
        - May see osseous metaplasia ("lollipop bodies")
- Scleromyxedema: ↑ CD34+ dermal fibroblasts ("sclero"), ↑ mucin ("myxedema")
- Scleredema: thickened dermis w/ widened spaces between collagen bundles (more evident in the deep dermis) filled w/ mucin (can be subtle and mimics normal back skin)
- Stiff skin syndrome: **Sclerosis of the fascia** but unlike deep morphea, there is **no inflammation at the dermal/SQ junction, and it does not trap adnexal structures**
- Connective tissue nevus: thick dermis due to excessive collagen and/or elastic tissue
- Chronic radiation dermatitis: papillary dermal pallor **w/o vacuolar change or a lymphoid band (vs LS&A)**, large stellate fibroblasts, loss of adnexa and superficial dilated vessels

- Morphea/scleroderma: **thick, closely packed (hyalinized) collagen bundles**, loss of adventitial fat resulting in **"trapped" eccrine glands**, ↓ CD34+ fibroblasts in the dermis, **lymphoplasmacytic infiltrate at dermal-SQ junction (can mimic panniculitis)**
- GVHD (sclerodermoid): dermis is thickened and sclerotic w/ no adnexal structures
- Scar: fibroblasts w/ east-west orientation and blood vessels w/ north-south orientation
  - Hypertrophic scar: nodules or whorls of thickened collagen and fibroblasts
  - Keloid: hyalinized "bubble gum" collagen in a background of hypertrophic scar
- Back skin: thick dermis (extends deeper than other sites); **no mucin (vs scleredema)**

Other sclerotic conditions not in the square punch mnemonic

- Lichen sclerosis ("red, white, blue"): compact stratum corneum w/ follicular plugging ("red"), papillary dermal pallor ("white") overlying a deeper lymphoid band ("blue")
  - **Process is too superficial to cause a square punch biopsy**
  - LS&A can also occur after radiation and look like radiation dermatitis on path but **LS&A has a lymphoid band, subtle vacuolar changes and lacks large fibroblasts**
- Eosinophilic fasciitis: thick eosinophilic fascia due to fibrosis and hyalinized collagen
  - Process is too deep to make a square punch (it's more dumbbell shaped)

Perforating disorders ("Emergency Room PA")

- Elastosis perforans serpiginosa: tortuous (serpiginous) channel through an acanthotic epidermis w/ extrusion of pink (Eosinophilic) elastic fibers
  - Penicillamine-induced EPS shows "Bramble bush" elastic fibers (lateral buds)
- Reactive perforating collagenosis: broad (cup-shaped) vertical channel through an acanthotic epidermis w/ extrusion of basophilic collagen bundles
- Perforating calcific elastosis: keratotic plug w/ calcified elastic fibers (like PXE)
- Acquired perforating dermatoses (Kyrle's disease): most resembles RPC

Huge superficial mucin deposits ("Myxoid DFSP")

- Myxoid DFSP: like DFSP (bland spindled cells w/ storiform pattern) in a myxoid stroma
- Myxoid neurothekeoma (dermal nerve sheath myxoma): multinodular growth pattern w/ fibrous bands separating the nodules; nodules composed of cords of S100, GFAP and CD57-positive spindled or stellate cells in a background of abundant myxoid matrix
- Myxoid NF
- Mucocele: pseudocystic space containing mucin +/- adjacent salivary gland
- DM/tumid lupus: abundant mucin in the dermis w/ PV/PA lymphs; mild interface change
- Digital mucoid cyst: **dome-shaped papule containing a pool of mucin on acral skin**
- Focal cutaneous mucinosis: **resembles digital mucoid cyst but is not on acral skin**
- Spindle cell lipoma/pleomorphic lipoma: **bland CD34+ spindle cells**, mature **adipocytes**, **wiry/ropy collagen** in a myxoid stroma (pleomorphic lipoma has floret cells)
- Pretibial myxedema: **large amounts of mucin throughout the entire dermis (deep too) w/ attenuation of collagen fibers (reduced to thin wisps)**
- Scleredema is not listed since the mucin (between collagen bundles) is deep and subtle

Calcification

- Calciphlaxis: **calcified small vessels in the SQ fat** w/ overlying epidermal necrosis
- Subepidermal calcified nodule: **pseudoepitheliomatous hyperplasia** w/ calcium deposits
- Scrotal calcinosis: **calcium masses; smooth muscle and rugated epidermis** (genital skin)

Deposition of fissured amorphous pale pink dermal material ("The MALE is pale")

- Colloid Milium: **very fragmented/fissured**, pale pink material in the superficial dermis; atrophic epidermis and surrounding solar elastosis (vs Grenz zone in juvenile form)
    - Stains w/ amyloid stains (crystal violet, congo red, thioflavin t) except pagoda red
- Amyloid (nodular): large fissured pale pink nodules in the dermis; **many plasma cells**
    - Primary systemic amyloid: amorphous eosinophilic fissured material
    - Macular/lichen amyloid: small superficial dermal deposits (invisible dermatosis)
- Lipoid proteinosis: pale pink hyaline deposits (not as fissured) which tend to be arranged perpendicular to the epidermis and arrange around blood vessels, adnexal structures and eccrine glands (much deeper and more extensive than the others which corresponds to the clinically indurated skin seen)
    - Alcian blue+; PAS+ diastase resistant; usually negative for Amyloid stains
- Erythropoietic protoporphyria (EPP): PAS+ hyaline cuff (reduplicated basement membrane) **around superficial blood vessels** (post-capillary venules)
    - There will be no solar elastosis (vs colloid milium) because the pts avoid the sun
    - PCT has much smaller hyaline cuffs of MBZ material around superficial vessels
    - Cutaneous collagenous vasculopathy clinically looks like generalized essential telangiectasia but histologically looks like EPP (ectatic superficial blood vessels w/ PAS+ and diastase resistant hyalinization of blood vessel walls)

Disorders of skin appendages and panniculitides

Non-scarring alopecia (all have ↑ % of catagen/telogen hairs; "All tricky tracts tell the pattern")

- "All tricky tracts": huge ↑ (> 50%) of hairs in catagen/telogen phase
    - Alopecia areata: **peri-bulbar lymphs ("swarm of bees")**, lymphs/eos/melanin within fibrous tract remnants, follicular miniaturization (resembles vellus hairs)
        - Syphilitic alopecia may be identical to alopecia areata but w/ plasma cells
    - Trichotillomania: **trichomalacia + melanin in hair canal (pigment casts)**
    - Traction alopecia: resembles trichotillomania on path
- "Tell the pattern": mild ↑ (25-50%) of hairs in catagen/telogen phase
    - Telogen effluvium: normal hair density w/ mild ↑ of catagen/telogen hairs
    - Pattern (androgenetic) alopecia: **miniature** anagen **follicles** w/ variable diameter (**anisotrichosis**), normal hair density and ↑ sebaceous glands (↑ androgens)
        - The scalp biopsy looks like facial skin (↑ sebaceous glands & vellus hairs)
- All are non-inflammatory non-scarring alopecias except AA which is inflammatory
- Other non-inflammatory non-scarring alopecias include temporal triangular alopecia (no terminal hairs/only vellus hairs) and congenital atrichia w/ papules

- Other forms of inflammatory (perifollicular inflammation) non-scarring alopecia include alopecia mucinosa (mucin within follicular epithelium), folliculotropic MF (large atypical lymphs within follicular epithelium), tinea capitis/Majocchi's granuloma (fungal spores within hair shaft) and acne vulgaris (infundibulum filled w/ laminated keratin)
- Important to know that any non-scarring alopecia can become scarring if long-standing

Scarring/cicatricial alopecia (differentiate based on predominant cell type)

- Lymphocytes
  - CCCA: **Concentric lamellar fibrosis** + perifollicular lymphs (less dense than LPP)
    - Many dermatopathologists believe LPP and CCCA are on a spectrum
  - LPP (including FFA variant): **dense lichenoid interface at level of infundibulum**
    - Late stage: wedge-shaped scar (like wedge-shaped hypergranulosis in LP)
    - DIF: shaggy linear fibrin and cytoid bodies
  - Discoid lupus: dense lymphoid infiltrate (vacuolar or lichenoid) at level of isthmus (**deeper**), follicular hyperkeratosis, BMZ thickening, melanin pigment incontinence under DEJ, vertical columns of lymhs, PV/PA/PE lymphs, and mucin
    - Late stage: scar throughout entire dermis
    - DIF: continuous granular band of IgG/A/M and C3 at the follicular BMZ
- Neutrophils (2nd letter is an 'e')
  - Dissecting cellulitis: neutrophilic abscesses, sinus tracts and granulation tissue
    - Hidradenitis suppurativa looks like this but on axillary/genital skin
  - Folliculitis decalvans: dense neutrophilic infiltrate at level of infundibulum
- Mixed (lymphoplasmacytic and neutrophilic) infiltrate
  - Acne keloidalis nuchae: perifollicular inflammation, plasma cells (posterior neck location), hypertrophic scar (not keloid) w/ concentric lamellar fibrosis
- Epithelioid histiocytes
  - Acute Langerhans cell histiocytosis: perifollicular epithelioid histiocytes w/ kidney-shaped nuclei at level of infundibulum

Septal panniculitis (erythema nodosum)

- Septal panniculitis: thickened and inflamed septa w/ lipocytes similar in size and shape ("looks like Cheerios") vs "oil-in-water" appearance of fat cells in lobular panniculitis
- Septa: acute phase has neuts; chronic phase has granulomas (Miescher's granuloma)

Lobular panniculitis (just like scarring alopecia, differentiate based on cell type)

- Lymphocytes
  - Lupus panniculitis (lupus profundus): **hyalinized septae**, **nodular lymphoid aggregates** w/ plasma cells, overlying interface dermatitis and dermal mucin
    - Ddx: SQ panniculitis-like Lymphoma (both entities display a lobular lymph infiltrate w/ lymphs rimming fat cells but lupus panniculitis is more likely to show epidermal atrophy, interface changes w/ thickened BM, mucin, follicular plugging, hyaline lipomembraneous change, **B-cell follicles** and a mixed infiltrate of plasma cell & plasmacytoid dendritic cell aggregates

- Granulomatous (giant cells + histiocytes)
  - SQ fat necrosis of the newborn/post-steroid panniculitis/sclerema neonatorum: **fat crystallization** (radial crystals in fat); sclerema neonatorum lacks the infiltrate
  - Cytophagic histiocytic panniculitis: beanbag cells (giant cells engulf RBCs/WBCs)
- Eosinophils
  - Eosinophilic panniculitis: eosinophils in septum and lobule
- Neutrophils ("I'M A neutrophilic lobular panniculitis")
  - Infectious: mixed inflammation and necrosis of fat lobules (rule out with stains)
  - Myelodysplastic syndrome
  - Alpha$_1$-antitrypsin deficiency: neuts + liquefactive necrosis of septum and dermis
- Neutrophils w/ vasculitis
  - Nodular vasculitis/erythema induratum: LCV of all vessel sizes (in dermis and fat)
- Others
  - Pancreatic panniculitis: calcium soap outlining necrotic lipocytes (saponification)
  - Lipodermatosclerosis (stasis panniculitis): **lipomembraneous change ("frost on a window") w/ overlying stasis change** (nodular angioplasia and hemosiderin)
  - Cold panniculitis (deep perniosis): lobular fat necrosis w/ mixed infiltrate (lymphs, histiocytes, neuts) and overlying perniosis (papillary dermal edema)

Infectious

Bacterial rods or cocci in the stratum corneum ("PIE")

- Pitted keratolysis: pit in stratum corneum w/ G+ rods and cocci at base of dell; acral skin
- Impetigo: **clusters of G+ cocci and neutrophilic crust in stratum corneum**
- Erythrasma: G+ rods forming vertical filaments in stratum corneum (normal skin ddx)
- Ecthyma ("ulcerated impetigo"): **G+ cocci (staph colonies) at the base of an ulcer**
  - Don's confuse w/ ecthyma gangrenosum: G- bacteria around vessels (light blue haze) + lack of inflammation (no dark blue lymphs) + pink necrosis of vessels

PEH w/ neutrophilic intraepidermal pustule ("Here Come Big Green Leafy Veggies")

- Halogenoderma: pseudoepitheliomatous hyperplasia (PEH) + pus but **no organisms**
- Chromomycosis: **round copper-colored Medlar bodies ("Copper pennies")**
- Blastomycosis: organisms have an asymmetric refractile wall w/ dark Blue nucleus; if Budding is present it is Broad-Based ("looks like a snowman")
- Granuloma inguinale: organisms within histiocytes (Donovan bodies)
- Leishmaniasis: parasitized organisms at periphery of histiocytes (marquee sign)
- Pemphigus vegetans: PEH w/ **eosinophilic pustules;** no organisms

Parasitized histiocytes/organisms within histiocytes ("pH GIRL")

- Penicillium marneffei: looks like histoplasmosis on path but patient is from SE Asia
- Histoplasmosis: small uniform dots evenly spaces within histiocytes; no kinetoplasts
  - African histoplasmosis: large clear yeast evenly spaced within histiocytes
- Granuloma inguinale: organisms in histiocytes (Donovan bodies)

- Rhinoscleroma: **sheets of plasma cells**, Russell bodies (plasma cells filled with bright pink immunoglobin), Mikulicz cells (histiocytes with large round collections of bacilli)
- Leishmania: organisms at periphery of histiocytes (marquee sign) +/- kinetoplasts
- **Lepromatous leprosy: sheets of histiocytes containing globi (amphophilic collections of mycobacteria which stain w/ Fite stain); grenz zone**
    - Tuberculoid leprosy: epithelioid granulomas ("sausages") running east-west in dermis (tracks along nerves) and lacks globi (so Fite stain is negative)

Deep fungal infections not in PEH + pus or pH GIRL mnemonics
- Coccidioidomycosis: **large spherules** (much larger than surrounding cells) w/ gray lacy and granular cytoplasm; uniform size and shape; refractile wall +/- PEH + pus
- Cryptococcus: **yeast w/ clear gelatinous capsule** (around same size as surrounding cells)
    - Capsule stains red w/ mucicarmine; organisms stain w/ Fontana-Masson
    - Ddx: round yeast are same size and shape to blastomycosis but blasto has PEH + pus and yeast in crypto have a clear gelatinous capsule
- Paracoccidioides (South-American Blasto): **Mariner's wheel pattern** of budding yeast
- Lobomycosis: **organisms in chains** ("resembling a child's pop-beads")
- Sporotrichosis: palisaded neutrophilic abscess +/- **cigar-shaped yeasts** (not usually seen)
- Zygomycosis (Rhizopus): **Red 90º (Right-angle) non-septate hyphae** ("Red Ribbon")
- Hyalohyphomycosis (Aspergillosis/fusariosis): non-pigmented **septate** hyphae that **invade vessels, branch at a 45º angle (Acute)**; surrounded by bright pink immunoglobin (Splendore-Hoeppli phenomenon but not in BEAN mnemonic as it does not coat grains)
- Protothecosis: non-pigmented algae which demonstrate **soccer ball-like Murula**
- Rhinosporidiosis: **huge sporangia (even larger than spherules in coccidioidomycosis)** w/ many endospores; may resemble coccidiodes but **w/ central small nucleus**

Splendore-Hoeppli phenomenon (deposition of pink immunoglobin at the edge of grain/BEAN)
- Botryomycosis: grains composed of huge staph colonies; neuts/sinus tracts
- Eumycetoma: grains composed of fungal hyphae (may be pigmented); neuts/sinus tracts
    - Pigmented eumycetoma ("My Lousy Greasy Jeans are black"): Madurella mycetomatis, Leptosphaeria spp, M. grisea, Exophiala jeanselmei)
    - The MC eumycetoma is the non-pigmented Pseudoallescheria boydii (white boy)
- Actinomycetoma: grains composed of filamentous bacteria; neuts/sinus tracts
    - Nocardia brasiliensis (white grains; "white note cards"), Actinomyces israelii (yellow), Actinomadura pelletieri (red), Actinomadura madurae (pink/cream)
- Nocardia: MC type of actinomycetoma

Pigmented fungi ("the dark art of CPT code manipulation")
- Chromoblastomycosis: PEH + pus; round pigmented medlar bodies in the dermis
- Phaeohyphomycosis: long thin **pigmented hyphae in the dermis**
    - Pheaeomycotic cysts have these hyphae in cyst walls (MC organism is Exophiala jeanselmei and unlike phaeohyphomycosis, occurs in immunocompetent people)
- Tinea nigra: **pigmented hyphae in thickened (acral skin) stratum corneum**

HPV infections

- Verruca vulgaris: papillomatous exophytic lesion w/ hypergranulosis and koilocytes (keratinocytes w/ clear cytoplasm and shrunken "raisin-like" nuclei)
- Myrmecia: endophytic lesion w/ course red cytoplasmic inclusions
- Verruca plana: flatter lesion w/ rounded koilocytes ("bird-eyes") w/o shriveled nuclei
- Epidermodysplasia verruciformis (β HPV types): verruca plana w/ **βlue cytoplasm**
  - The truncal tinea versicolor-like macules also look like this
- Condyloma acuminatum: **path looks like SK in genitals** w/ vacuolated keratinocytes, round parakeratosis and course hypergranulosis
- **Bowenoid papulosis: can look like CA but w/ Bowenoid full thickness atypia**
- Verrucous cyst: cyst with wart-like lining
- Heck's disease: rounded parakeratosis (red) atop epithelial pallor (white)

HSV infections

- HSV 1-3: **Multinucleated keratinocytes w/ nuclear Molding and Margination of chromatin to the periphery** (looks like a cluster of grapes or egg-shells)
  - LCV more typically seen in HSV3 (VZV > chickenpox)
- HSV5 (CMV): large swollen endothelial cells w/ intranuclear inclusions ("owl's eye")

Poxvirus

- Smallpox: epidermal necrosis w/ reticular degeneration and eosinophilic inclusions
- Monkeypox: like smallpox but w/ Multinucleated keratinocytes
- Molluscum contagiosum: cup-shaped lesion w/ Henderson-Paterson bodies
- Parapox (Orf and Milker's nodules): → **"ENBREL"**
  - **Epidermal Necrosis (late lesions)**
  - **Bodies (eosinophilic cytoplasmic viral inclusion bodies in keratinocytes)**
  - **Reticular degeneration (intraepidermal vesiculation)**
  - **Eosinophilic viral inclusions are characteristic of parapox virus infection**
  - **Long slender epidermal acanthosis**

Other viruses

- Coxsackievirus (hand-foot-mouth): **reticular or ballooning degeneration and necrotic keratinocytes** affecting the upper half of the epidermis ("cox-half-ie"); often acral skin
- Gianotti-Crosti syndrome: non-specific but shows papillary dermal edema, spongiosis and a superficial perivascular lymphoid infiltrate

Parasites ("My there are LOTS of parasites")

- Myiasis: fly larva w/ thick undulating chitinous wall w/ pigmented setae
  - "My undulating chitinous wall"
- Larva currens: larval nematode within the dermis
- Cutaneous Larva migrans: may see worm in the epidermis
- Onchocerciasis: microfilaria within lymphatics

- Tungiasis: chitinous gravid flea w/ striated muscle and guts ("looks Trippy") on acral skin
- Tick: organism w/ chitinous wall, striated muscle and blood-filled guts attached to skin
- Sparganosis: Subcutaneous cestode w/ Smooth muscle and calcareous bodies
- Scabies: chitinous pigtails within the Stratum corneum

Miscellaneous dermpath differentials

Papillary dermal edema ("PASTE"): differentiate this based on cells type

- PMLE: superficial **lymphocytic** infiltrate; **not on acral skin** +/- dyskeratotic keratinocytes
- Perniosis: superficial and deep **lymphocytic** infiltrate which damaged vessels; **acral skin**
    - The deep lymphocytes are primarily around eccrine glands (vs follicles in lupus)
- Arthropod assault/Well's syndrome: superficial and deep **eosinophilic** infiltrate
- Sweet's syndrome: diffuse **neutrophilic** infiltrate
- Tinea (bullous): variable infiltrate w/ PAS+ hyphae in the epidermis
- Erythema multiforme: vacuolar infiltrate w/ keratinocyte death; mostly lymphocytes
    - Biopsy on edge looks like FDE (keratinocyte death, interface, eos, melanophages)
- Early EED: looks like Sweets (later lesions show onion skin fibrosis)

Invisible dermatosis: "TRUMPS ALIVE but we can't see him because he's invisible (cancelled)"

- TMEP: **subtle increase in perivascular spindle-shaped mast cells in the upper dermis**
- Tinea: non-pigmented hyphae in stratum corneum that is compact just above granular layer and normal basket weave on the very top (may see parakeratosis or neutrophils)
- Tinea versicolor: short curved hyphae and round spores (ziti and meatballs) in corneum
- Regressing melanoma: may see a subtle lichenoid inflammation w/ melanophages
- Urticaria: intravascular and perivascular neutrophilic infiltrate
- Ulerythema ophryogenes (and all KP variants): plugging of hair follicles + hyperkeratosis
- Macular amyloid: subtle pink material in the papillary dermis outlined by melanophages
- PXE: **curly calcified elastic fibers ("twigs on the bramble bush")** in the dermis
- Psoriasis (guttate): mounds of parakeratosis w/ neutrophils and mild spongiosis
- Scleredema: **mucin separating collagen in the deep dermis** (looks like normal back skin)
- Argyria: **tiny gray/black granules concentrated around eccrine glands**
- Atrophoderma (superficial morphea): dermis is less thick and individual collagen bundles are smaller (H&E changes subtle unless compared against normal adjacent skin)
- Anetoderma: elastic fibers are absent in the superficial and mid-dermis (normal H&E)
    - Mid-dermal elastolysis: absent elastic fibers in the mid-dermis only
    - Cutis laxa: absent elastic fibers at all levels of the dermis
- Lichen amyloid: same as macular amyloid but with overlying acanthosis + hyperkeratosis
- Ichthyosis vulgaris: **compact hyper-orthokeratosis with a paradoxically diminished granular layer** (pityriasis rotunda looks identical on path)
- Vitiligo: absence of melanocytes in basal layer
- Erythrasma: very small/thin G+ rods forming vertical filaments in the stratum corneum

Note, the mnemonic above are things that resemble normal skin at scan, **nevus anemicus** however is histologically and ultrastructurally identical to normal skin

Busy dermis ("Busy Dermis Can Kill Grandmas Sweet Niece")

- Blue nevus: pigmented dendritic melanocytes in a sclerotic dermis; pigment extends to the base but there are no deep mitoses or necrosis
- Desmoplastic melanoma: busy dermis w/ spindled cell proliferation in a sclerotic dermis that looks scar-like but cells are more hyperchromatic; **lymph aggregates** is a good clue
- Dermal spitz: enlarged epithelioid cells intersecting through collagen bundles; pigment
- Dermatofibroma: interstitial spindle cell proliferation, collagen trapping towards the periphery, overlying plate-like acanthosis w/ follicular induction
- Cutaneous metastasis: varies based on type for example breast carcinoma, carcinoma en cuirasse and leukemia cutis have large atypical cells lining up single file (Indian filing); RCC shows tubules of clear cells w/ prominent blood vessels and RBC extravasation
- Kaposi's (patch/plaque stage): dermal vessels gives a busy appearance; **promontory sign**
- Granuloma annulare (interstitial): patchy interstitial histiocytes, lymphocytes and **mucin**
- Scleromyxedema: ↑ dermal fibroblasts, fine collagen fibers and interstitial mucin
- Neurofibroma: comma-shaped spindle cells in a pale mucinous stroma w/ mast cells